Mechanical Criticality in Jamming Transitions

Edan Lerner

Institute for Theoretical Physics
University of Amsterdam

Mark Kac seminar, May 2026

acknowledgments

Matthieu Wyart

Gustavo Düring

Eran Bouchbinder

JOHNS HOPKINS
KRIEGER SCHOOL
of ARTS & SCIENCES

PONTIFICIA
UNIVERSIDAD
CATÓLICA
DE CHILE

WEIZMANN
INSTITUTE
OF SCIENCE

who am I?

Edan Lerner

who am I?

Edan Lerner

(pronunciation: **idaan**)

who am I?

Edan Lerner

(pronunciation: idaan)

(as in 'E-mail' → 'E-dan')

who am I?

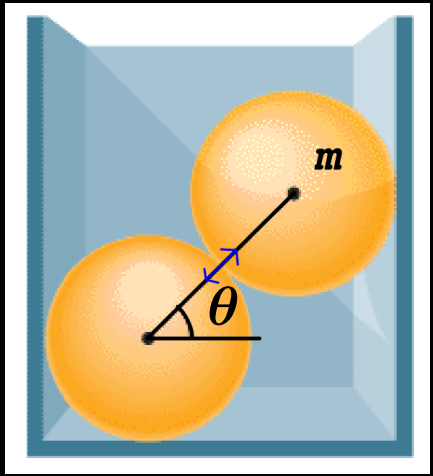
Edan Lerner

(pronunciation: idaan)

(as in 'E-mail' → 'E-dan')

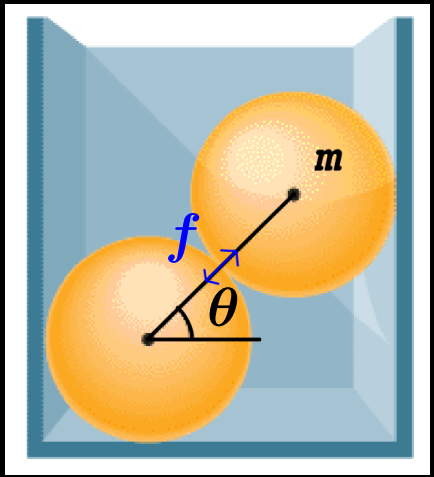
research field: classical stat. mech. of disordered systems

consider this system with **no friction**, subject to gravity g:



consider this system with **no friction**, subject to gravity g:

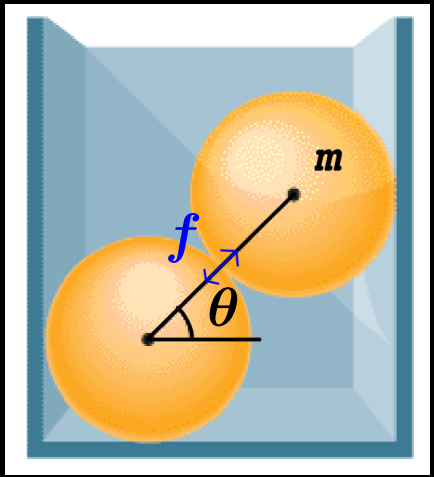
how does the **force** f between the spheres depend on the angle θ?



consider this system with **no friction**, subject to gravity g:

how does the **force** f between the spheres depend on the angle θ?

$$f \sim \frac{mg}{\theta}$$

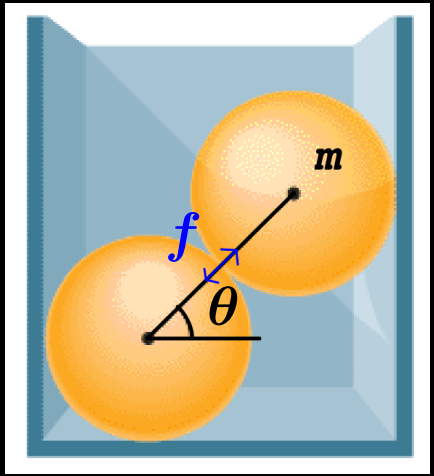


consider this system with **no friction**, subject to gravity g:

how does the **force** f between the spheres depend on the angle θ?

$$f \sim \frac{mg}{\theta}$$

pull of gravity is **finite**,
but response is **singular**
by virtue of **geometry**

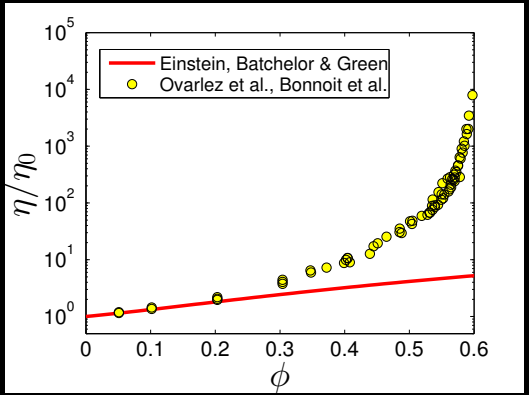


singular response by virtue of **geometry** (i)

immersed glass beads

singular response by virtue of **geometry** (i)

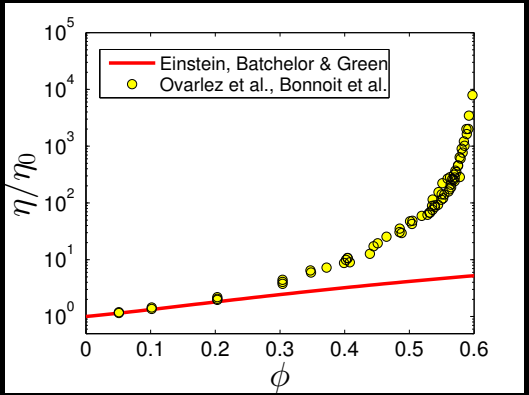
viscosity of **non-Brownian suspensions**



immersed glass beads

singular response by virtue of **geometry** (i)

viscosity of **non-Brownian suspensions**



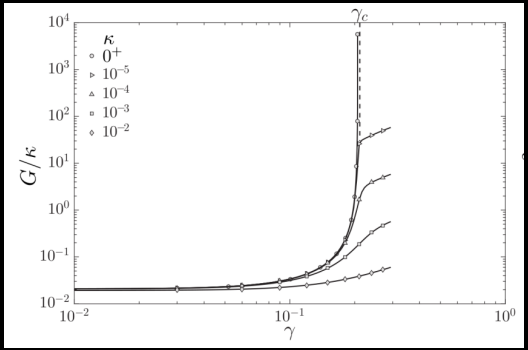
1/2 of 2nd hour story...

immersed glass beads

singular response by virtue of **geometry** (ii)

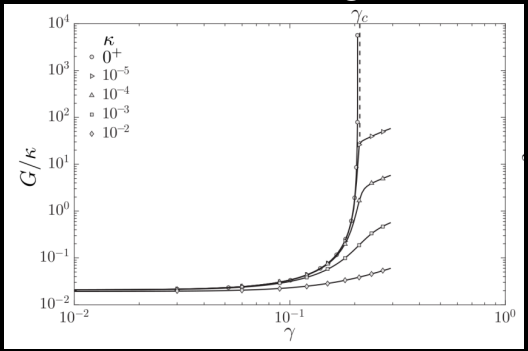
singular response by virtue of **geometry** (ii)

‘strain stiffening’



singular response by virtue of **geometry** (ii)

'strain stiffening'



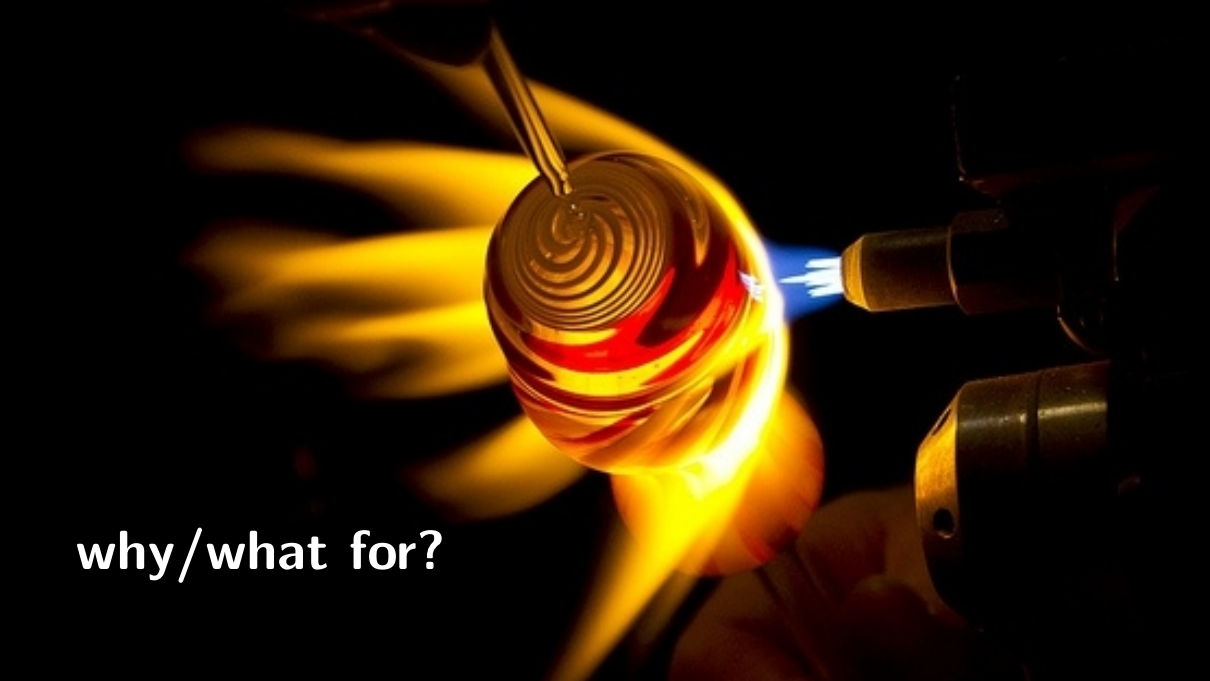
2/2 of 2nd hour story...

glass

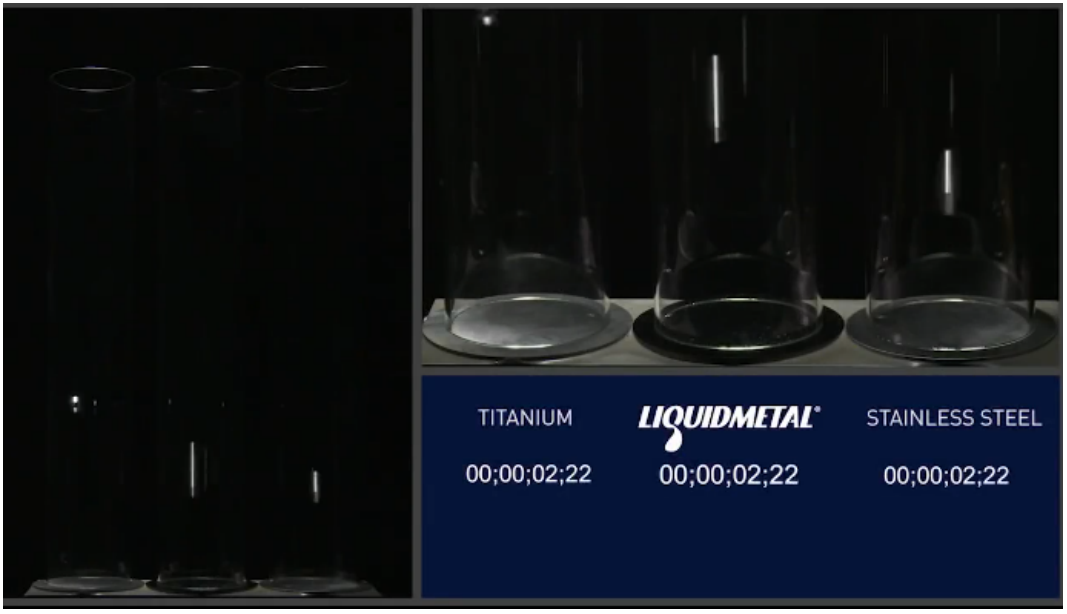
glass

mayo

metallic glass


why/what for?

making better materials (source: youtube)

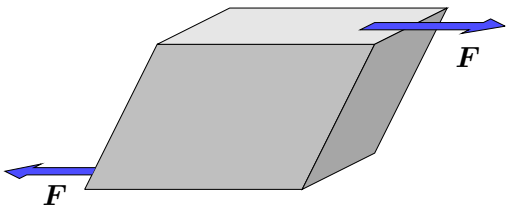


what are glasses?

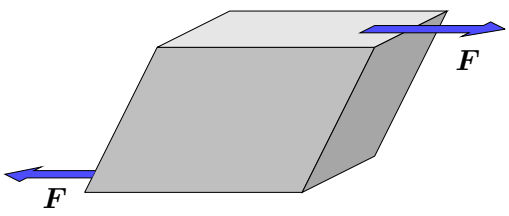
what are glasses?

(1) glasses are **solids** (on the timescale of the experiment)

(1) glasses are **solids** (on the timescale of the experiment)



(1) glasses are **solids** (on the timescale of the experiment)

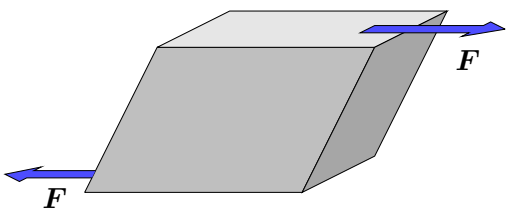


if the system:

— moves a bit* and stops moving**

→ **it's a solid**

(1) glasses are **solids** (on the timescale of the experiment)



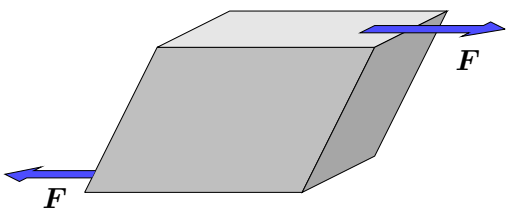
if the system:

— moves a bit* and stops moving**

→ **it's a solid**

*the force $\boldsymbol{F}$ must be "small enough"

(1) glasses are **solids** (on the timescale of the experiment)



if the system:

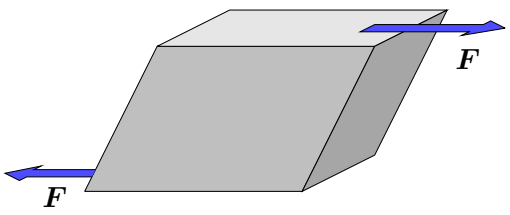
— moves a bit* and stops moving**

→ **it's a solid**

*the force $\boldsymbol{F}$ must be "small enough"

**over the experiment duration

(1) glasses are **solids** (on the timescale of the experiment)



if the system:

— moves a bit* and stops moving**

$\rightarrow$ **it's a solid**

— moves, and never stops moving

$\rightarrow$ **it's a liquid**

*the force $\boldsymbol{F}$ must be "small enough"

**over the experiment duration

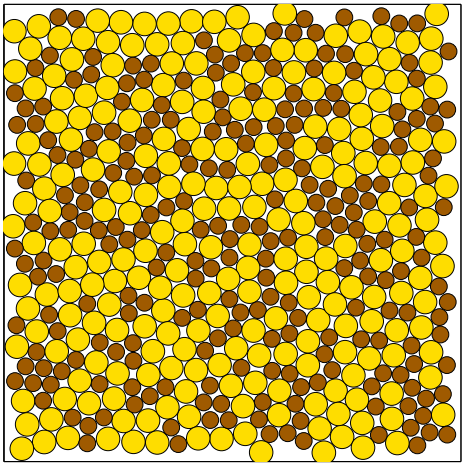
(2) glasses are **disordered** on **atomistic** scales

(2) glasses are **disordered** on **atomistic** scales

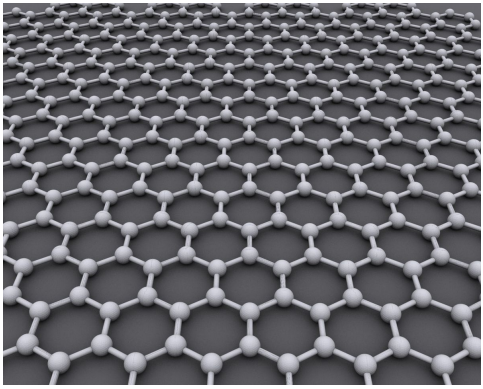
atomistic $\equiv$ on the scale of particle sizes

(2) glasses are **disordered** on **atomistic** scales

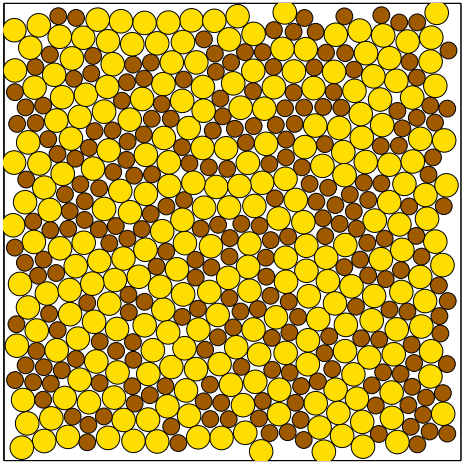
2D "computer glass"



(2) glasses are **disordered** on **atomistic** scales



most solids in nature are crystalline



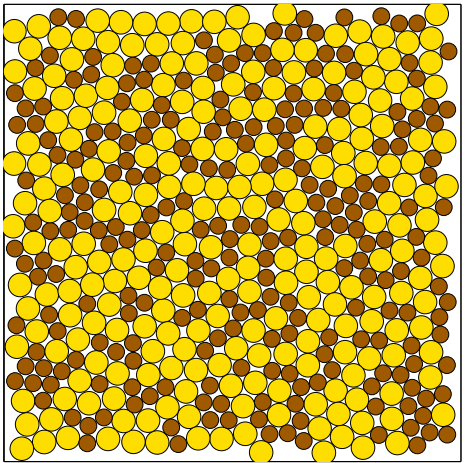
glasses

(2) glasses are **disordered** on **atomistic** scales

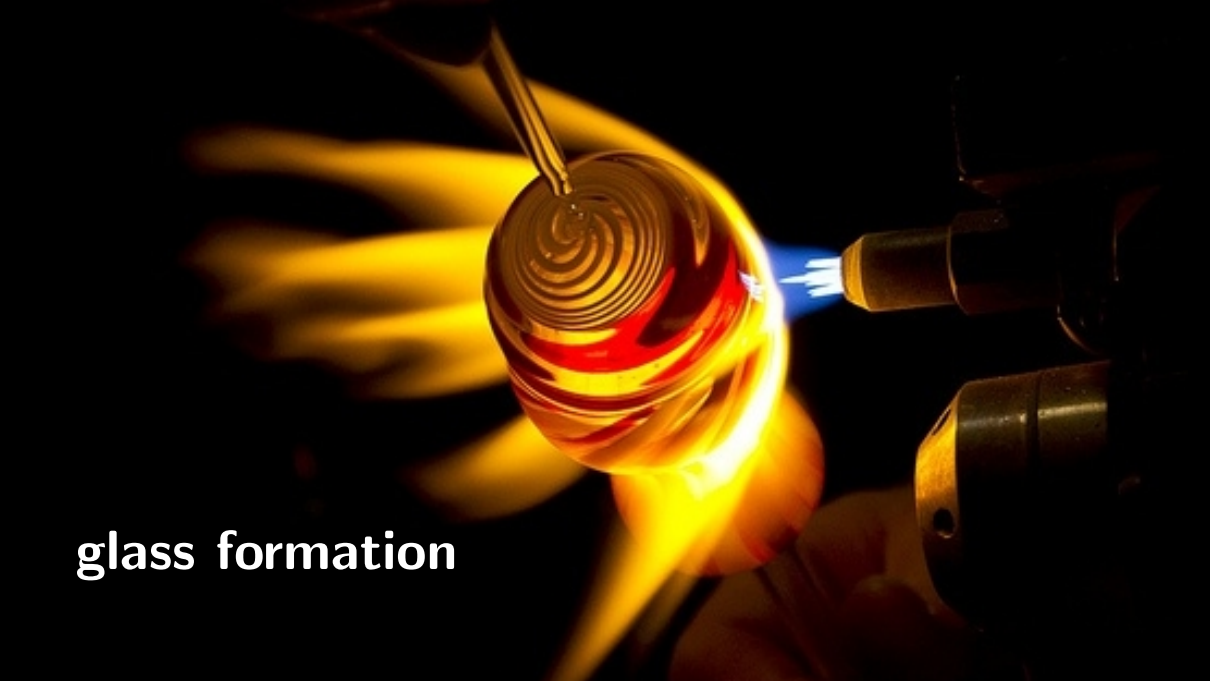
solids in thermodynamic-equilibrium

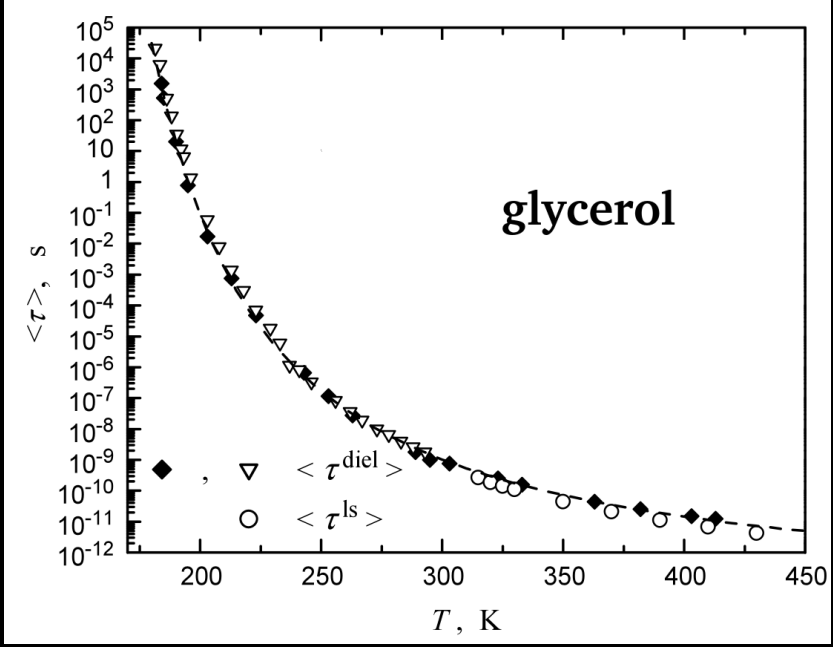
are crystals, therefore **glasses** are

intrinsically **nonequilibrium** systems



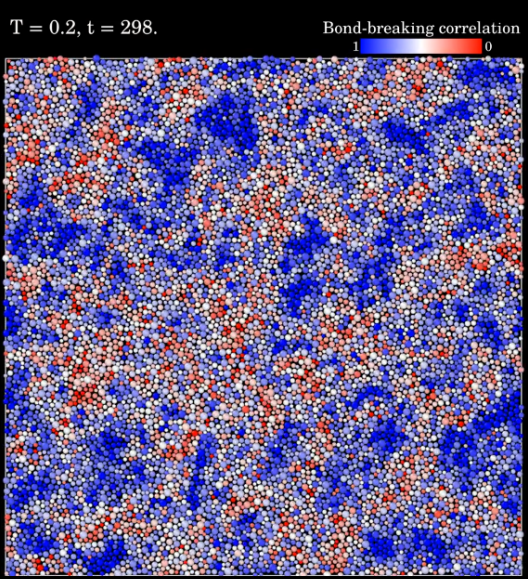
glasses

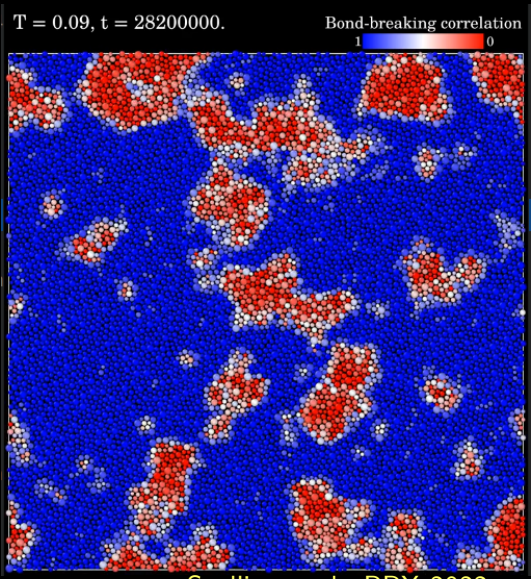

glass formation



Brodin & Rössler, EPJB 2005

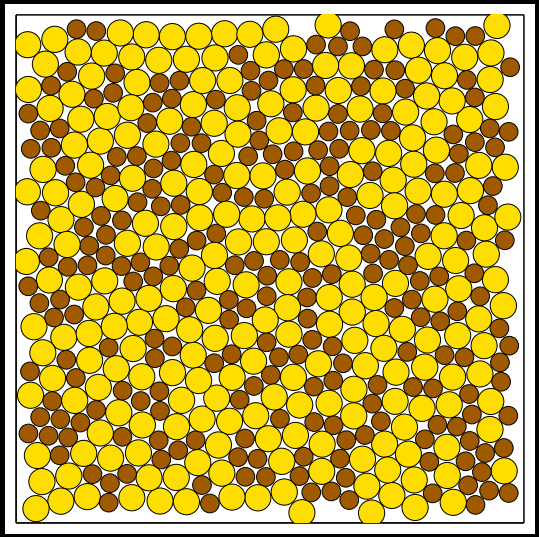
glass formation





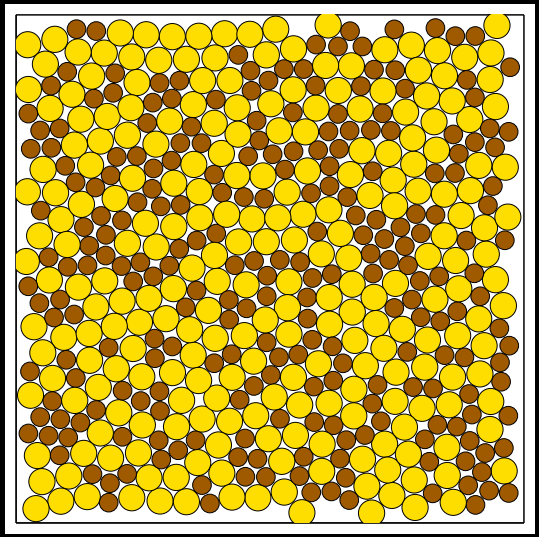
Scalliet et al., PRX 2022

glasses are solids whose microscopic structure is **disordered**



glasses are solids whose microscopic structure is **disordered**

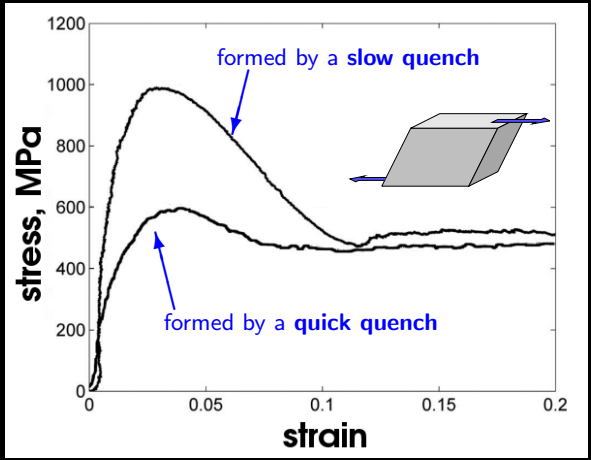
how does glass disorder
affect its **mechanical** properties?



glasses are solids whose microscopic structure is **disordered**

how does glass disorder
affect its **mechanical** properties?

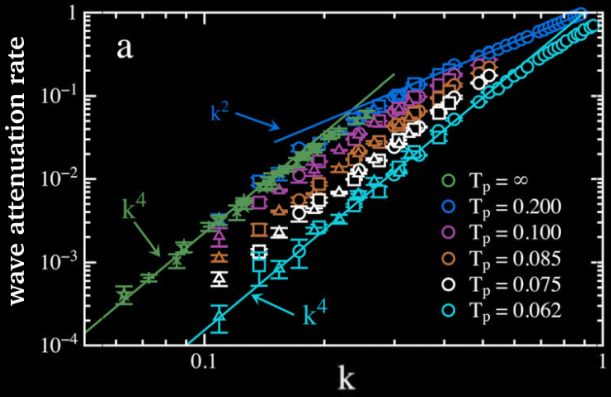
mechanical experiments on metallic glasses



glasses are solids whose microscopic structure is **disordered**

how does glass disorder
affect its **mechanical** properties?

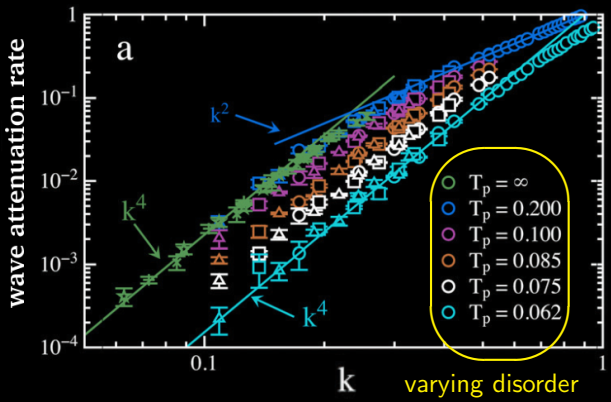
simulation measuring wave attenuation rates



glasses are solids whose microscopic structure is **disordered**

how does glass disorder
affect its **mechanical** properties?

simulation measuring wave attenuation rates

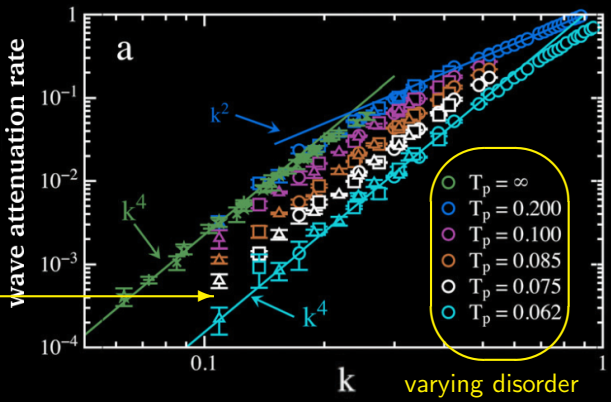


varying disorder

glasses are solids whose microscopic structure is **disordered**

how does glass disorder
affect its **mechanical** properties?

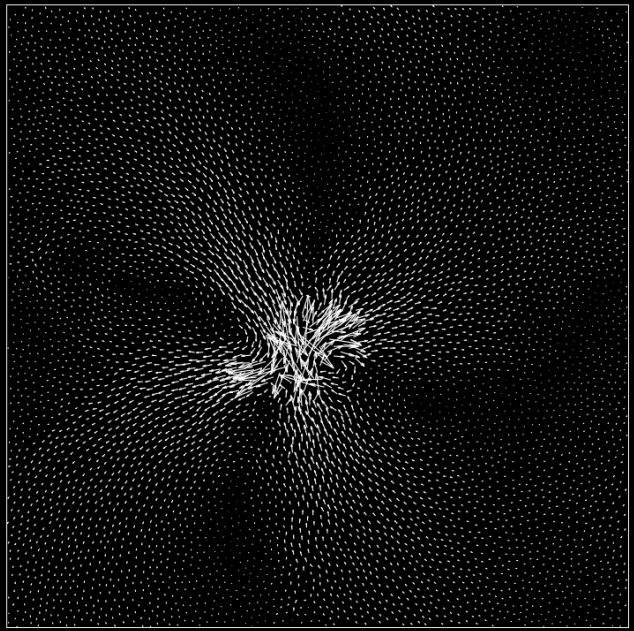
simulation measuring wave attenuation rates



over a decade in variability

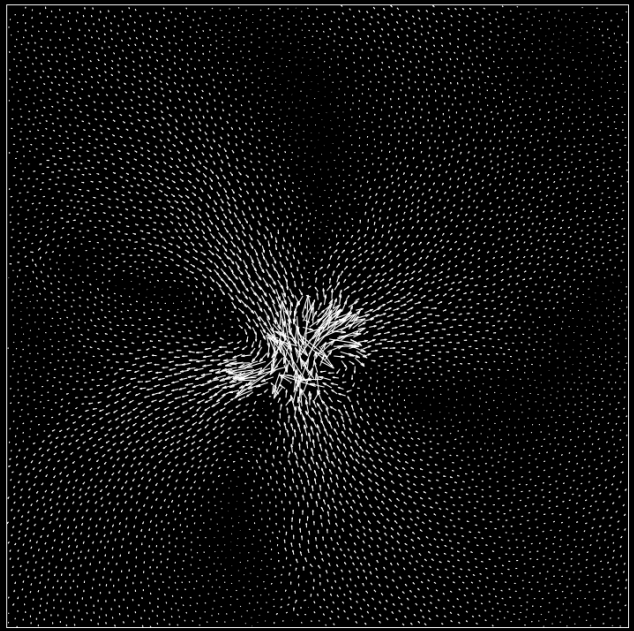
varying disorder

glasses have **defects**

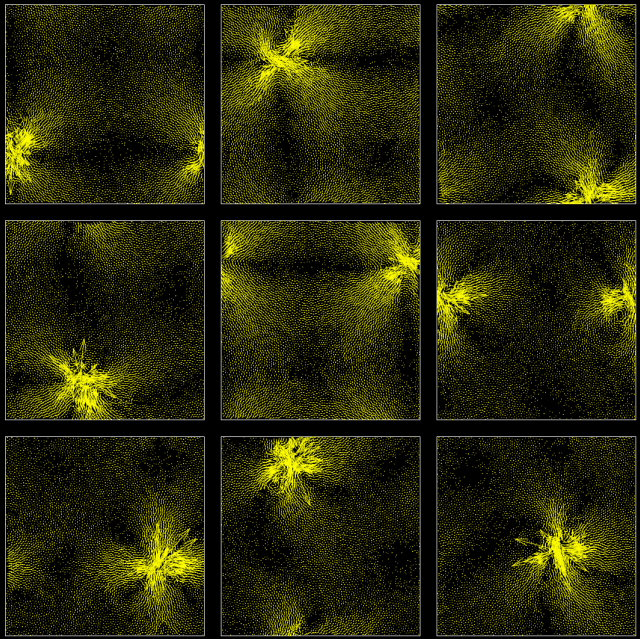


glasses have **defects**

these defects are
low-energy excitations
that emerge from the
glass's disordered nature

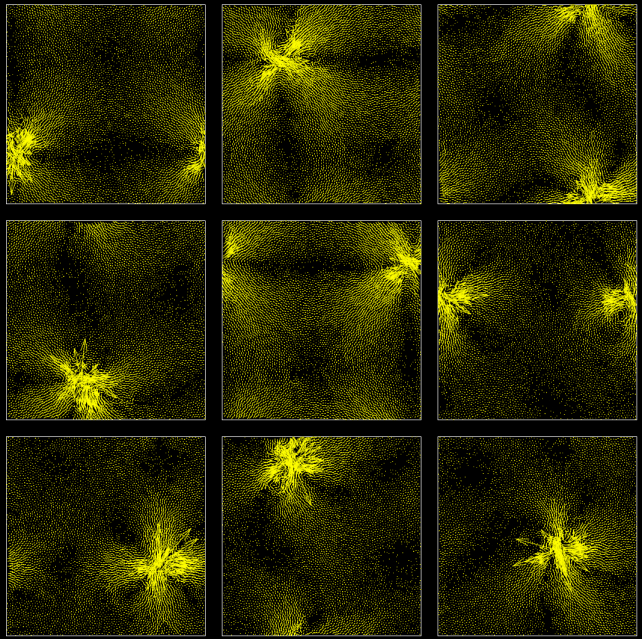


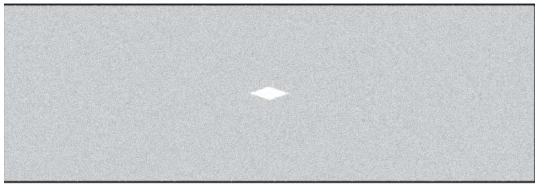
glasses have **defects**



glasses have **defects**

these defects control
how glasses break

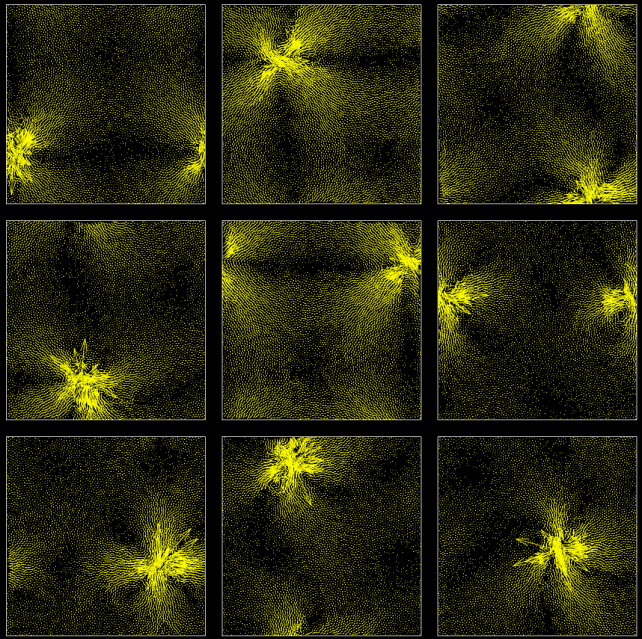


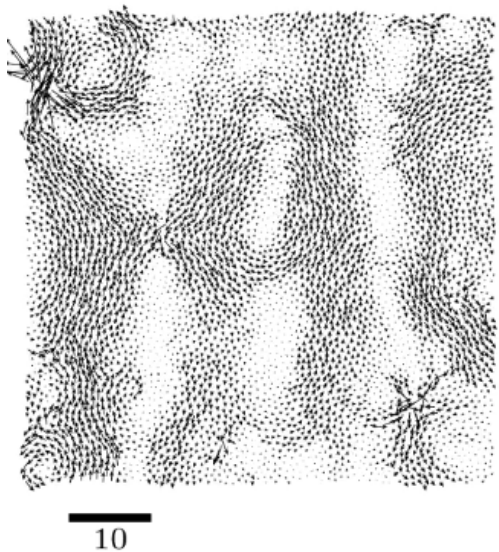


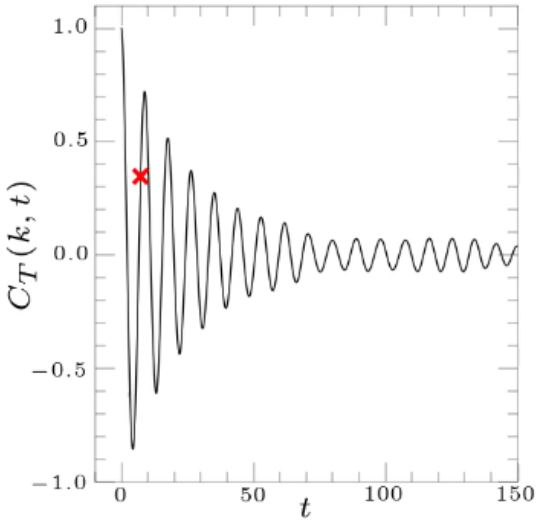
videos by David Richard

glasses have **defects**

these defects control
how glasses conduct heat



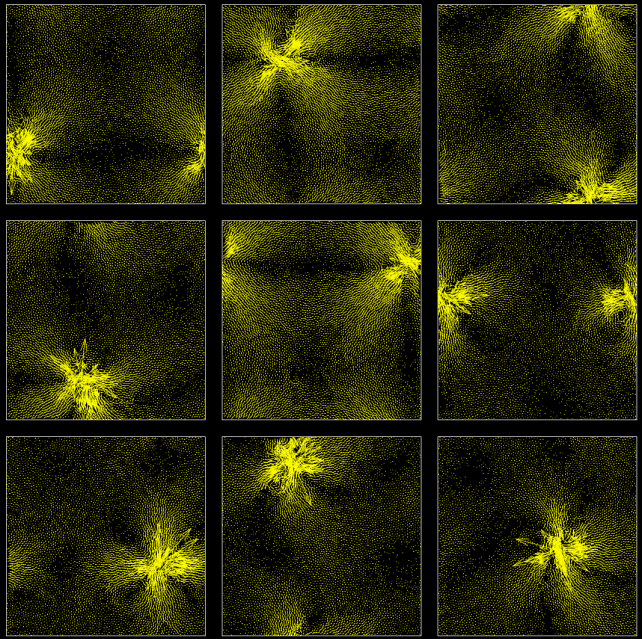




video by Anael Lemaitre

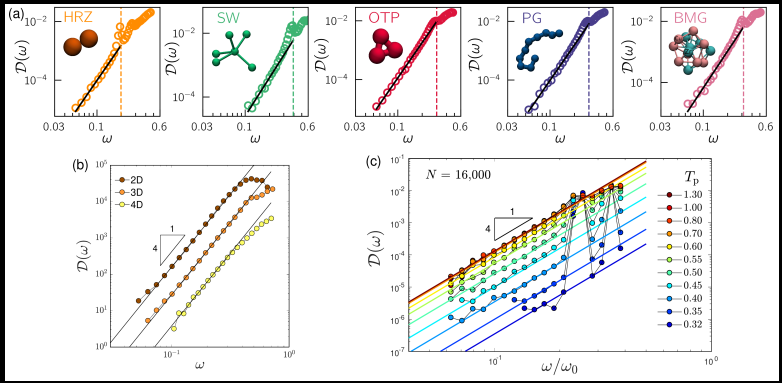
glasses have **defects**

these defects abide by
a universal distribution
over frequencies

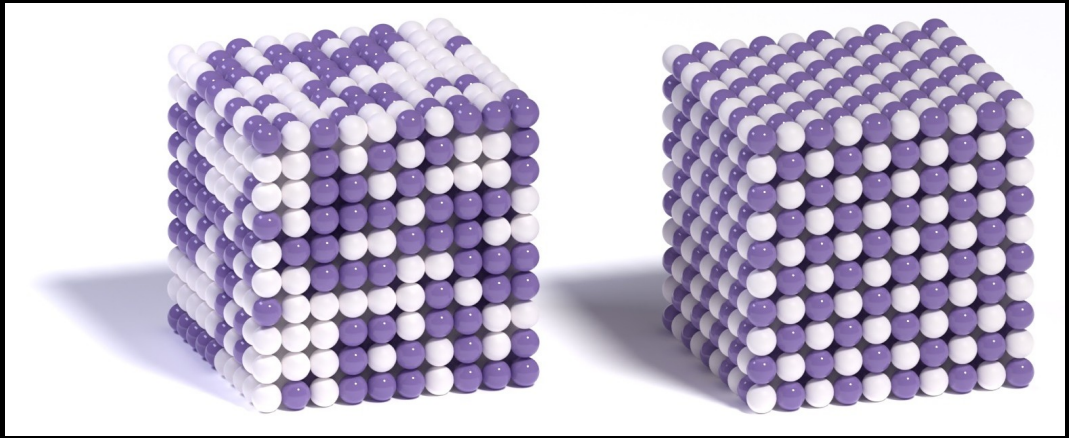


glasses have **defects**

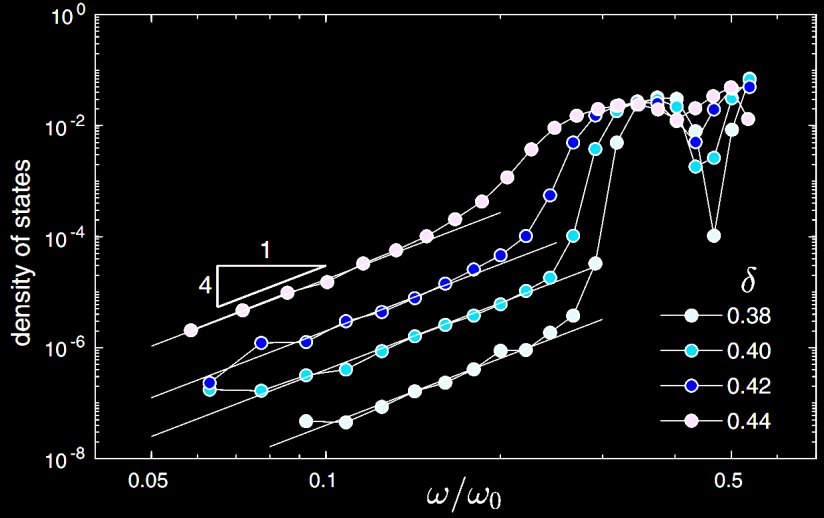
these defects abide by a **universal** distribution over frequencies



disordered crystals



disordered crystals have the *same* glassy defects & universal spectrum

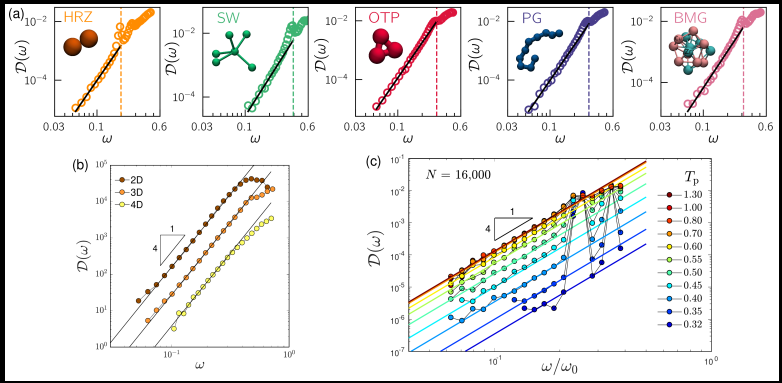


EL and Bouchbinder, PRL 2022

glasses have **defects**

still an open question!

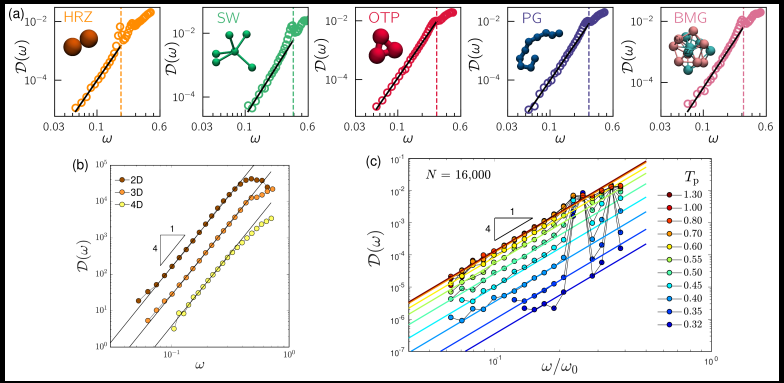
these defects abide by a **universal** distribution over frequencies



glasses have **defects**

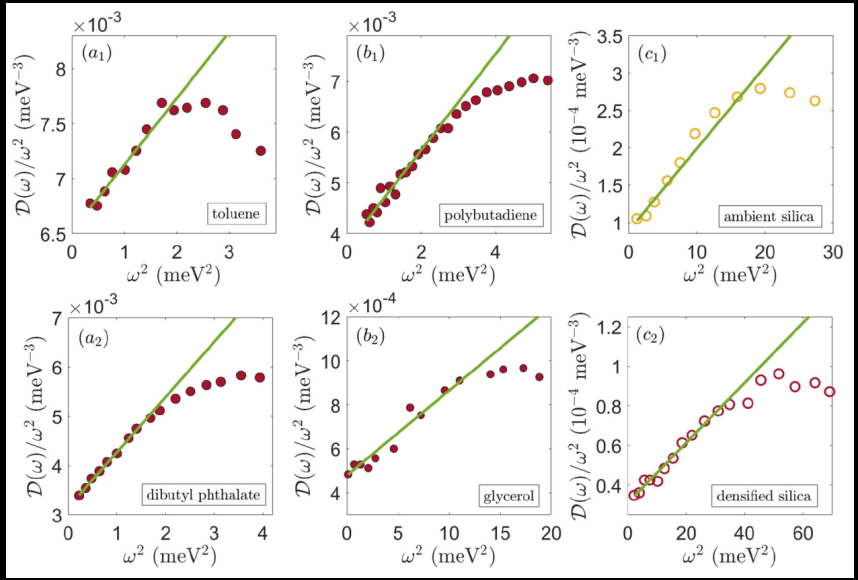
still an open question!

these defects abide by a **universal** distribution over frequencies



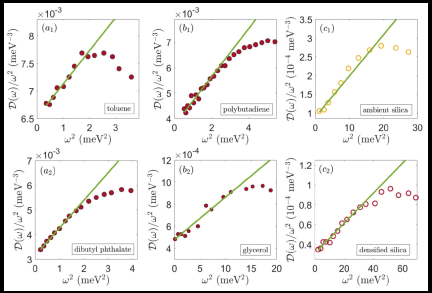
what about experiments?

experimental evidence for universal $\sim \omega^4$ nonphononic spectra:



Moriel, EL, and Bouchbinder, JAP 2024

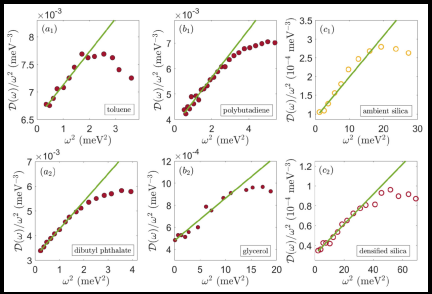
experimental evidence for universal $\sim \omega^4$ nonphononic spectra:



Moriel, EL, and Bouchbinder, JAP 2024

experimental evidence for universal $\sim \omega^4$ nonphononic spectra:

$$\mathcal{D}(\omega) = \frac{\hbar}{\omega_{\mathrm{D}}}\omega^2 + A_{\mathrm{g}}\omega^4 \qquad \text{assume additive structure}$$

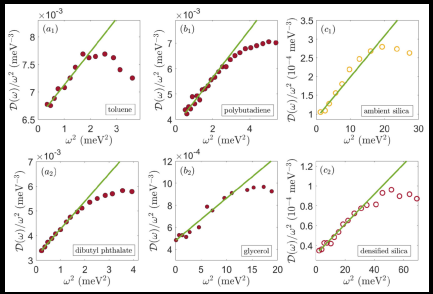


Moriel, EL, and Bouchbinder, JAP 2024

experimental evidence for universal $\sim \omega^4$ nonphononic spectra:

$$\mathcal{D}(\omega) = \frac{đ}{\omega_{\mathrm{D}}}\omega^2 + A_{\mathrm{g}}\omega^4$$ assume additive structure

$\Rightarrow$ extract the **dimensionless ratio** of frequencies $\dfrac{A_{\mathrm{g}}^{-1/5}}{\omega_{\mathrm{D}}}$

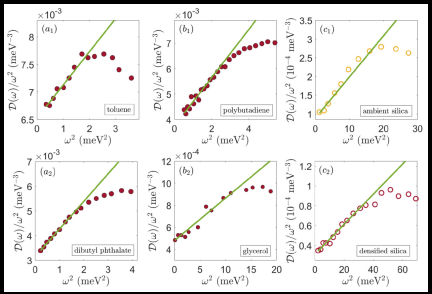


Moriel, EL, and Bouchbinder, JAP 2024

experimental evidence for universal $\sim \omega^4$ nonphononic spectra:

$$\mathcal{D}(\omega) = \frac{đ}{\omega_{\mathrm{D}}}\omega^2 + A_{\mathrm{g}}\omega^4$$ assume additive structure

$\Rightarrow$ extract the **dimensionless ratio** of frequencies $\dfrac{A_{\mathrm{g}}^{-1/5}}{\omega_{\mathrm{D}}}$



the experimentally-extracted ratio spans 0.30-0.57

the numerically extracted ratio spans 0.22-0.73

Moriel, EL, and Bouchbinder, JAP 2024

today: ‘jamming $\leftrightarrow$ unjamming’ transitions

today: ‘jamming $\leftrightarrow$ unjamming’ transitions

nonequilibrium mechanical phase transitions

between rigid/solid $\leftrightarrow$ floppy/liquid states of matter

‘jamming $\leftrightarrow$ unjamming’ transitions

today (2nd hour):

viscosity of non-Brownian suspensions

elasticity of fibrous matter

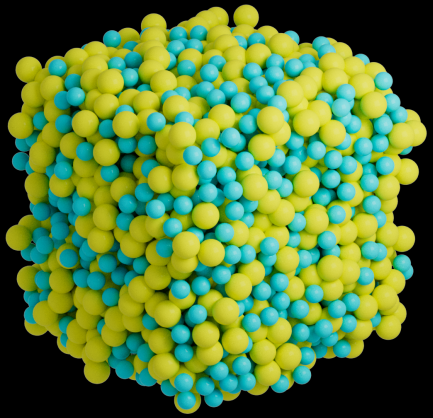
today (2nd hour):

viscosity of non-Brownian suspensions

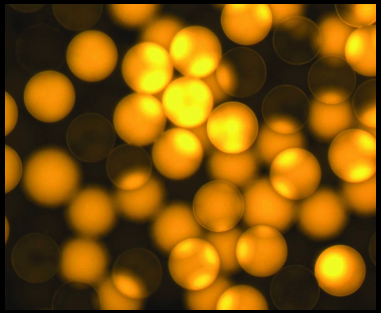
elasticity of fibrous matter

approach: develop scaling theories by perturbing critical state

‘jamming $\leftrightarrow$ unjamming’ transitions



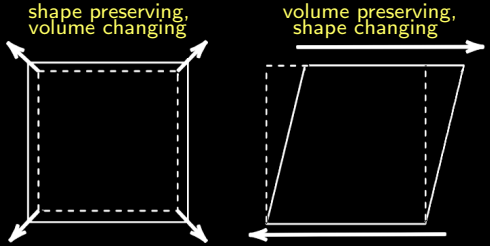
dense packings



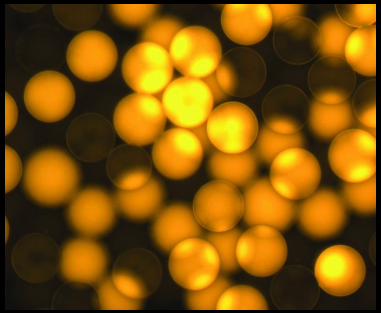
emulsions
(dense droplets)

foams
(dense bubbles)

'jamming $\leftrightarrow$ unjamming' transitions



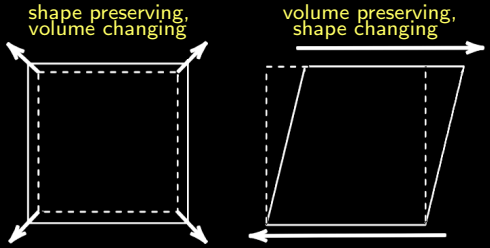
bulk modulus K shear modulus G



emulsions
(dense droplets)

foams
(dense bubbles)

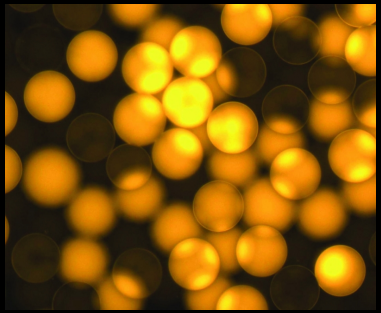
'jamming $\leftrightarrow$ unjamming' transitions



bulk modulus K shear modulus G

key dimensionless control parameter:

$$\frac{\text{pressure}}{\text{bulk modulus}} = \frac{p}{K}$$

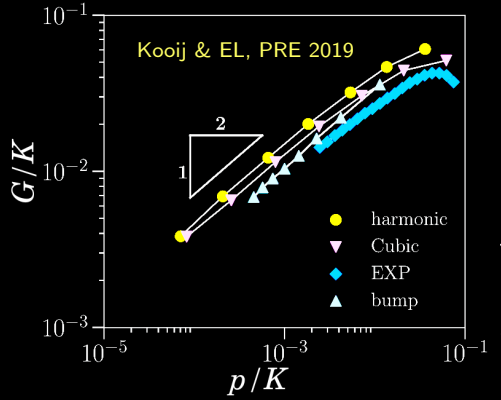


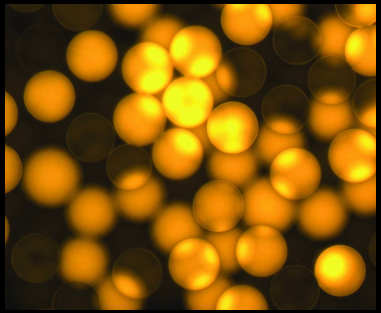
emulsions
(dense droplets)

foams
(dense bubbles)

'jamming ↔ unjamming' transitions

→ elastic moduli scaling: $G/K \sim \sqrt{p/K}$



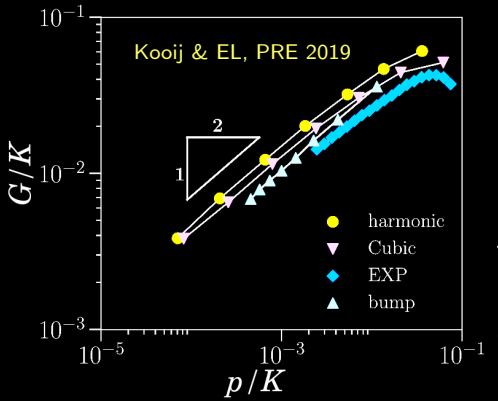


emulsions
(dense droplets)

foams
(dense bubbles)

'jamming ↔ unjamming' transitions

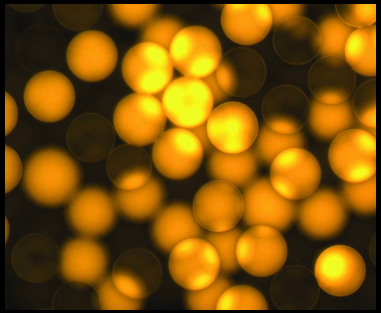
→ elastic moduli scaling: $G/K \sim \sqrt{p/K}$



implication on the Poisson's ratio:

$$\nu = \frac{3 - 2G/K}{6 + 2G/K} \sim \tfrac{1}{2}\left(1 - \sqrt{p/K}\right)$$

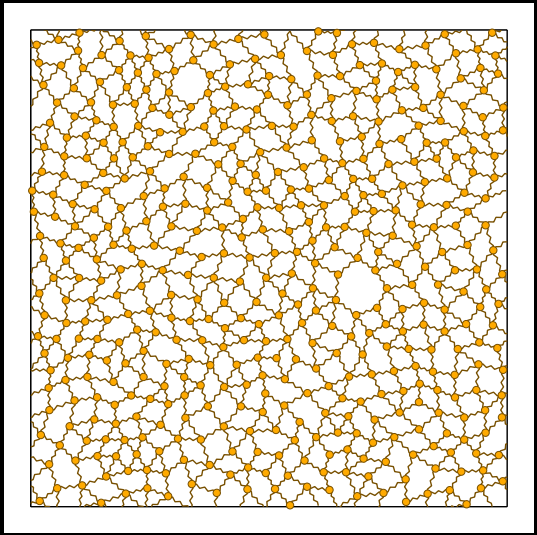
approach the incompressible limit ($\nu \to 1/2$) at unjamming



emulsions
(dense droplets)

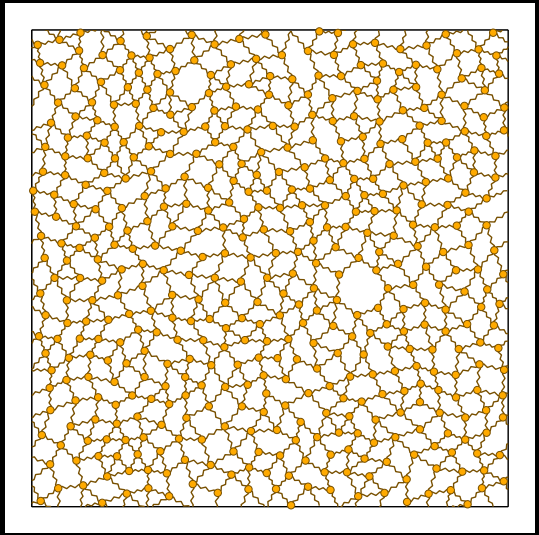
foams
(dense bubbles)

‘jamming $\leftrightarrow$ unjamming’ transitions

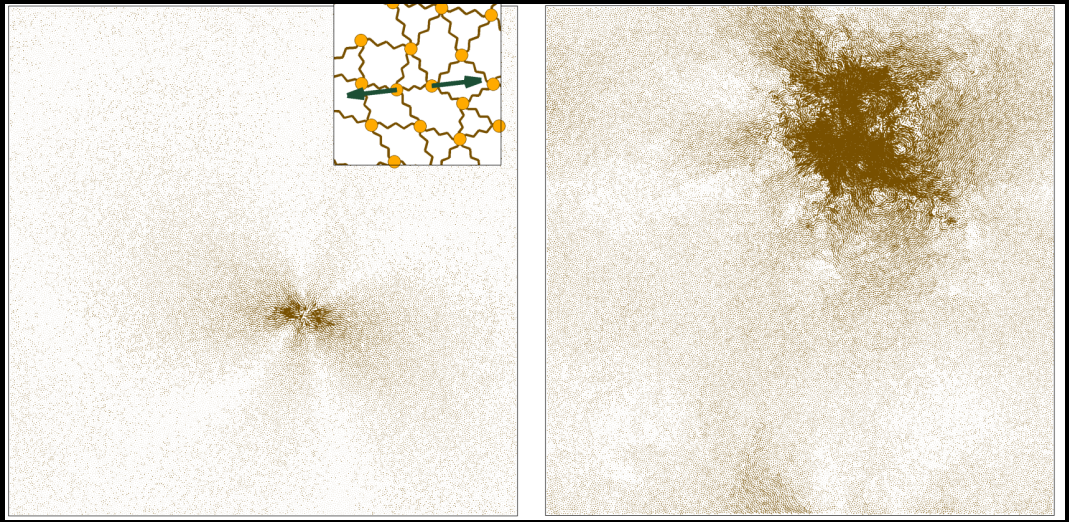


‘jamming $\leftrightarrow$ unjamming’ transitions

coordination z
controls mechanics



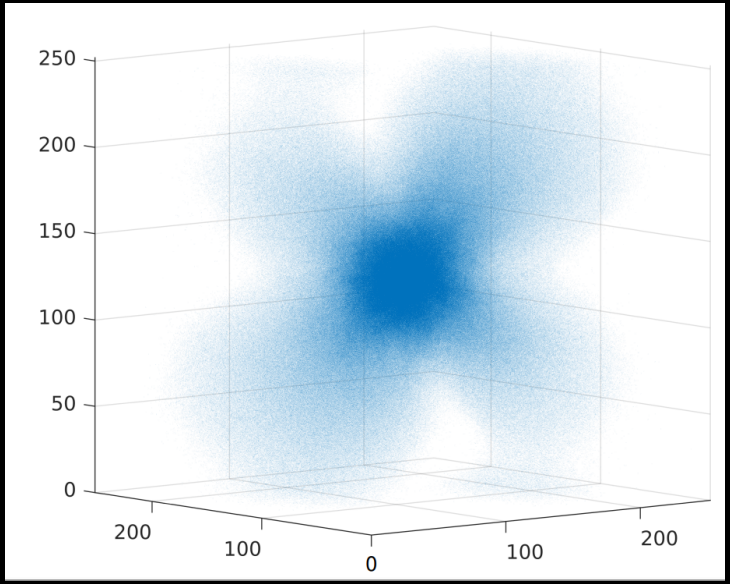
‘jamming ↔ unjamming’ transitions



high z low z

EL, Düring, Wyart, Soft Matter 2014

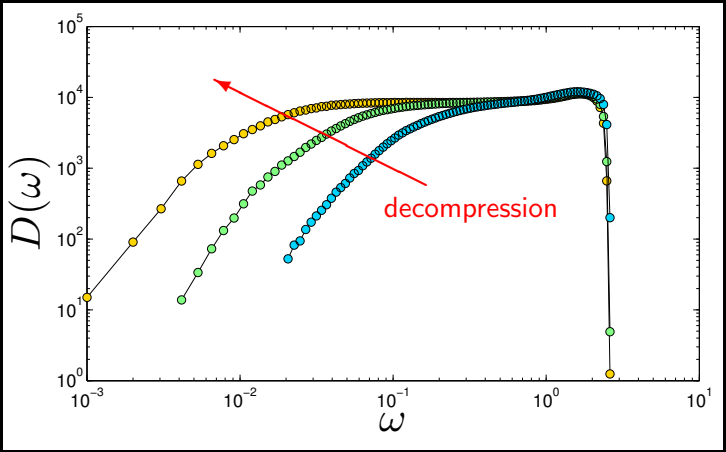
‘jamming $\leftrightarrow$ unjamming’ transitions

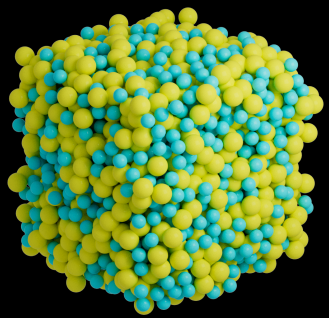


EL, Bouchbinder, Wyart, Soft Matter

'jamming ↔ unjamming' transitions

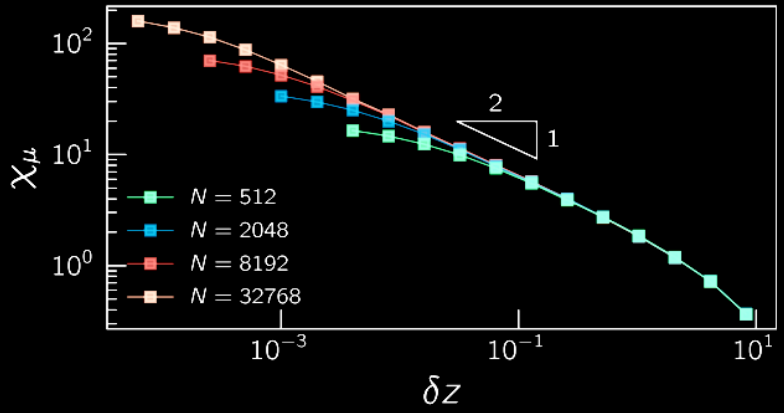
density of states (elastic vibrations)

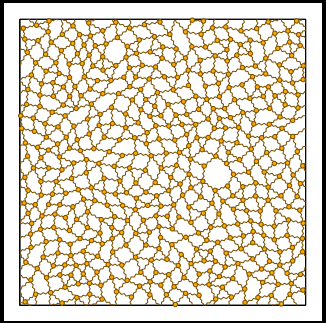




'jamming ↔ unjamming' transitions

sample-to-sample fluctuations of elastic moduli

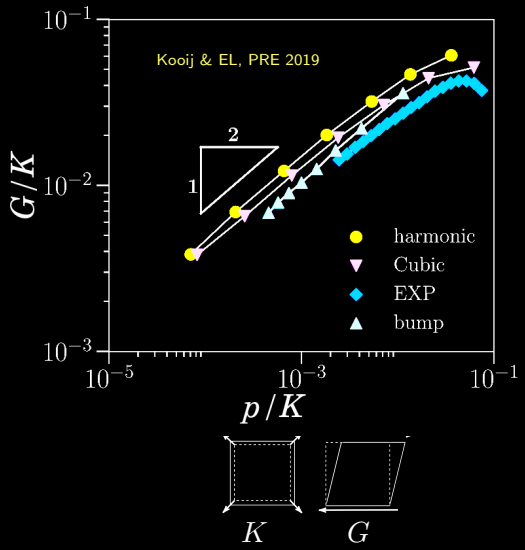


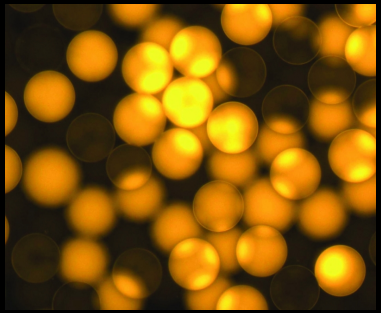


Giannini, EL, Zamponi, Manning, JCP 2024

today (before break):

shear/bulk modulus scaling: $G/K \sim \sqrt{p/K}$

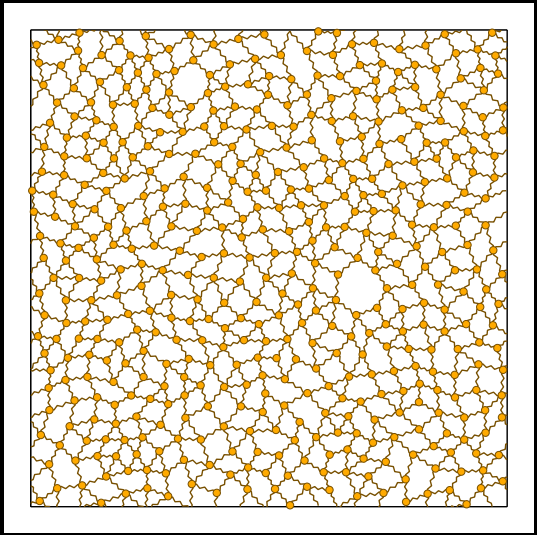




emulsions
(dense droplets)

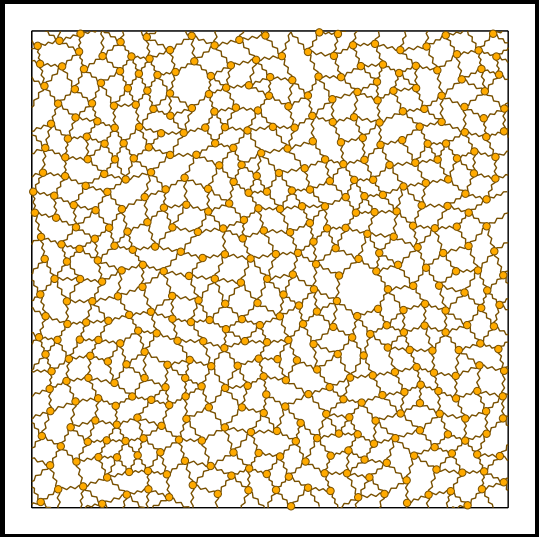
foams
(dense bubbles)

consider disordered bead-spring networks of **relaxed** Hookean springs of stiffness $k=1$



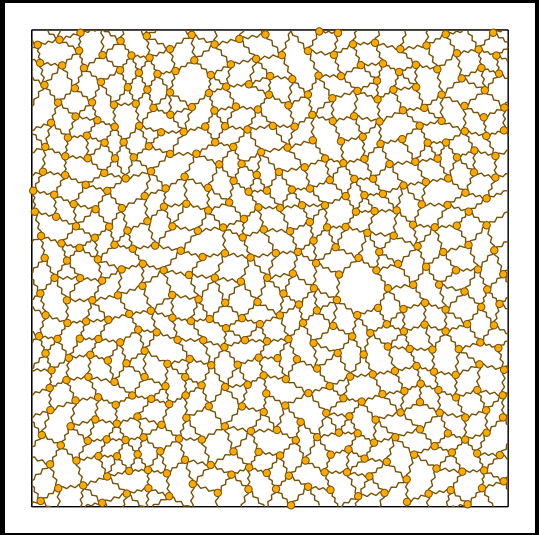
consider disordered bead-spring networks of **relaxed** Hookean springs of stiffness $k=1$

coordination z
controls mechanics



consider disordered bead-spring networks of **relaxed** Hookean springs of stiffness $k=1$

coordination z
controls mechanics

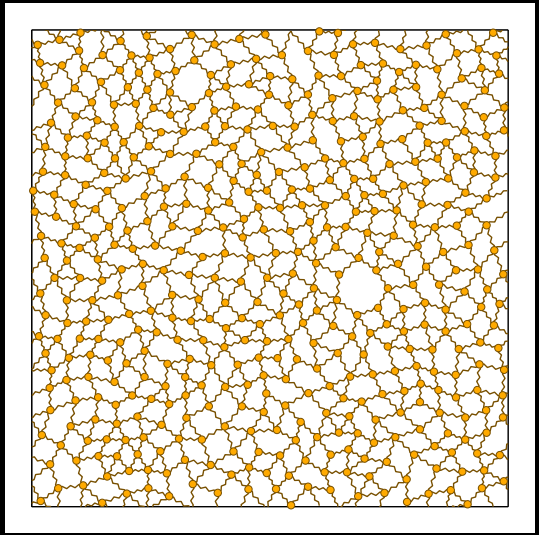


goal:

how do elastic moduli depend on z?

consider disordered bead-spring networks of **relaxed** Hookean springs of stiffness $k=1$

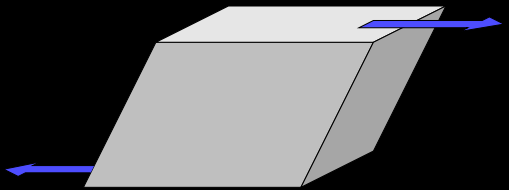
coordination z
controls mechanics



goal:

how do **elastic moduli** depend on z?

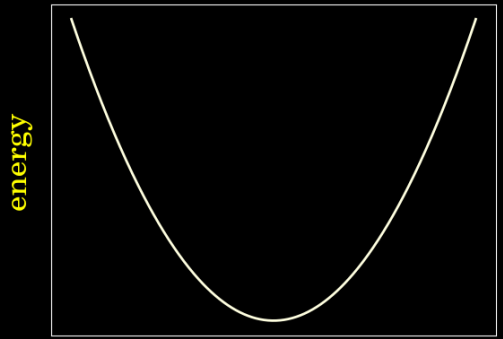
elastic moduli



goal:

how do **elastic moduli** depend on z?

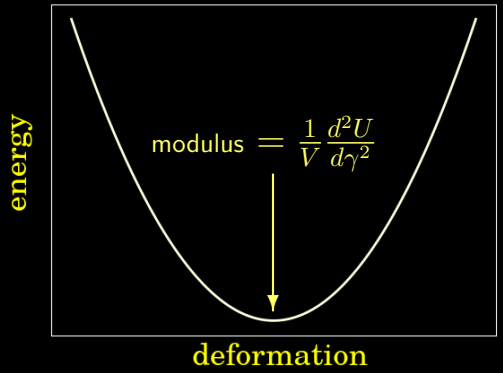
elastic moduli



goal:

how do **elastic moduli** depend on z?

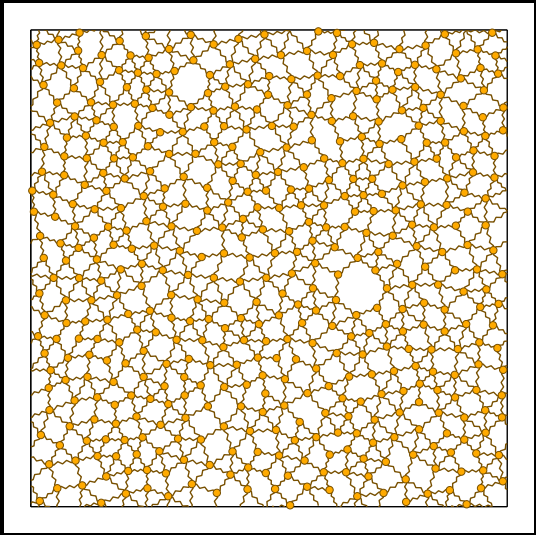
elastic moduli



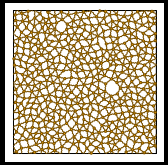
goal:

how do **elastic moduli** depend on z?

mechanics of disordered networks of **relaxed** Hookean springs



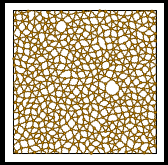
mechanics of disordered networks of **relaxed** Hookean springs



$\rightarrow$ the energy is simply $U = \sum_{i \leftrightarrow j} \frac{1}{2} k (r_{ij} - \ell_{ij})^2$,

where ℓ_{ij} is the **rest length** of the $i \leftrightarrow j$ spring

mechanics of disordered networks of **relaxed** Hookean springs

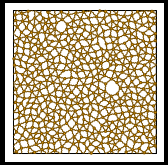


→ the energy is simply $U = \sum_{i \leftrightarrow j} \frac{1}{2} k (r_{ij} - \ell_{ij})^2$,

where ℓ_{ij} is the **rest length** of the $i \leftrightarrow j$ spring

→ since we take all $r_{ij} = \ell_{ij}$, the energy is **zero**, and there are **no forces** in the springs

mechanics of disordered networks of **relaxed** Hookean springs



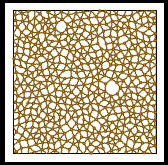
→ the energy is simply $U = \sum_{i \leftrightarrow j} \frac{1}{2} k (r_{ij} - \ell_{ij})^2$,

where ℓ_{ij} is the **rest length** of the $i \leftrightarrow j$ spring

→ since we take all $r_{ij} = \ell_{ij}$, the energy is **zero**, and there are **no forces** in the springs

→ choose units such that $k = 1$, now write the hessian matrix

mechanics of disordered networks of **relaxed** Hookean springs



→ the energy is simply $U = \sum_{i \leftrightarrow j} \frac{1}{2} k (r_{ij} - \ell_{ij})^2$,

where ℓ_{ij} is the **rest length** of the $i \leftrightarrow j$ spring

→ since we take all $r_{ij} = \ell_{ij}$, the energy is **zero**, and there are **no forces** in the springs

→ choose units such that $k = 1$, now write the hessian matrix

$$\mathcal{H}_{k\ell} \equiv \frac{\partial^2 U}{\partial \boldsymbol{x}_k \partial \boldsymbol{x}_\ell} = \sum_{i \leftrightarrow j} \frac{\partial r_{ij}}{\partial \boldsymbol{x}_k} \frac{\partial r_{ij}}{\partial \boldsymbol{x}_\ell} = \mathcal{S}^T \mathcal{S}$$

→ what is the geometric meaning of the object $\mathcal{S}_{ij,k} \equiv \frac{\partial r_{ij}}{\partial \boldsymbol{x}_k}$?

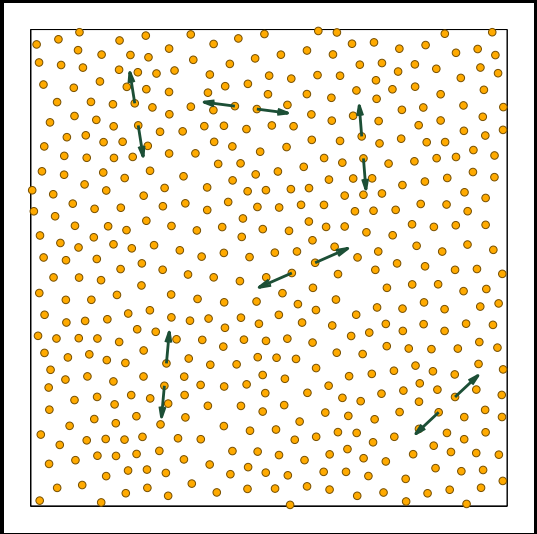
the matrix $\boldsymbol{\mathcal{S}}_{ij,k} \equiv \dfrac{\partial r_{ij}}{\partial \boldsymbol{x}_k} = (\delta_{jk} - \delta_{ik})\dfrac{\boldsymbol{x}_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik})\boldsymbol{n}_{ij}$

the matrix $\boldsymbol{\mathcal{S}}_{ij,k} \equiv \dfrac{\partial r_{ij}}{\partial \boldsymbol{x}_k} = (\delta_{jk} - \delta_{ik})\dfrac{\boldsymbol{x}_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik})\boldsymbol{n}_{ij}$

$\boldsymbol{\mathcal{S}}_{ij,k}$ is a **vector** that lives (in fact, it is a **dipole**) only on nodes i, j

the matrix $\boldsymbol{\mathcal{S}}_{ij,k} \equiv \dfrac{\partial r_{ij}}{\partial \boldsymbol{x}_k} = (\delta_{jk} - \delta_{ik})\dfrac{\boldsymbol{x}_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik})\boldsymbol{n}_{ij}$

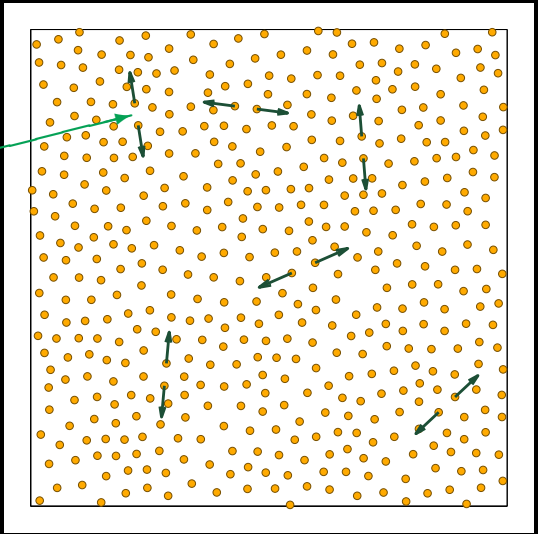
$\boldsymbol{\mathcal{S}}_{ij,k}$ is a **vector** that lives (in fact, it is a **dipole**) only on nodes i, j



the matrix $\boldsymbol{\mathcal{S}}_{ij,k} \equiv \dfrac{\partial r_{ij}}{\partial \boldsymbol{x}_k} = (\delta_{jk} - \delta_{ik})\dfrac{\boldsymbol{x}_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik})\boldsymbol{n}_{ij}$

$\boldsymbol{\mathcal{S}}_{ij,k}$ is a **vector** that lives (in fact, it is a **dipole**) only on nodes i, j

each dipole is a different $i \leftrightarrow j$



the matrix $\boldsymbol{\mathcal{S}}_{ij,k} \equiv \dfrac{\partial r_{ij}}{\partial \boldsymbol{x}_k} = (\delta_{jk} - \delta_{ik})\dfrac{\boldsymbol{x}_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik})\boldsymbol{n}_{ij}$

$\boldsymbol{\mathcal{S}}_{ij,k}$ is a **vector** that lives (in fact, it is a **dipole**) only on nodes i, j

$\boldsymbol{\mathcal{S}}$ can be thought of as a **linear operator** that takes vectors from particles' coordinate space, to the space of **springs/interactions/bonds/contacts**:

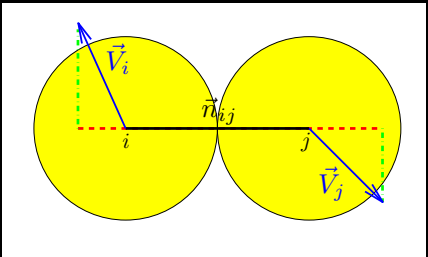
$$\mathcal{S}|V\rangle = |v\rangle$$

the matrix $\boldsymbol{\mathcal{S}}_{ij,k} \equiv \dfrac{\partial r_{ij}}{\partial \boldsymbol{x}_k} = (\delta_{jk} - \delta_{ik})\dfrac{\boldsymbol{x}_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik})\boldsymbol{n}_{ij}$

$\boldsymbol{\mathcal{S}}_{ij,k}$ is a **vector** that lives (in fact, it is a **dipole**) only on nodes i, j

$\boldsymbol{\mathcal{S}}$ can be thought of as a **linear operator** that takes vectors from particles' coordinate space, to the space of **springs/interactions/bonds/contacts**:

$$\mathcal{S}|V\rangle = |v\rangle$$

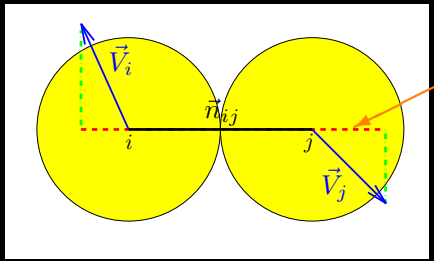


the matrix $\boldsymbol{\mathcal{S}}_{ij,k} \equiv \dfrac{\partial r_{ij}}{\partial \boldsymbol{x}_k} = (\delta_{jk} - \delta_{ik})\dfrac{\boldsymbol{x}_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik})\boldsymbol{n}_{ij}$

$\boldsymbol{\mathcal{S}}_{ij,k}$ is a **vector** that lives (in fact, it is a **dipole**) only on nodes i, j

$\boldsymbol{\mathcal{S}}$ can be thought of as a **linear operator** that takes vectors from particles' coordinate space, to the space of **springs/interactions/bonds/contacts**:

$$\mathcal{S}|V\rangle = |v\rangle$$



v_{ij} = **sum** of the red dashed lines

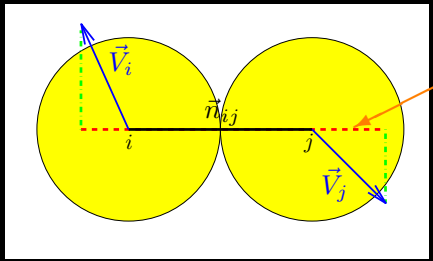
the matrix $\boldsymbol{\mathcal{S}}_{ij,k} \equiv \dfrac{\partial r_{ij}}{\partial \boldsymbol{x}_k} = (\delta_{jk} - \delta_{ik})\dfrac{\boldsymbol{x}_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik})\boldsymbol{n}_{ij}$

$\boldsymbol{\mathcal{S}}_{ij,k}$ is a **vector** that lives (in fact, it is a **dipole**) only on nodes i, j

$\mathcal{S}$ can be thought of as a **linear operator** that takes vectors from particles' coordinate space, to the space of **springs/interactions/bonds/contacts**:

$$\mathcal{S}|V\rangle = |v\rangle$$

notice that $\mathcal{S}$ is not necessarily square!



v_{ij} = **sum** of the red dashed lines

the matrix $\boldsymbol{\mathcal{S}}_{ij,k} \equiv \dfrac{\partial r_{ij}}{\partial \boldsymbol{x}_k} = (\delta_{jk} - \delta_{ik})\dfrac{\boldsymbol{x}_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik})\boldsymbol{n}_{ij}$

$\boldsymbol{\mathcal{S}}_{ij,k}$ is a **vector** that lives (in fact, it is a **dipole**) only on nodes i, j

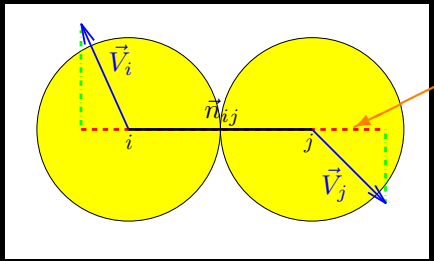
$\mathcal{S}$ can be thought of as a **linear operator** that takes vectors from particles' coordinate space, to the space of **springs/interactions/bonds/contacts**:

$$\mathcal{S}|V\rangle = |v\rangle$$

notice that $\mathcal{S}$ is not necessarily square!

$N \times d$ columns

N_c ($\#$ of contacts/springs) rows



v_{ij} = **sum** of the red dashed lines

the matrix $\boldsymbol{\mathcal{S}}_{ij,k} \equiv \dfrac{\partial r_{ij}}{\partial \boldsymbol{x}_k} = (\delta_{jk} - \delta_{ik})\dfrac{\boldsymbol{x}_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik})\boldsymbol{n}_{ij}$

$\boldsymbol{\mathcal{S}}_{ij,k}$ is a **vector** that lives (in fact, it is a **dipole**) only on nodes i, j

$\mathcal{S}$ can be thought of as a **linear operator** that takes vectors from particles' coordinate space, to the space of **springs/interactions/bonds/contacts**:

$$\mathcal{S}|V\rangle = |v\rangle$$

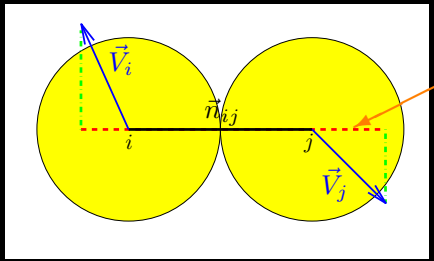
notice that $\mathcal{S}$ is not necessarily square!

$N \times d$ columns

N_c ($\#$ of contacts/springs) rows

square when $2N_c = N \times d$

i.e. when $z \equiv \dfrac{2N_c}{N} = 2d \equiv z_c$



v_{ij} = **sum** of the red dashed lines

the matrix $\boldsymbol{\mathcal{S}}_{ij,k} \equiv \dfrac{\partial r_{ij}}{\partial \boldsymbol{x}_k} = (\delta_{jk} - \delta_{ik})\dfrac{\boldsymbol{x}_{ij}}{r_{ij}} = (\delta_{jk} - \delta_{ik})\boldsymbol{n}_{ij}$

$\boldsymbol{\mathcal{S}}_{ij,k}$ is a **vector** that lives (in fact, it is a **dipole**) only on nodes i, j

$\mathcal{S}$ can be thought of as a **linear operator** that takes vectors from particles' coordinate space, to the space of **springs/interactions/bonds/contacts**:

$$\mathcal{S}|V\rangle = |v\rangle$$

notice that $\mathcal{S}$ is not necessarily square!

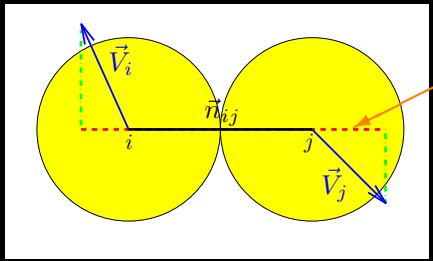
$N \times \check{d}$ columns

N_c (# of contacts/springs) rows

square when $2N_c = N \times \check{d}$

i.e. when $z \equiv \dfrac{2N_c}{N} = 2\check{d} \equiv z_c$

$z_c = 2\check{d}$ is the **Maxwell threshold**



v_{ij} = **sum** of the red dashed lines

under what conditions can **homogeneous solutions** exist?

$$\mathcal{S}|V\rangle = |0\rangle$$

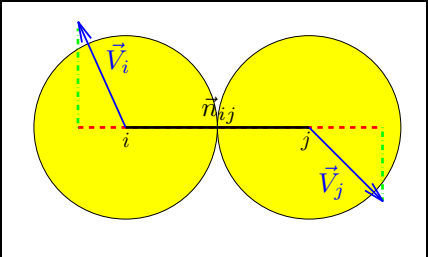
notice that $\mathcal{S}$ is not necessarily square!

$N \times \vec{d}$ columns

N_c (# of contacts/springs) rows

square when $2N_c = N \times \vec{d}$

i.e. when $z \equiv \dfrac{2N_c}{N} = 2\vec{d} \equiv z_c$



under what conditions can **homogeneous solutions** exist?

$\rightarrow$ when $2N_c < N \times \hat{d}$

or when $z < z_c = 2\hat{d}$

$$\mathcal{S}|V\rangle = |0\rangle$$

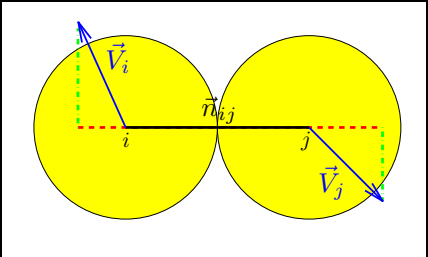
notice that $\mathcal{S}$ is not necessarily square!

$N \times \hat{d}$ columns

N_c (# of contacts/springs) rows

square when $2N_c = N \times \hat{d}$

i.e. when $z \equiv \dfrac{2N_c}{N} = 2\hat{d} \equiv z_c$



under what conditions can **homogeneous solutions** exist?

$$\mathcal{S}|V\rangle = |0\rangle$$

$\rightarrow$ when $2N_c < N \times \check{d}$

or when $z < z_c = 2\check{d}$

we expect $\sim N(z_c - z)$ of them

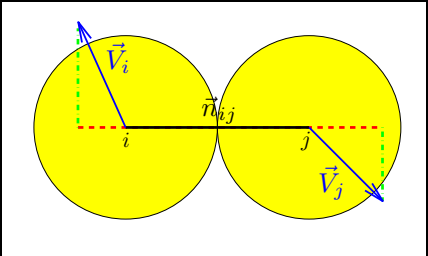
notice that $\mathcal{S}$ is not necessarily square!

$N \times \check{d}$ columns

N_c (# of contacts/springs) rows

square when $2N_c = N \times \check{d}$

i.e. when $z \equiv \dfrac{2N_c}{N} = 2\check{d} \equiv z_c$



under what conditions can **homogeneous solutions** exist?

$$\mathcal{S}|V\rangle = |0\rangle$$

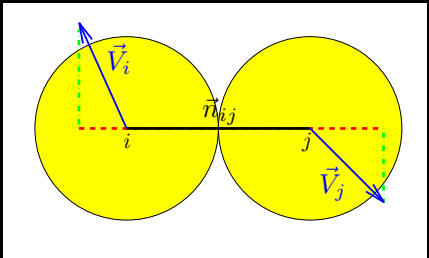
notice that $\mathcal{S}$ is not necessarily square!

$N \times \check{d}$ columns

N_c (# of contacts/springs) rows

square when $2N_c = N \times \check{d}$

i.e. when $z \equiv \dfrac{2N_c}{N} = 2\check{d} \equiv z_c$



$\rightarrow$ when $2N_c < N \times \check{d}$

or when $z < z_c = 2\check{d}$

we expect $\sim N(z_c - z)$ of them

then we call the system a

FLOPPY MATERIAL

under what conditions can **homogeneous solutions** exist?

$$\mathcal{S}|V\rangle = |0\rangle$$

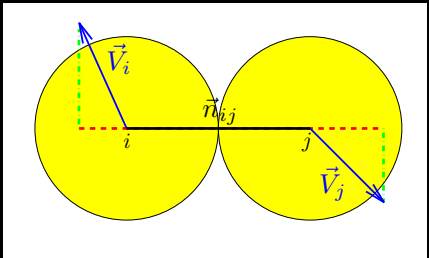
notice that $\mathcal{S}$ is not necessarily square!

$N \times \vec{d}$ columns

N_c (# of contacts/springs) rows

square when $2N_c = N \times \vec{d}$

i.e. when $z \equiv \dfrac{2N_c}{N} = 2\vec{d} \equiv z_c$



$\rightarrow$ when $2N_c < N \times \vec{d}$

or when $z < z_c = 2\vec{d}$

we expect $\sim N(z_c - z)$ of them

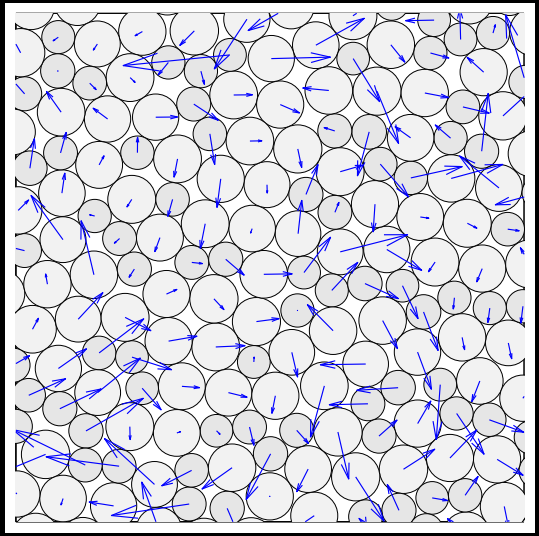
then we call the system a

FLOPPY MATERIAL

such homogeneous solutions are called

FLOPPY MODES

example of a FLOPPY MODE

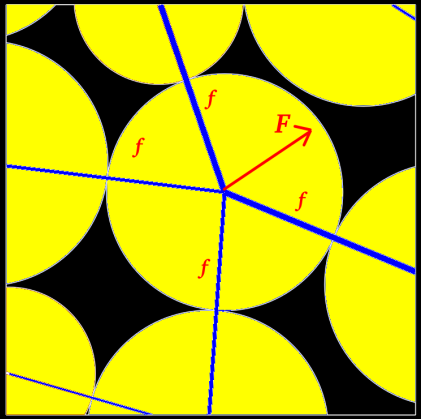


the matrix $\mathcal{S}^T$

the matrix $\mathcal{S}^T$

$\rightarrow$ imagine we sum over **bonds** $i \leftrightarrow j$:

$$\boldsymbol{F}_k = \sum_{i \leftrightarrow j} \boldsymbol{\mathcal{S}}_{k,ij} f_{ij}$$

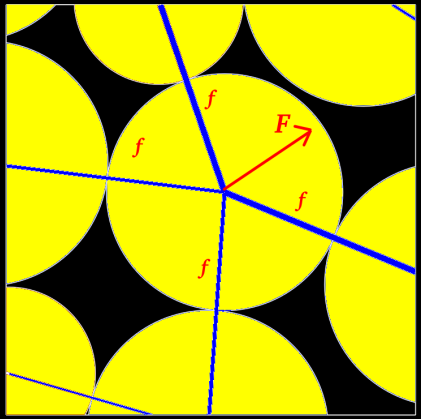


the matrix $\mathcal{S}^T$

$\rightarrow$ imagine we sum over **bonds** $i \leftrightarrow j$:

$$\boldsymbol{F}_k = \sum_{i \leftrightarrow j} \boldsymbol{\mathcal{S}}_{k,ij} f_{ij}$$

$\rightarrow$ $\mathcal{S}^T$ takes the **vector sum** of pairwise (bond-space) quantities



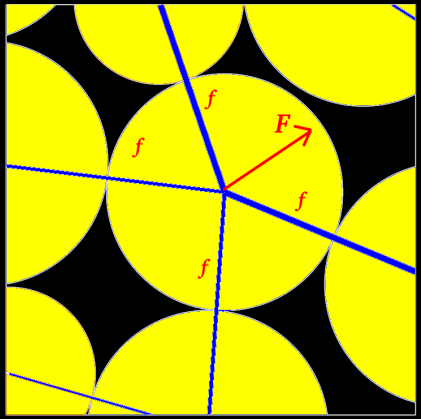
the matrix $\mathcal{S}^T$

$\rightarrow$ imagine we sum over **bonds** $i \leftrightarrow j$:

$$\boldsymbol{F}_k = \sum_{i \leftrightarrow j} \boldsymbol{\mathcal{S}}_{k,ij} f_{ij}$$

$\rightarrow$ $\mathcal{S}^T$ takes the **vector sum** of pairwise (bond-space) quantities

$\rightarrow$ $\mathcal{S}^T$ takes vectors from bond/pair-space to particles' coordinates space



the matrix $\mathcal{S}^T$

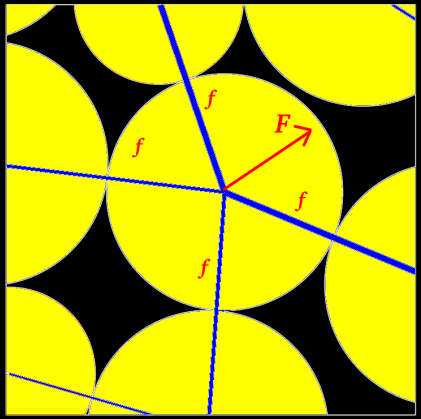
→ imagine we sum over **bonds** $i \leftrightarrow j$:

$$\boldsymbol{F}_k = \sum_{i \leftrightarrow j} \boldsymbol{\mathcal{S}}_{k,ij} f_{ij}$$

→ $\mathcal{S}^T$ takes the **vector sum** of pairwise (bond-space) quantities

→ $\mathcal{S}^T$ takes vectors from bond/pair-space to particles' coordinates space

→ in bra-ket notation, we'd write $\mathcal{S}^T|f\rangle = |F\rangle$



the matrix $\mathcal{S}^T$

$\rightarrow$ imagine we sum over **bonds** $i \leftrightarrow j$:

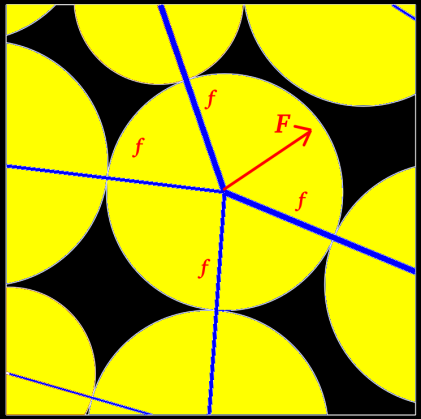
$$\boldsymbol{F}_k = \sum_{i \leftrightarrow j} \boldsymbol{\mathcal{S}}_{k,ij} f_{ij}$$

$\rightarrow$ $\mathcal{S}^T$ takes the **vector sum** of pairwise (bond-space) quantities

$\rightarrow$ $\mathcal{S}^T$ takes vectors from bond/pair-space to particles' coordinates space

$\rightarrow$ in bra-ket notation, we'd write $\mathcal{S}^T|f\rangle = |F\rangle$

$\rightarrow$ what spans the kernel of $\mathcal{S}^T$?



the matrix $\mathcal{S}^T$

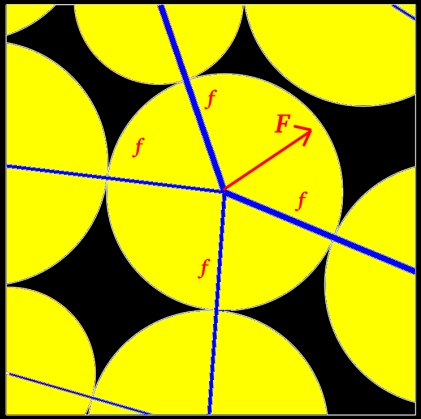
→ imagine we sum over **bonds** $i \leftrightarrow j$:

$$\boldsymbol{F}_k = \sum_{i \leftrightarrow j} \boldsymbol{\mathcal{S}}_{k,ij} f_{ij}$$

→ $\mathcal{S}^T$ takes the **vector sum** of pairwise (bond-space) quantities

→ $\mathcal{S}^T$ takes vectors from bond/pair-space to particles' coordinates space

→ in bra-ket notation, we'd write $\mathcal{S}^T|f\rangle = |F\rangle$



→ what spans the kernel of $\mathcal{S}^T$?

→ sets of forces that **self-balance** (i.e. produce mech. equil.). These are known as **states-of-self-stress** $|\phi\rangle$

the matrix $\mathcal{S}^T$

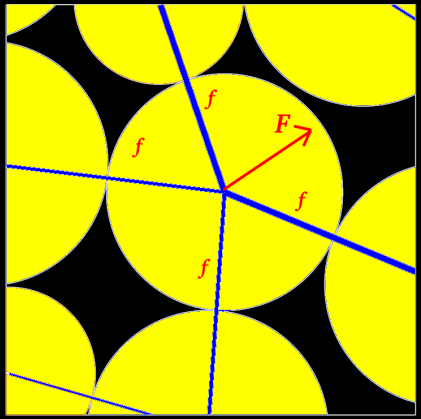
→ imagine we sum over **bonds** $i \leftrightarrow j$:

$$\boldsymbol{F}_k = \sum_{i \leftrightarrow j} \boldsymbol{\mathcal{S}}_{k,ij} f_{ij}$$

→ $\mathcal{S}^T$ takes the **vector sum** of pairwise (bond-space) quantities

→ $\mathcal{S}^T$ takes vectors from bond/pair-space to particles' coordinates space

→ in bra-ket notation, we'd write $\mathcal{S}^T|f\rangle = |F\rangle$



→ what spans the kernel of $\mathcal{S}^T$?

→ sets of forces that **self-balance** (i.e. produce mech. equil.). These are known as **states-of-self-stress** $|\phi\rangle$

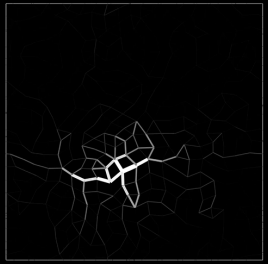
$$\mathcal{S}^T|\phi\rangle = |0\rangle$$

states-of-self-stress $\mathcal{S}^T|\phi\rangle = |0\rangle$

EL, EPJE 2018

states-of-self-stress $\mathcal{S}^T|\phi\rangle = |0\rangle$

Q: when do we expect to find states-of-self-stress?

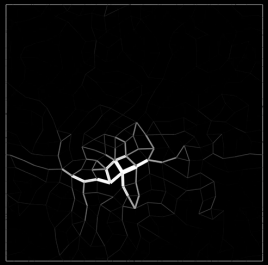


EL, EPJE 2018

states-of-self-stress $\mathcal{S}^T|\phi\rangle = |0\rangle$

Q: when do we expect to find states-of-self-stress?

A: when $z > z_c = 2\check{d}$ (more constraints than dof...)



EL, EPJE 2018

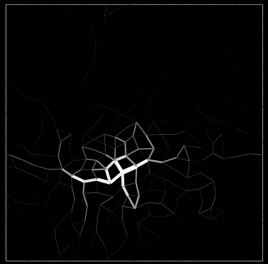
states-of-self-stress $\mathcal{S}^T|\phi\rangle = |0\rangle$

Q: when do we expect to find states-of-self-stress?

A: when $z > z_c = 2\check{d}$ (more constraints than dof...)

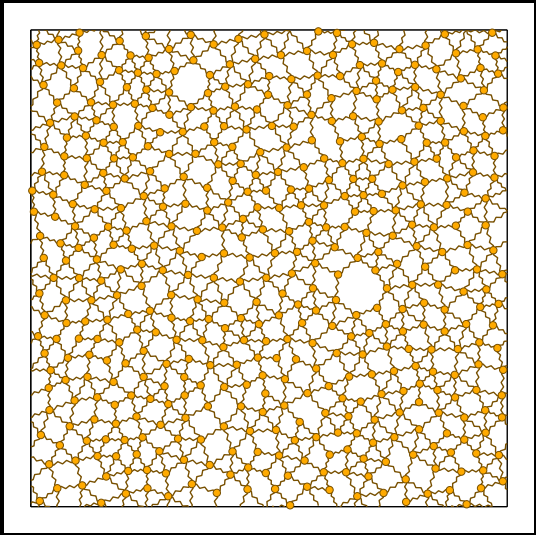
Q: how many orthonormal states-of-self-stress do we expect to find?

A: we should find $\sim N(z - z_c)$ orthonormal states-of-self-stress

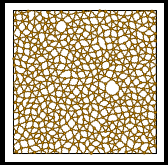


EL, EPJE 2018

...back to the mechanics of disordered networks of **relaxed** Hookean springs

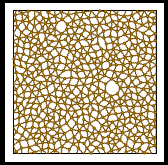


mechanics of disordered networks of **relaxed** Hookean springs



$$G = \frac{1}{V}\left(\frac{\partial^2 U}{\partial \gamma^2} - \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}}\right)$$

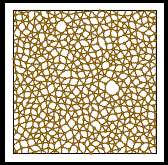
mechanics of disordered networks of **relaxed** Hookean springs



$$G = \frac{1}{V}\left(\frac{\partial^2 U}{\partial \gamma^2} - \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}}\right)$$

'Born / Affine term'

mechanics of disordered networks of **relaxed** Hookean springs

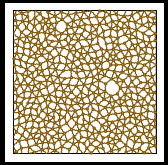


$$G = \frac{1}{V}\left(\frac{\partial^2 U}{\partial \gamma^2} - \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}}\right)$$

'Born / Affine term'

'nonaffine term'

mechanics of disordered networks of **relaxed** Hookean springs



$$G = \frac{1}{V}\left(\frac{\partial^2 U}{\partial \gamma^2} - \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}}\right)$$

'Born / Affine term'

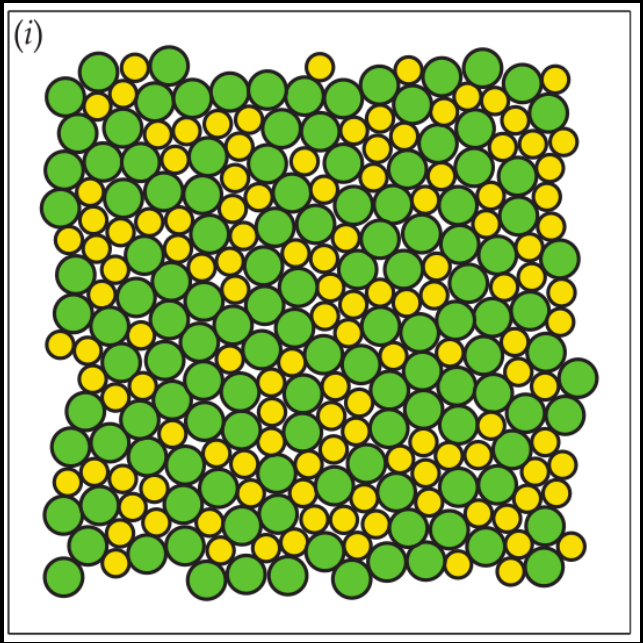
'nonaffine term'

stems from structural disorder

nonaffine displacements

nonaffine displacements

(i) start with $\boldsymbol{f} = \mathbf{0}$ state

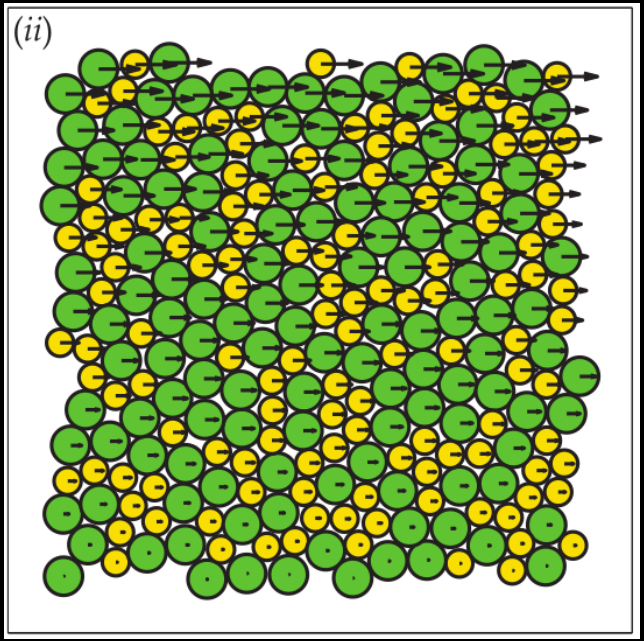


nonaffine displacements

(i) start with $\boldsymbol{f} = \boldsymbol{0}$ state

(ii) impose affine transformation

$$\boldsymbol{x} \rightarrow \boldsymbol{F} \cdot \boldsymbol{x}$$



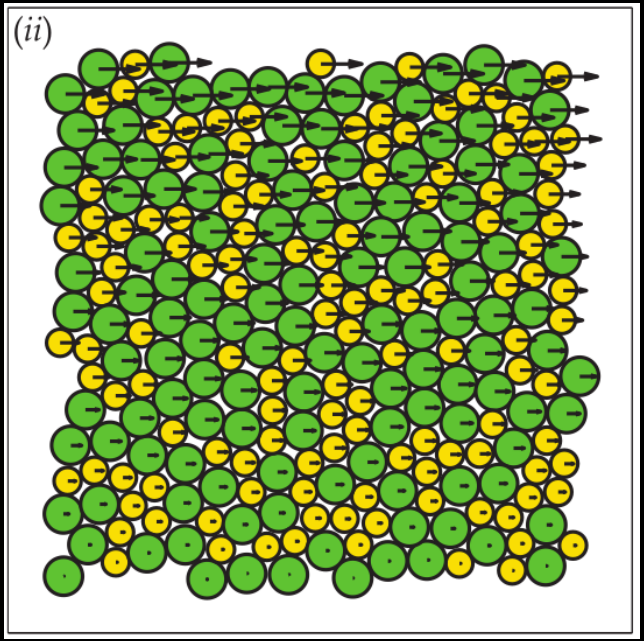
nonaffine displacements

(i) start with $\boldsymbol{f} = \boldsymbol{0}$ state

(ii) impose affine transformation

$$\boldsymbol{x} \rightarrow \boldsymbol{F} \cdot \boldsymbol{x}$$

(iii) in new state $\boldsymbol{f} \neq \boldsymbol{0}$



nonaffine displacements

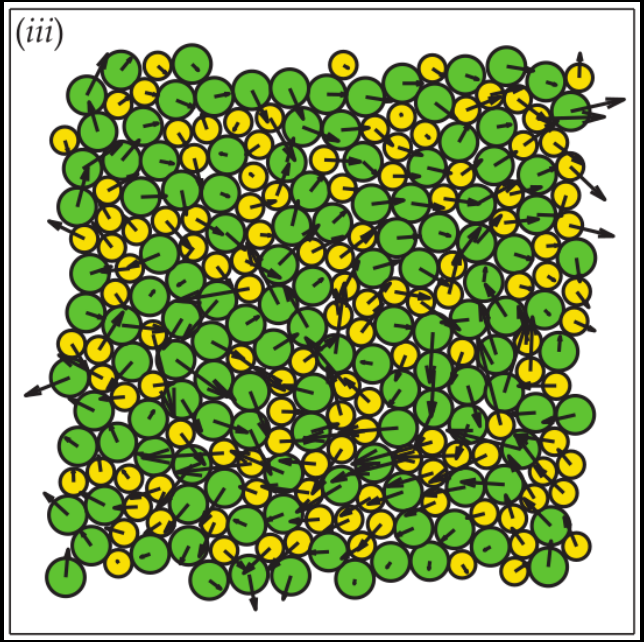
(i) start with $\boldsymbol{f} = \boldsymbol{0}$ state

(ii) impose affine transformation

$$\boldsymbol{x} \rightarrow \boldsymbol{F} \cdot \boldsymbol{x}$$

(iii) in new state $\boldsymbol{f} \neq \boldsymbol{0}$

make additional (nonaffine) displacements



nonaffine displacements

(i) start with $\boldsymbol{f} = \boldsymbol{0}$ state

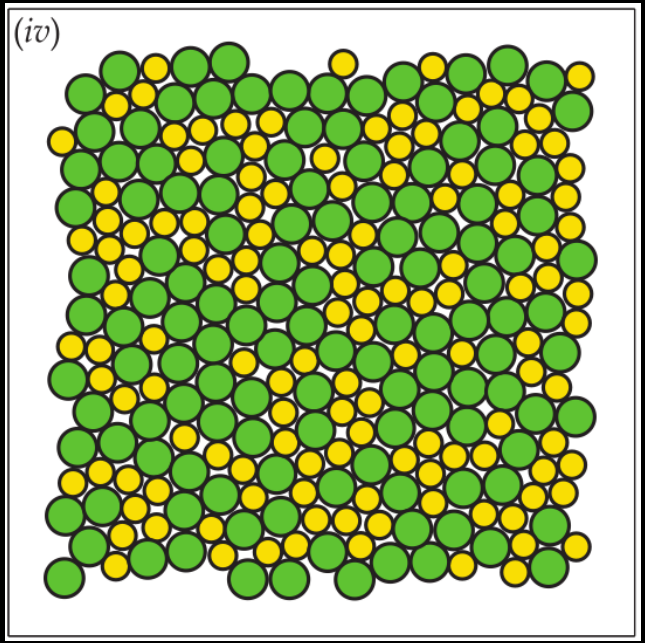
(ii) impose affine transformation

$$\boldsymbol{x} \rightarrow \boldsymbol{F} \cdot \boldsymbol{x}$$

(iii) in new state $\boldsymbol{f} \neq \boldsymbol{0}$

make additional (nonaffine) displacements

(iv) reach deformed $\boldsymbol{f} = \boldsymbol{0}$ state



nonaffine displacements

(i) start with $\boldsymbol{f} = \boldsymbol{0}$ state

(ii) impose affine transformation

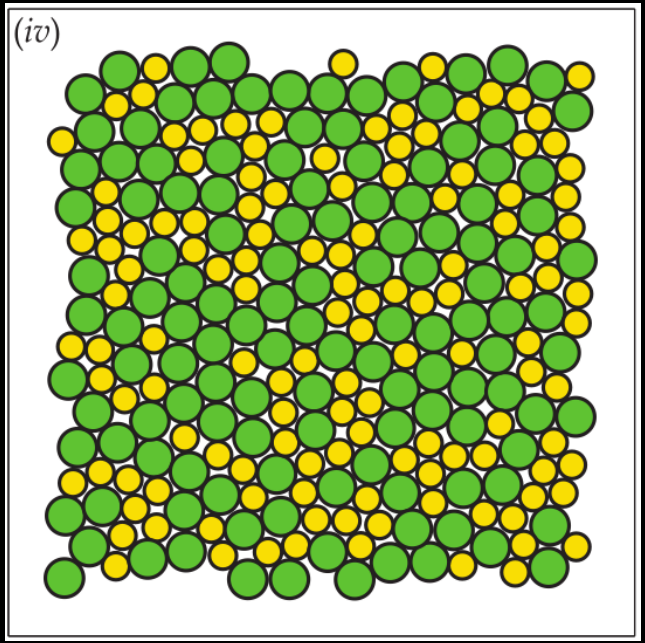
$$\boldsymbol{x} \to \boldsymbol{F} \cdot \boldsymbol{x}$$

(iii) in new state $\boldsymbol{f} \neq \boldsymbol{0}$

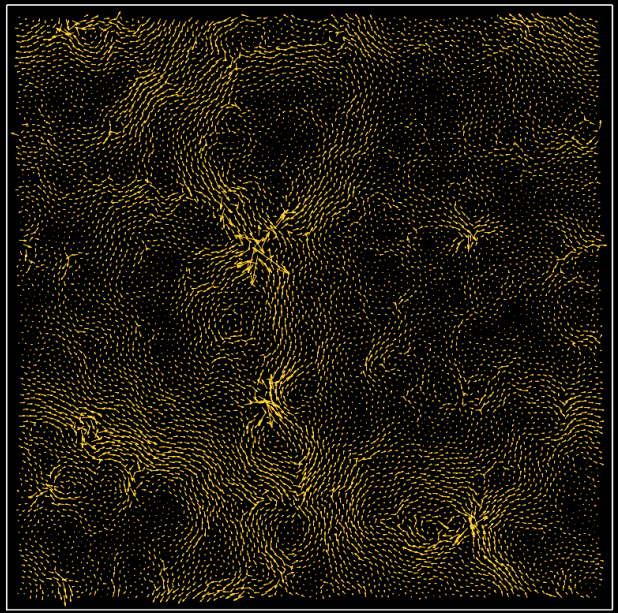
make additional (nonaffine) displacements

(iv) reach deformed $\boldsymbol{f} = \boldsymbol{0}$ state

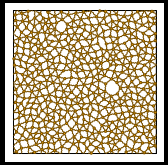
how do nonaffine displacements look like?



nonaffine displacements



mechanics of disordered networks of **relaxed** Hookean springs

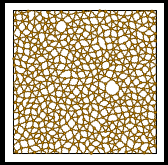


$$G = \frac{1}{V}\left(\frac{\partial^2 U}{\partial \gamma^2} - \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}}\right)$$

'Born / Affine term'

'nonaffine term'

mechanics of disordered networks of **relaxed** Hookean springs

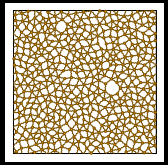


$$G = \frac{1}{V}\left(\frac{\partial^2 U}{\partial \gamma^2} - \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}}\right)$$

$$\left\langle \frac{\partial r}{\partial \gamma} \middle| \frac{\partial r}{\partial \gamma} \right\rangle$$

'nonaffine term'

mechanics of disordered networks of **relaxed** Hookean springs

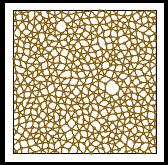


$$G = \frac{1}{V}\left(\frac{\partial^2 U}{\partial \gamma^2} - \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}}\right)$$

$$\left\langle \frac{\partial r}{\partial \gamma} \middle| \frac{\partial r}{\partial \gamma} \right\rangle$$

$$\left\langle \frac{\partial r}{\partial \gamma} \middle| \mathcal{S}\left(\mathcal{S}^T \mathcal{S}\right)^{-1} \mathcal{S}^T \middle| \frac{\partial r}{\partial \gamma} \right\rangle$$

mechanics of disordered networks of **relaxed** Hookean springs



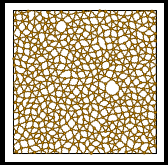
$$G = \frac{1}{V}\left(\frac{\partial^2 U}{\partial \gamma^2} - \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}}\right)$$

$$\left\langle \frac{\partial r}{\partial \gamma} \middle| \frac{\partial r}{\partial \gamma} \right\rangle \qquad \left\langle \frac{\partial r}{\partial \gamma} \middle| \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \middle| \frac{\partial r}{\partial \gamma} \right\rangle$$

$$\Rightarrow \quad G = \frac{1}{V}\left\langle \frac{\partial r}{\partial \gamma} \middle| \mathcal{I} - \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \middle| \frac{\partial r}{\partial \gamma} \right\rangle$$

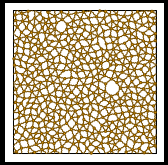
for the private case of disordered networks of relaxed Hookean springs

mechanics of disordered networks of **relaxed** Hookean springs



$$\Rightarrow \quad G = \frac{1}{V}\left\langle \frac{\partial r}{\partial \gamma}\middle| \mathcal{I} - \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \middle| \frac{\partial r}{\partial \gamma}\right\rangle$$

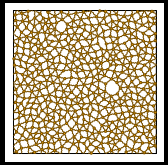
mechanics of disordered networks of **relaxed** Hookean springs



$$\Rightarrow \quad G = \frac{1}{V}\left\langle \frac{\partial r}{\partial \gamma}\middle| \mathcal{I} - \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \middle| \frac{\partial r}{\partial \gamma}\right\rangle$$

$\rightarrow$ notice that $\mathcal{P} \equiv \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T$ is a **projection operator** $\mathcal{P}^2 = \mathcal{P}$

mechanics of disordered networks of **relaxed** Hookean springs

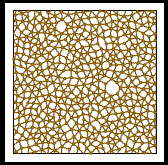


$$\Rightarrow \quad G = \frac{1}{V}\left\langle \frac{\partial r}{\partial \gamma}\middle| \mathcal{I} - \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \middle| \frac{\partial r}{\partial \gamma}\right\rangle$$

$\rightarrow$ notice that $\mathcal{P} \equiv \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T$ is a **projection operator** $\mathcal{P}^2 = \mathcal{P}$

$$\mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T\mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T = \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \quad \checkmark$$

mechanics of disordered networks of **relaxed** Hookean springs



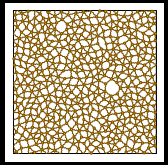
$$\Rightarrow \quad G = \frac{1}{V}\left\langle \frac{\partial r}{\partial \gamma}\middle| \mathcal{I} - \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \middle| \frac{\partial r}{\partial \gamma}\right\rangle$$

$\rightarrow$ notice that $\mathcal{P} \equiv \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T$ is a **projection operator** $\mathcal{P}^2 = \mathcal{P}$

$$\mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T\mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T = \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \quad \checkmark$$

$\rightarrow$ consequently, $\mathcal{I} - \mathcal{P} = \sum_\phi |\phi\rangle\langle\phi|$ is **also** a projection operator

mechanics of disordered networks of **relaxed** Hookean springs



$$\Rightarrow \quad G = \frac{1}{V}\left\langle \frac{\partial r}{\partial \gamma}\middle| \mathcal{I} - \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \middle| \frac{\partial r}{\partial \gamma}\right\rangle$$

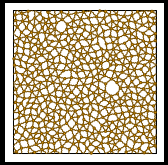
→ notice that $\mathcal{P} \equiv \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T$ is a **projection operator** $\mathcal{P}^2 = \mathcal{P}$

$$\mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T\mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T = \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \quad \checkmark$$

→ consequently, $\mathcal{I} - \mathcal{P} = \sum_\phi |\phi\rangle\langle\phi|$ is **also** a projection operator

→ $\mathcal{I} - \mathcal{P}$ projects onto the **space of states-of-self-stress** $\mathcal{S}^T|\phi\rangle = |0\rangle$

mechanics of disordered networks of **relaxed** Hookean springs



$$\Rightarrow \quad G = \frac{1}{V}\left\langle \frac{\partial r}{\partial \gamma}\right| \mathcal{I} - \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \left|\frac{\partial r}{\partial \gamma}\right\rangle$$

$\rightarrow$ notice that $\mathcal{P} \equiv \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T$ is a **projection operator** $\mathcal{P}^2 = \mathcal{P}$

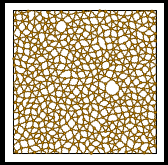
$$\mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T\mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T = \mathcal{S}\left(\mathcal{S}^T\mathcal{S}\right)^{-1}\mathcal{S}^T \quad \checkmark$$

$\rightarrow$ consequently, $\mathcal{I} - \mathcal{P} = \sum_\phi |\phi\rangle\langle\phi|$ is **also** a projection operator

$\rightarrow$ $\mathcal{I} - \mathcal{P}$ projects onto the **space of states-of-self-stress** $\quad \mathcal{S}^T|\phi\rangle = |0\rangle$

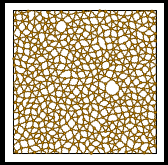
$\rightarrow \quad G = \frac{1}{V}\sum_\phi \left\langle \frac{\partial r}{\partial \gamma}|\phi\right\rangle^2$ is a sum of projections sqrd over states-of-self-stress!

mechanics of disordered networks of **relaxed** Hookean springs



$$\Rightarrow \quad G = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \quad \phi \equiv \text{states-of-self-stress}$$

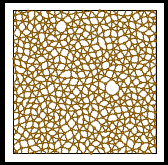
mechanics of disordered networks of **relaxed** Hookean springs



$$\Rightarrow \quad G = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \quad \phi \equiv \text{states-of-self-stress}$$

$\rightarrow$ recall there are $N(z - z_c)$ orthonormal ϕ's, assume ϕ's are **random**

mechanics of disordered networks of **relaxed** Hookean springs

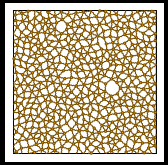


$$\Rightarrow \quad G = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \quad \phi \equiv \text{states-of-self-stress}$$

$\rightarrow$ recall there are $N(z - z_c)$ orthonormal ϕ's, assume ϕ's are **random**

$$\Rightarrow \quad G \sim z - z_c, \quad \text{if } z > z_c$$

mechanics of disordered networks of **relaxed** Hookean springs



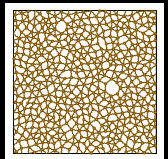
$$\Rightarrow \quad G = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \quad \phi \equiv \text{states-of-self-stress}$$

$\rightarrow$ recall there are $N(z - z_c)$ orthonormal ϕ's, assume ϕ's are **random**

$$\Rightarrow \quad G \sim z - z_c, \quad \text{if } z > z_c$$

$$G = 0 \text{ if } z < z_c, \text{ because there are no states-of-self-stress when } z < z_c!$$

mechanics of disordered networks of **relaxed** Hookean springs

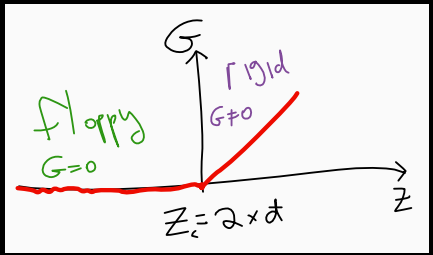


$$\Rightarrow \quad G = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \quad \phi \equiv \text{states-of-self-stress}$$

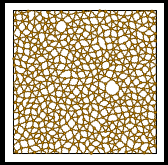
$\rightarrow$ recall there are $N(z - z_c)$ orthonormal ϕ's, assume ϕ's are **random**

$$\Rightarrow \quad G \sim z - z_c, \quad \text{if } z > z_c$$

$$G = 0 \text{ if } z < z_c, \text{ because there are no states-of-self-stress when } z < z_c!$$

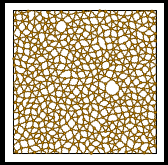


mechanics of disordered networks of **relaxed** Hookean springs

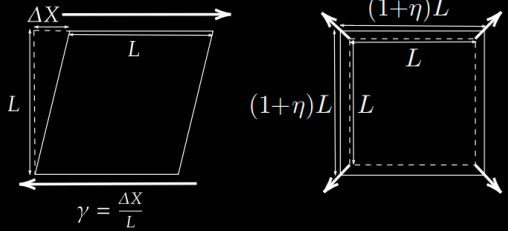


$$\Rightarrow \quad G = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \quad \phi \equiv \text{states-of-self-stress}$$

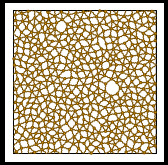
mechanics of disordered networks of **relaxed** Hookean springs



$$\Rightarrow \quad G = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \qquad \phi \equiv \text{states-of-self-stress}$$

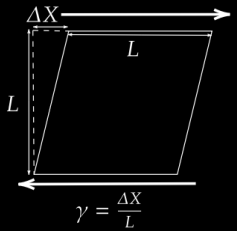


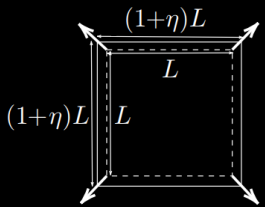
mechanics of disordered networks of **relaxed** Hookean springs



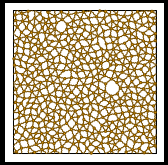
$$\Rightarrow \quad G = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \qquad \phi \equiv \text{states-of-self-stress}$$

$$K = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$



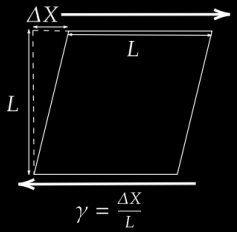


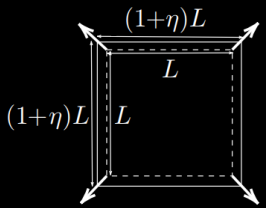
mechanics of disordered networks of **relaxed** Hookean springs



$$\Rightarrow \quad G = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \qquad \phi \equiv \text{states-of-self-stress}$$

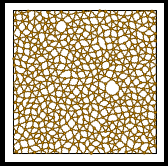
$$K = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$





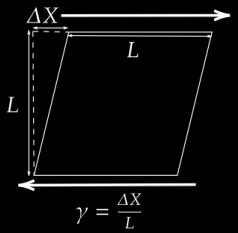
$$\Rightarrow \quad K \sim G \sim z - z_{\mathrm{c}}, \qquad \text{if } z > z_{\mathrm{c}}$$

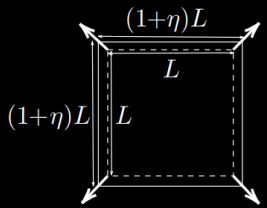
mechanics of disordered networks of **relaxed** Hookean springs



$$\Rightarrow \quad G = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \qquad \phi \equiv \text{states-of-self-stress}$$

$$K = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$

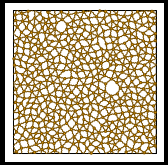




$$\Rightarrow \quad K \sim G \sim z - z_{\mathrm{c}}, \qquad \text{if } z > z_{\mathrm{c}}$$

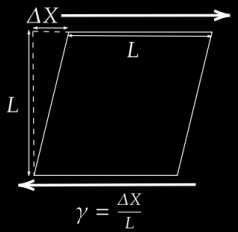
$K = 0$ if $z < z_{\mathrm{c}}$, because there are no states-of-self-stress when $z < z_{\mathrm{c}}$!

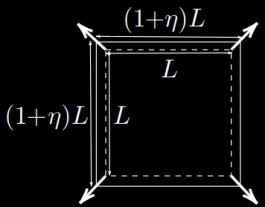
mechanics of disordered networks of **relaxed** Hookean springs



$$\Rightarrow \quad G = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \qquad \phi \equiv \text{states-of-self-stress}$$

$$K = \frac{1}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$



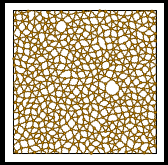


$$\Rightarrow \quad K \sim G \sim z - z_{\mathrm{c}}, \qquad \text{if } z > z_{\mathrm{c}}$$

$K = 0$ if $z < z_{\mathrm{c}}$, because there are no states-of-self-stress when $z < z_{\mathrm{c}}$!

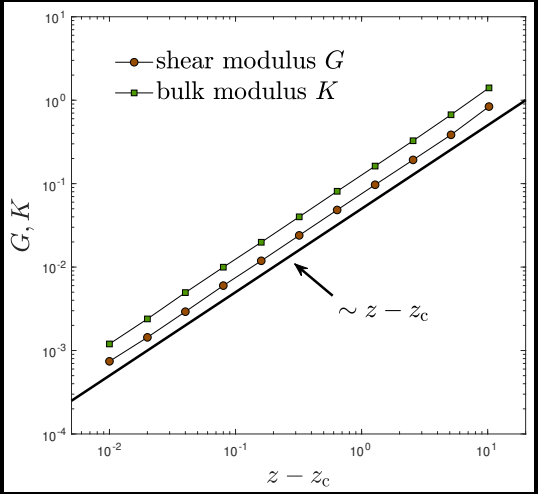
SAME SCALING AS $G \sim z - z_{\mathrm{c}}$

mechanics of disordered networks of **relaxed** Hookean springs

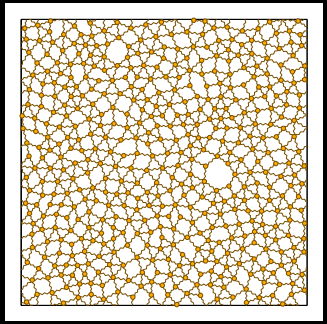


numerical validation

✓

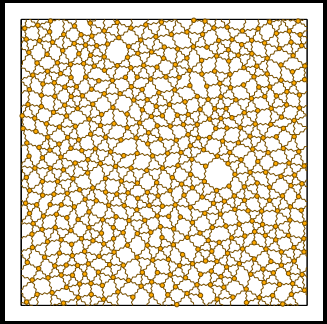


$$G \sim K \sim z - z_c$$



$$G \sim z - z_{\mathrm{c}}$$

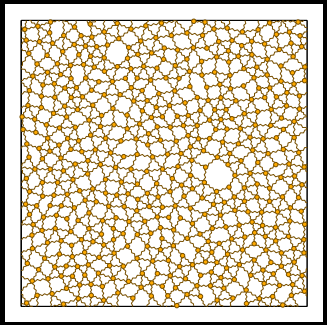
$$K \sim z - z_{\mathrm{c}}$$



$$G \sim z - z_{\mathrm{c}}$$

$$K \sim z - z_{\mathrm{c}}$$

$$\Rightarrow \quad G/K \sim \text{const}$$

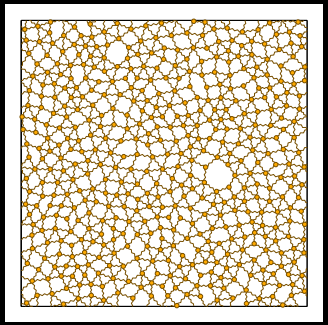


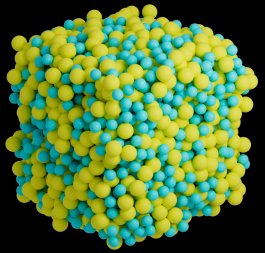
$$G \sim z - z_{\mathrm{c}}$$

$$K \sim z - z_{\mathrm{c}}$$

$$\Rightarrow \quad G/K \sim \text{const}$$

HOWEVER...



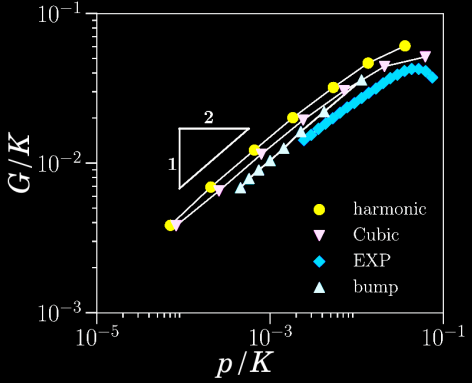


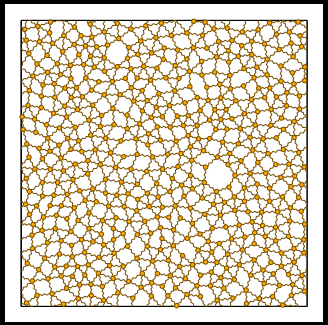
$$G \sim z - z_c$$

$$K \sim z - z_c$$

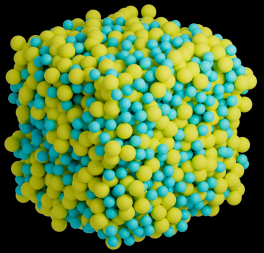
$$\Rightarrow \quad G/K \sim \text{const}$$

HOWEVER...





how can we understand this difference?

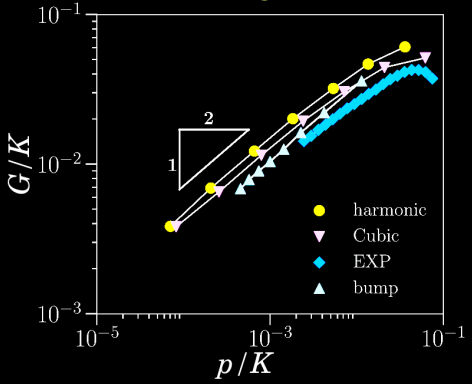


$$G \sim z - z_c$$

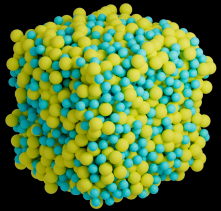
$$K \sim z - z_c$$

$$\Rightarrow \quad G/K \sim \text{const}$$

HOWEVER...

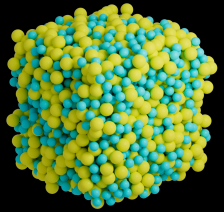


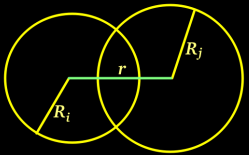
soft-spheres contact potentials



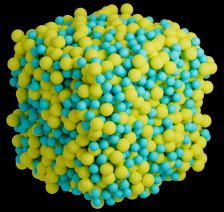
soft-spheres contact potentials

$\rightarrow$ consider purely-repulsive pairwise potentials $\varphi(r)$, i.e. with $\varphi'' > 0$ and $\varphi' < 0$



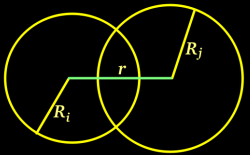


soft-spheres contact potentials

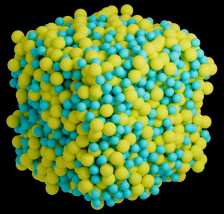


→ consider purely-repulsive pairwise potentials $\varphi(r)$, i.e. with $\varphi'' > 0$ and $\varphi' < 0$

→ require that $\varphi'' \gg -\varphi'/r$ as $p \to 0$



soft-spheres contact potentials



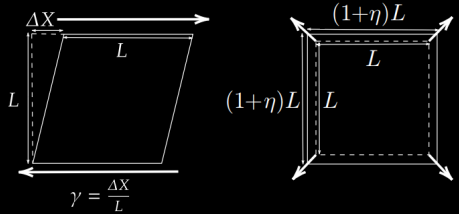
→ consider purely-repulsive pairwise potentials $\varphi(r)$, i.e. with $\varphi'' > 0$ and $\varphi' < 0$

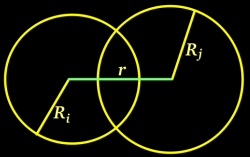
→ require that $\varphi'' \gg -\varphi'/r$ as $p \to 0$

→ elastic moduli:

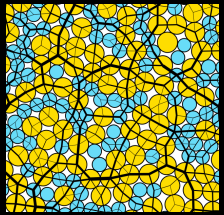
$$G \sim \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \qquad \phi \equiv \text{states-of-self-stress}$$

$$K \sim \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$





packings' 'special' state-of-self-stress

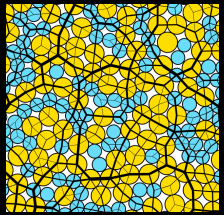


$$G \sim \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2 \qquad K = \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$

$\phi \equiv$ states-of-self-stress

packings' 'special' state-of-self-stress

$\rightarrow$ in packings all interparticle forces f are **positive** (have the same sign)



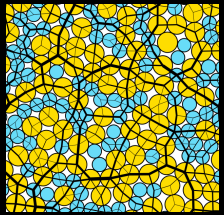
$$G \sim \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2 \qquad K = \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$

$\phi \equiv$ states-of-self-stress

packings' 'special' state-of-self-stress

$\rightarrow$ in packings all interparticle forces f are **positive** (have the same sign)

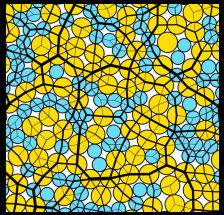
$\rightarrow$ by construction, contact forces form a state-of-self-stress: $\mathcal{S}^T|f\rangle = |0\rangle$



$$G \sim \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2 \qquad K = \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$

$\phi \equiv$ states-of-self-stress

packings' 'special' state-of-self-stress



$\rightarrow$ in packings all interparticle forces f are **positive** (have the same sign)

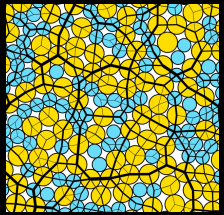
$\rightarrow$ by construction, contact forces form a state-of-self-stress: $\mathcal{S}^T|f\rangle = |0\rangle$

$\rightarrow$ for G, doesn't matter because $\partial r/\partial\gamma$ have random signs

$$G \sim \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2 \qquad K = \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$

$\phi \equiv$ states-of-self-stress

packings' 'special' state-of-self-stress



→ in packings all interparticle forces f are **positive** (have the same sign)

→ by construction, contact forces form a state-of-self-stress: $\mathcal{S}^T|f\rangle = |0\rangle$

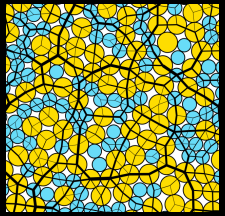
→ for G, doesn't matter because $\partial r/\partial\gamma$ have random signs

→ for K, matters because $\partial r/\partial\eta = r$ (all positive!), then $\left\langle \frac{\partial r}{\partial \eta} \middle| f \right\rangle^2 \sim N$

$$G \sim \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2 \qquad K = \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$

$\phi \equiv$ states-of-self-stress

packings' 'special' state-of-self-stress



→ in packings all interparticle forces f are **positive** (have the same sign)

→ by construction, contact forces form a state-of-self-stress: $\mathcal{S}^T|f\rangle = |0\rangle$

→ for G, doesn't matter because $\partial r/\partial\gamma$ have random signs

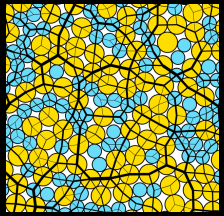
→ for K, matters because $\partial r/\partial\eta = r$ (all positive!), then $\left\langle \frac{\partial r}{\partial \eta} \middle| f \right\rangle^2 \sim N$

$$\Rightarrow \quad G \sim \varphi''(z - z_c),$$

$$G \sim \frac{\varphi''}{V} \sum_\phi \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2 \qquad K = \frac{\varphi''}{V} \sum_\phi \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$

$\phi \equiv$ states-of-self-stress

packings' 'special' state-of-self-stress



→ in packings all interparticle forces f are **positive** (have the same sign)

→ by construction, contact forces form a state-of-self-stress: $\mathcal{S}^T|f\rangle = |0\rangle$

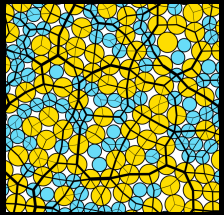
→ for G, doesn't matter because $\partial r/\partial\gamma$ have random signs

→ for K, matters because $\partial r/\partial\eta = r$ (all positive!), then $\left\langle \frac{\partial r}{\partial \eta} \middle| f \right\rangle^2 \sim N$

$$\Rightarrow \quad G \sim \varphi''(z - z_c), \quad \text{however} \quad K \sim \varphi''!$$

$$G \sim \frac{\varphi''}{V} \sum_\phi \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2 \qquad K = \frac{\varphi''}{V} \sum_\phi \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$

$\phi \equiv$ states-of-self-stress



packings' 'special' state-of-self-stress

→ in packings all interparticle forces f are **positive** (have the same sign)

→ by construction, contact forces form a state-of-self-stress: $\mathcal{S}^T|f\rangle = |0\rangle$

→ for G, doesn't matter because $\partial r/\partial\gamma$ have random signs

→ for K, matters because $\partial r/\partial\eta = r$ (all positive!), then $\left\langle \frac{\partial r}{\partial \eta}\middle| f\right\rangle^2 \sim N$

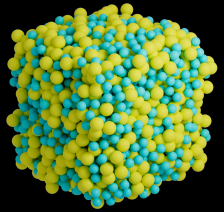
$$\Rightarrow \quad G \sim \varphi''(z - z_c), \quad \text{however} \quad K \sim \varphi''!$$

finally, we find $G/K \sim z - z_c$

$$G \sim \frac{\varphi''}{V}\sum_\phi \left\langle \frac{\partial r}{\partial \gamma}\middle|\phi\right\rangle^2 \qquad K = \frac{\varphi''}{V}\sum_\phi \left\langle \frac{\partial r}{\partial \eta}\middle|\phi\right\rangle^2$$

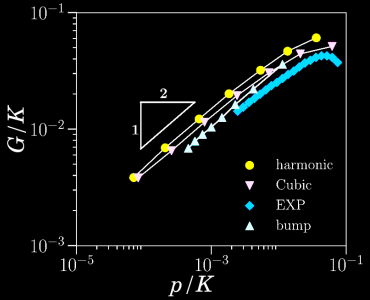
$\phi \equiv$ states-of-self-stress

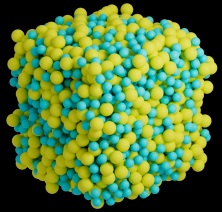
last piece of the puzzle



finally, we find $G/K \sim z - z_{\mathrm{c}}$

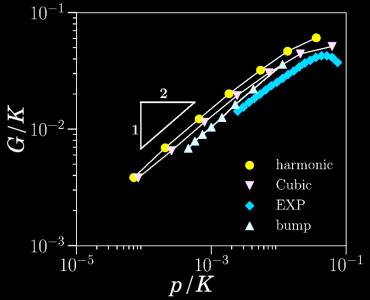
last piece of the puzzle



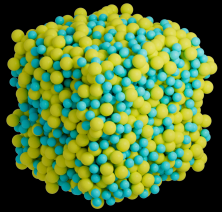


finally, we find $G/K \sim z - z_c$

last piece of the puzzle



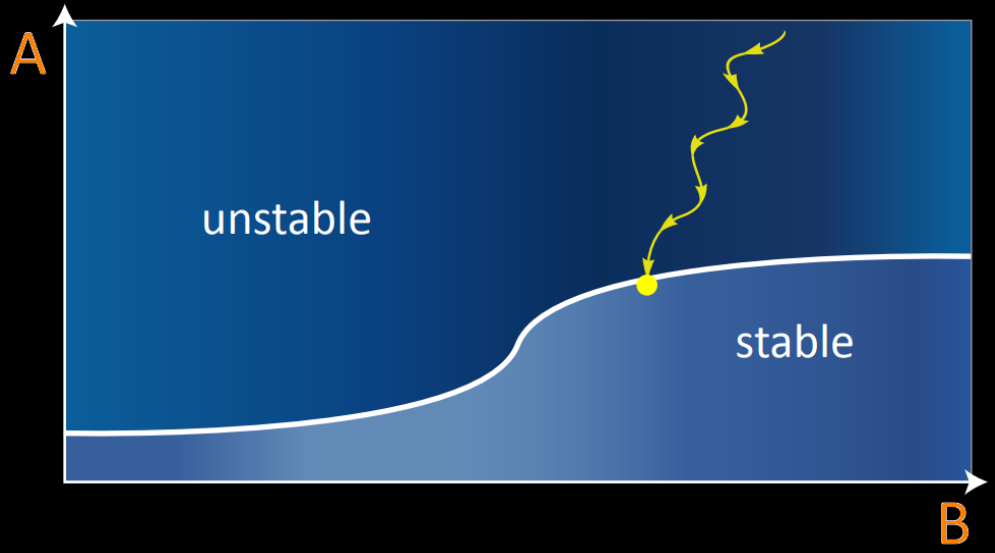
need to show $z - z_c \sim \sqrt{p/K}$



finally, we find $G/K \sim z - z_c$

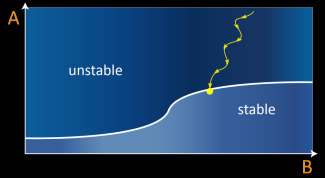
the principle of marginal stability

the principle of marginal stability



the principle of marginal stability

→ when packing is created, it passes through unstable states, until finally finding a stable state, and then it **jams/arrests** *exactly* on the boundary between stable and unstable states

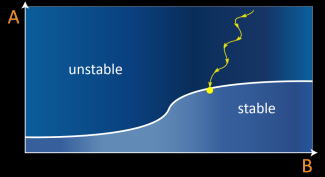


the principle of marginal stability

→ when packing is created, it passes through unstable states, until finally finding a stable state, and then it **jams/arrests** *exactly* on the boundary between stable and unstable states

→ consider the energetic stability of a displacement $\boldsymbol{u}$:

$$\boldsymbol{u} \cdot \boldsymbol{\mathcal{H}} \cdot \boldsymbol{u} \sim \varphi'' \langle u|\mathcal{S}^T\mathcal{S}|u\rangle + \frac{\varphi'}{r}\langle u|(\mathcal{I} - \mathcal{S}^T\mathcal{S})|u\rangle$$



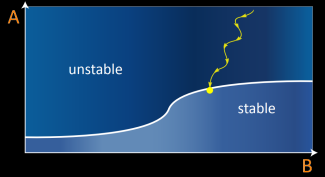
the principle of marginal stability

→ when packing is created, it passes through unstable states, until finally finding a stable state, and then it **jams/arrests** *exactly* on the boundary between stable and unstable states

→ consider the energetic stability of a displacement $\boldsymbol{u}$:

$$\boldsymbol{u} \cdot \boldsymbol{\mathcal{H}} \cdot \boldsymbol{u} \sim \varphi'' \langle u|\mathcal{S}^T\mathcal{S}|u\rangle + \frac{\varphi'}{r}\langle u|(\mathcal{I} - \mathcal{S}^T\mathcal{S})|u\rangle$$

→ substitute $\varphi'' \sim K$ and $\varphi' \sim -p$

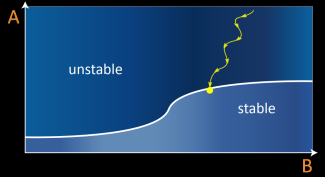


the principle of marginal stability

→ when packing is created, it passes through unstable states, until finally finding a stable state, and then it **jams/arrests** *exactly* on the boundary between stable and unstable states

→ consider the energetic stability of a displacement $\boldsymbol{u}$:

$$\boldsymbol{u} \cdot \boldsymbol{\mathcal{H}} \cdot \boldsymbol{u} \sim K\langle u|\mathcal{S}^T\mathcal{S}|u\rangle - p\langle u|(\mathcal{I} - \mathcal{S}^T\mathcal{S})|u\rangle$$



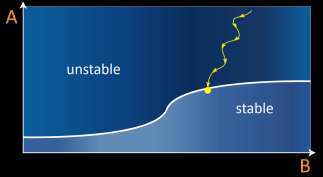
the principle of marginal stability

→ when packing is created, it passes through unstable states, until finally finding a stable state, and then it **jams/arrests** *exactly* on the boundary between stable and unstable states

→ consider the energetic stability of a displacement $\boldsymbol{u}$:

$$\boldsymbol{u}\cdot\boldsymbol{\mathcal{H}}\cdot\boldsymbol{u} \sim K\underbrace{\langle u|\mathcal{S}^T\mathcal{S}|u\rangle}_{\sim (z-z_c)^2} - p\langle u|(\mathcal{I}-\mathcal{S}^T\mathcal{S})|u\rangle$$

according to **random matrix theory**,
assuming $\mathcal{S}^T\mathcal{S}$ is a member of the **'Wishart ensemble'**



the principle of marginal stability

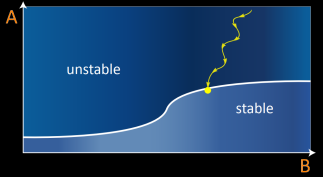
→ when packing is created, it passes through unstable states, until finally finding a stable state, and then it **jams/arrests** *exactly* on the boundary between stable and unstable states

→ consider the energetic stability of a displacement $\boldsymbol{u}$:

$$\boldsymbol{u}\cdot\boldsymbol{\mathcal{H}}\cdot\boldsymbol{u} \sim K\underbrace{\langle u|\mathcal{S}^T\mathcal{S}|u\rangle}_{\sim (z-z_c)^2} - p\langle u|(\mathcal{I}-\mathcal{S}^T\mathcal{S})|u\rangle$$

according to **random matrix theory**,
assuming $\mathcal{S}^T\mathcal{S}$ is a member of the **'Wishart ensemble'**

→ assuming displacement u is **marginally stable**, the two terms must be of the **same order**



the principle of marginal stability

→ when packing is created, it passes through unstable states, until finally finding a stable state, and then it **jams/arrests** *exactly* on the boundary between stable and unstable states

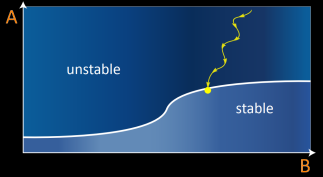
→ consider the energetic stability of a displacement $\boldsymbol{u}$:

$$\boldsymbol{u}\cdot\boldsymbol{\mathcal{H}}\cdot\boldsymbol{u} \sim K\underbrace{\langle u|\mathcal{S}^T\mathcal{S}|u\rangle}_{\sim (z-z_c)^2} - p\langle u|(\mathcal{I}-\mathcal{S}^T\mathcal{S})|u\rangle$$

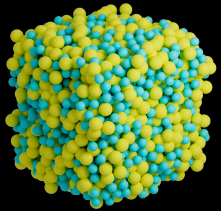
according to **random matrix theory**,
assuming $\mathcal{S}^T\mathcal{S}$ is a member of the **'Wishart ensemble'**

→ assuming displacement u is **marginally stable**, the two terms must be of the **same order**

$$\Rightarrow \quad (z-z_c)^2 \sim p/K$$

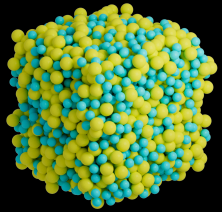


putting it all together



putting it all together

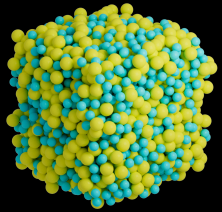
$\rightarrow$ elastic moduli depend on the solid's **states-of-self-stress** $\mathcal{S}|\phi\rangle = 0$



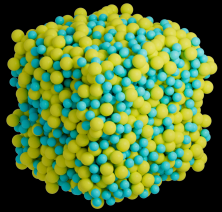
putting it all together

$\rightarrow$ elastic moduli depend on the solid's **states-of-self-stress** $\mathcal{S}|\phi\rangle = 0$

$\rightarrow$ from counting arguments there are $N(z - z_{\mathrm{c}})$ of them



putting it all together



→ elastic moduli depend on the solid's **states-of-self-stress** $\mathcal{S}|\phi\rangle = 0$

→ from counting arguments there are $N(z - z_c)$ of them

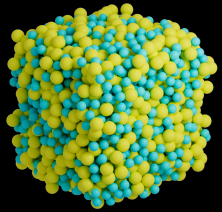
$$G \sim \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2, \quad K = \frac{\varphi''}{V} \sum_{\phi} \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$

putting it all together

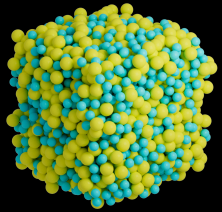
→ elastic moduli depend on the solid's **states-of-self-stress** $\mathcal{S}|\phi\rangle = 0$

→ from counting arguments there are $N(z - z_c)$ of them

$$G \sim \frac{\varphi''}{V} \underbrace{\sum_\phi \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2}_{\sim N(z-z_c)}, \quad K = \frac{\varphi''}{V} \sum_\phi \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2$$



putting it all together



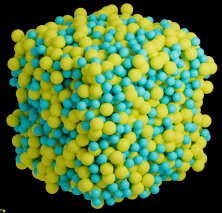
→ elastic moduli depend on the solid's **states-of-self-stress** $\mathcal{S}|\phi\rangle = 0$

→ from counting arguments there are $N(z - z_c)$ of them

$$G \sim \frac{\varphi''}{V} \underbrace{\sum_\phi \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2}_{\sim N(z-z_c)}, \quad K = \frac{\varphi''}{V} \underbrace{\sum_\phi \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2}_{\sim N}$$

due to purely-repulsive contact forces
that form a 'special' state-of-self-stress!

putting it all together



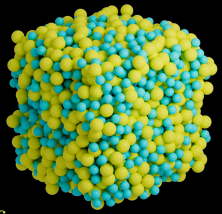
→ elastic moduli depend on the solid's **states-of-self-stress** $\mathcal{S}|\phi\rangle = 0$

→ from counting arguments there are $N(z - z_c)$ of them

$$G \sim \frac{\varphi''}{V} \underbrace{\sum_\phi \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2}_{\sim N(z - z_c)}, \quad K = \frac{\varphi''}{V} \underbrace{\sum_\phi \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2}_{\sim N} \quad \Rightarrow \quad G/K \sim z - z_c$$

due to purely-repulsive contact forces
that form a 'special' state-of-self-stress!

putting it all together



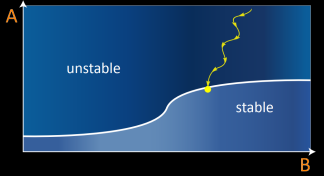
→ elastic moduli depend on the solid's **states-of-self-stress** $\mathcal{S}|\phi\rangle = 0$

→ from counting arguments there are $N(z - z_c)$ of them

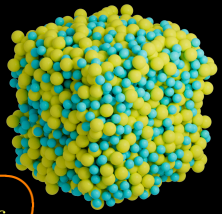
$$G \sim \frac{\varphi''}{V} \underbrace{\sum_\phi \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2}_{\sim N(z-z_c)}, \quad K = \frac{\varphi''}{V} \underbrace{\sum_\phi \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2}_{\sim N} \quad \Rightarrow \quad G/K \sim z - z_c$$

due to purely-repulsive contact forces that form a 'special' state-of-self-stress!

→ assume marginal stability towards displacements $\Rightarrow \quad (z - z_c)^2 \sim p/K$



putting it all together



→ elastic moduli depend on the solid's **states-of-self-stress** $\mathcal{S}|\phi\rangle = 0$

→ from counting arguments there are $N(z - z_c)$ of them

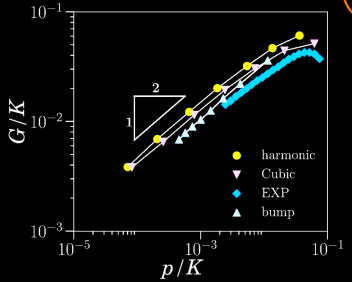
$$G \sim \frac{\varphi''}{V} \underbrace{\sum_\phi \left\langle \frac{\partial r}{\partial \gamma} \middle| \phi \right\rangle^2}_{\sim N(z-z_c)}, \quad K = \frac{\varphi''}{V} \underbrace{\sum_\phi \left\langle \frac{\partial r}{\partial \eta} \middle| \phi \right\rangle^2}_{\sim N} \quad \Rightarrow \quad G/K \sim z - z_c$$

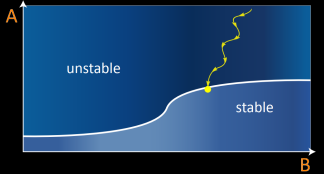
due to purely-repulsive contact forces that form a 'special' state-of-self-stress!

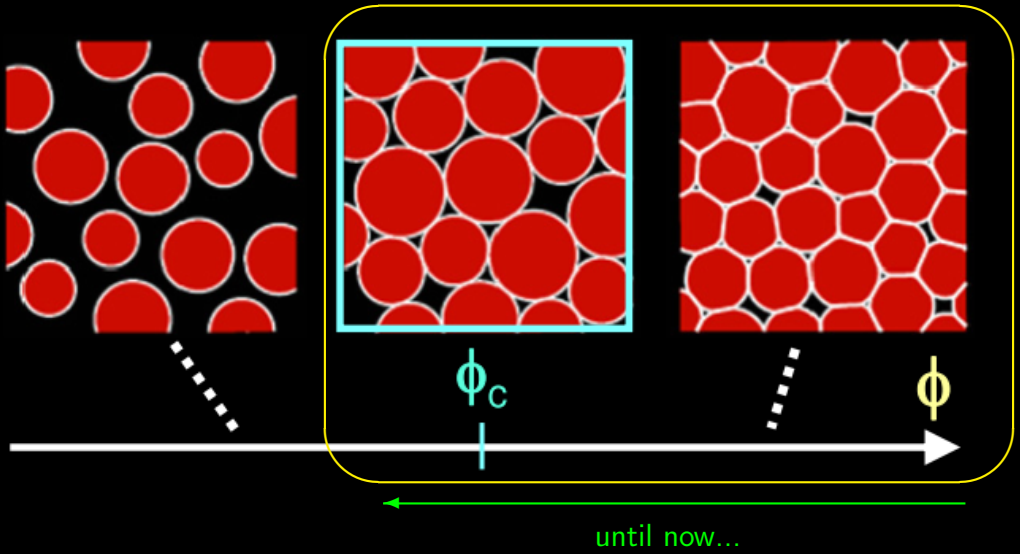
→ assume marginal stability towards displacements ⇒ $(z - z_c)^2 \sim p/K$

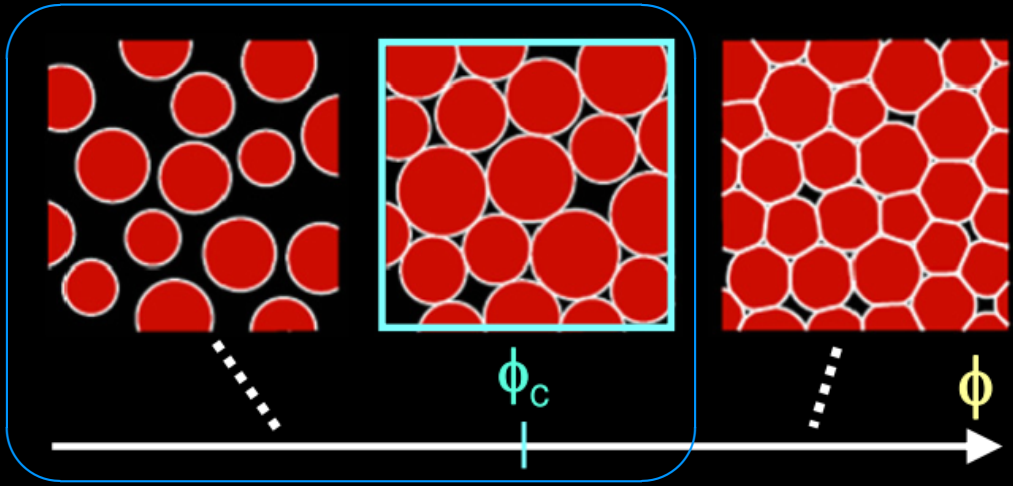
→ obtain universal scaling

$$G/K \sim \sqrt{p/K}$$






ϕ_c
ϕ
until now...



up next...

introduction – non-Brownian flows

- driven assemblies of particles interacting via finite-range, **purely repulsive** interactions (grains, bubbles, droplets, ...)
- **athermal** **(≡ non-Brownian)**

introduction – non-Brownian flows

- driven assemblies of particles interacting via finite-range, **purely repulsive** interactions (grains, bubbles, droplets, ...)
- **athermal** (≡ **non-Brownian**)

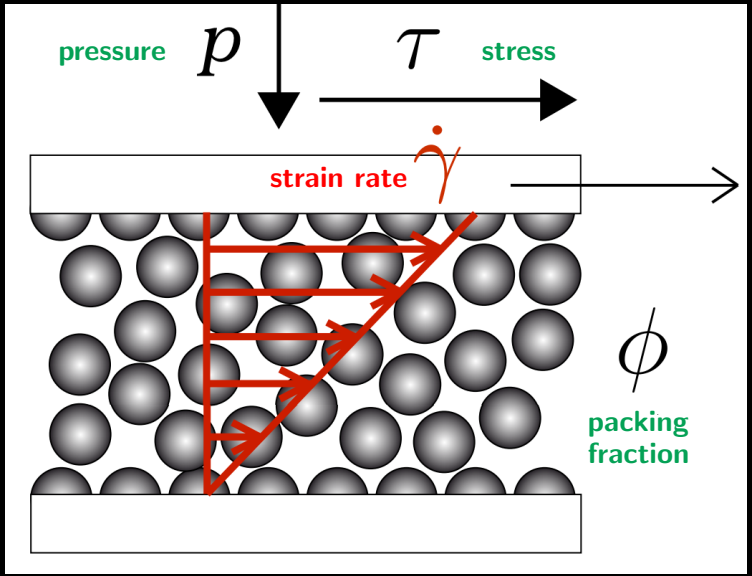
grains

emulsions

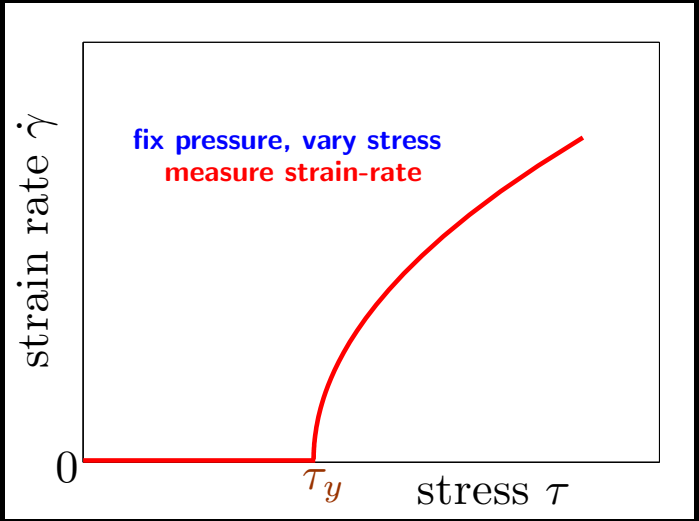
foams

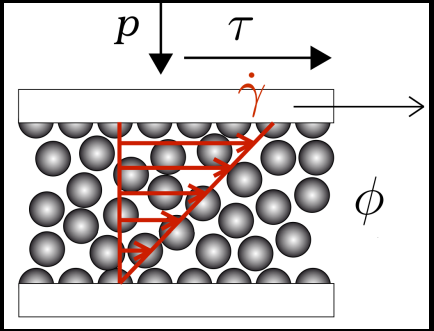
slurries/suspensions

rheology of non-Brownian flows – setup



rheology of non-Brownian flows – dynamic criticality





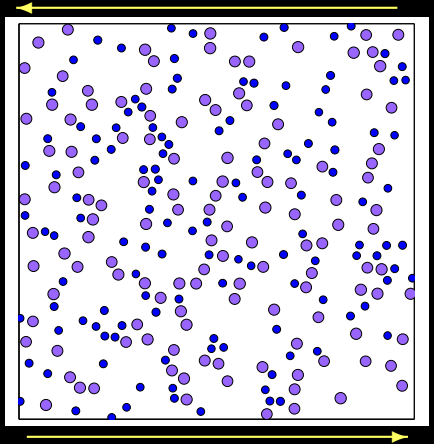
non-Brownian suspension viscosity – dynamic criticality

viscosity $\eta \equiv \dfrac{\text{stress}}{\text{strain-rate}}$

dilute limit $\phi \to 0$ was worked out by Einstein (1906):

$$\eta/\eta_0 = 1 + \tfrac{5}{2}\phi$$

($\phi \equiv$ packing fraction, $\eta_0 \equiv$ fluid viscosity)



non-Brownian suspension viscosity – dynamic criticality

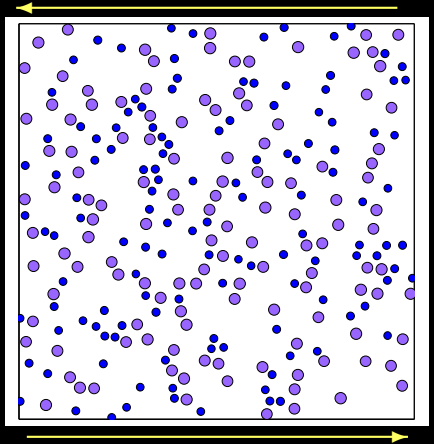
viscosity $\eta \equiv \dfrac{\text{stress}}{\text{strain-rate}}$

dilute limit $\phi \to 0$ **was worked out by Einstein (1906):**

$$\eta/\eta_0 = 1 + \tfrac{5}{2}\phi + \tfrac{38}{5}\phi^2$$

($\phi \equiv$ packing fraction, $\eta_0 \equiv$ fluid viscosity)

next order calculated by Batchelor & Green (1972)



non-Brownian suspension viscosity – dynamic criticality

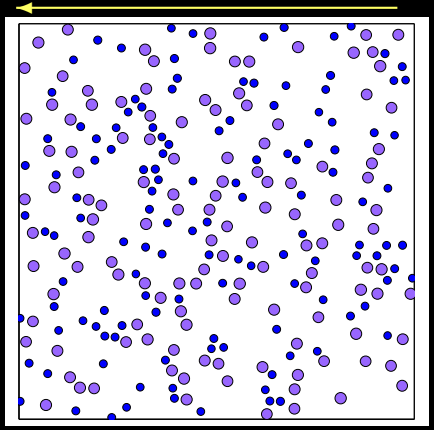
viscosity $\eta \equiv \dfrac{\text{stress}}{\text{strain-rate}}$

dilute limit $\phi \to 0$ **was worked out by Einstein (1906):**

$$\eta/\eta_0 = 1 + \tfrac{5}{2}\phi + \tfrac{38}{5}\phi^2$$

($\phi \equiv$ packing fraction, $\eta_0 \equiv$ fluid viscosity)

next order calculated by Batchelor & Green (1972)

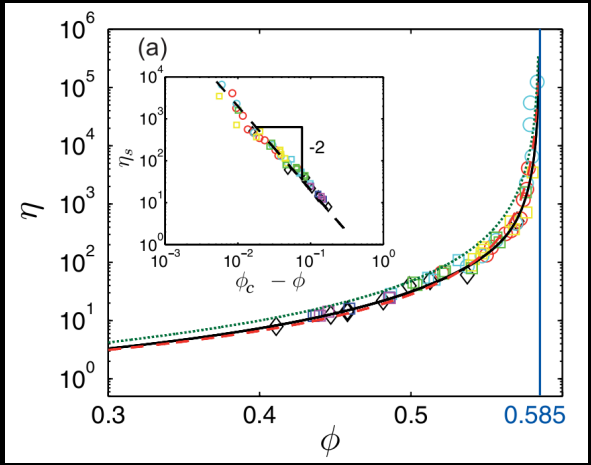


what happens in the dense limit?

non-Brownian suspension viscosity – dynamic criticality

viscosity appears to diverge as $\phi \to \phi_c$

Boyer, Guazzelli & Pouliquen PRL 107, 188301 (2011)
experiments, sheared suspensions



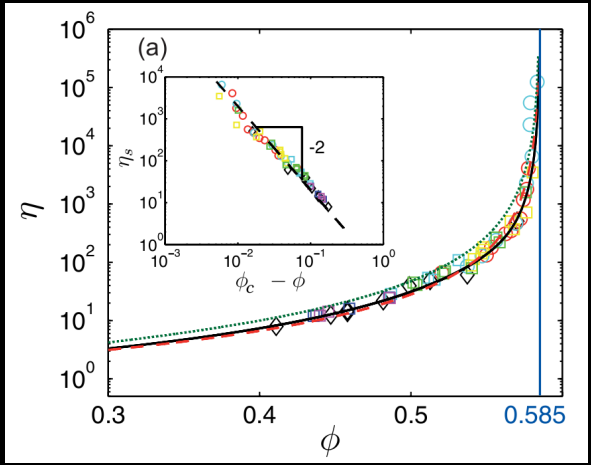
what happens in the dense limit?

non-Brownian suspension viscosity – dynamic criticality

Boyer, Guazzelli & Pouliquen PRL 107, 188301 (2011)
experiments, sheared suspensions

viscosity appears to diverge as $\phi \to \phi_c$

what is the microscopic origin of this singularity?

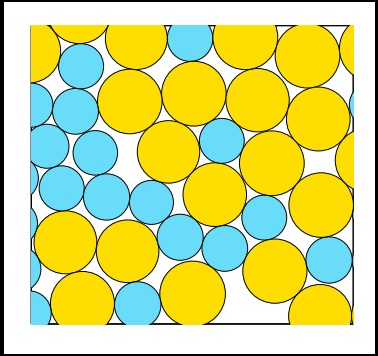


the ‘Affine-Solvent Model’ (ASM)

the ‘Affine-Solvent Model’ (ASM)

setup:

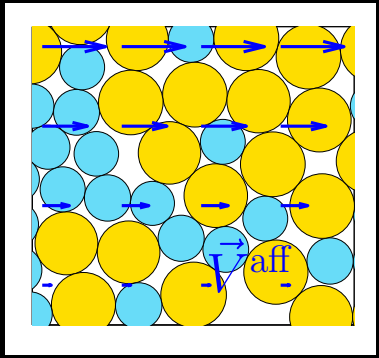
- N spherical particles immersed in a viscous fluid



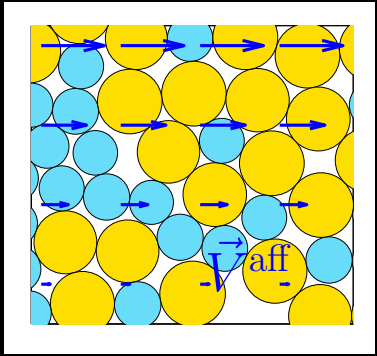
the ‘Affine-Solvent Model’ (ASM)

setup:

- N spherical particles immersed in a viscous fluid
- fluid flows with an affine velocity profile



the ‘Affine-Solvent Model’ (ASM)



setup:

- N spherical particles immersed in a viscous fluid
- fluid flows with an affine velocity profile

ingredients:

1) drag forces, but no hydrodynamic interactions

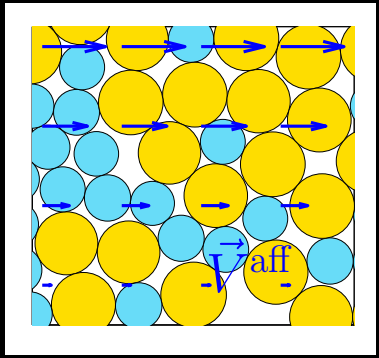
2) Overdamped dynamics – no inertial effects

3) Rigid, frictionless particles – no elasticity

the 'Affine-Solvent Model' (ASM)

setup:

- N spherical particles immersed in a viscous fluid
- fluid flows with an affine velocity profile



ingredients:

1) drag forces, but no hydrodynamic interactions
2) Overdamped dynamics – no inertial effects
3) Rigid, frictionless particles – no elasticity

$\Rightarrow$

- no energy scale, $\dot{\gamma}^{-1}$ is the only microscopic time scale
- $\phi < \phi_c$
- contacts!

ASM equations of motion

EL, G. Düring, M. Wyart, PNAS **109**, 4798 (2012)

ASM equations of motion

EL, G. Düring, M. Wyart, PNAS **109**, 4798 (2012)

from model ingredients we derive the EOM

$$|\dot{R}\rangle = \left(\mathcal{I} - \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S}\right) |V^{\mathrm{aff}}\rangle$$

ASM equations of motion

EL, G. Düring, M. Wyart, PNAS **109**, 4798 (2012)

from model ingredients we derive the EOM

$$|\dot{R}\rangle = \left(\mathcal{I} - \boldsymbol{\mathcal{S}}^T \boldsymbol{\mathcal{N}}^{-1} \boldsymbol{\mathcal{S}}\right) |V^{\mathrm{aff}}\rangle$$

encodes **deformation mode**

ASM equations of motion

EL, G. Düring, M. Wyart, PNAS **109**, 4798 (2012)

from model ingredients we derive the EOM

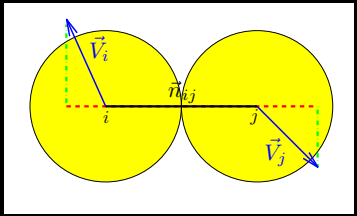
$$|\dot{R}\rangle = \left(\mathcal{I} - \boldsymbol{\mathcal{S}}^T \boldsymbol{\mathcal{N}}^{-1} \boldsymbol{\mathcal{S}}\right) |V^{\mathrm{aff}}\rangle$$

encodes **deformation mode**

where $\boldsymbol{\mathcal{S}}$ is the linear operator defined by:

$$|\dot{r}\rangle = \boldsymbol{\mathcal{S}}|V\rangle \quad \Leftrightarrow \quad \dot{r}_{ij} = \left(\vec{V}_j - \vec{V}_i\right) \cdot \vec{n}_{ij}$$

and $\boldsymbol{\mathcal{N}} \equiv \boldsymbol{\mathcal{S}}\boldsymbol{\mathcal{S}}^T \sim \vec{n}_{ij} \cdot \vec{n}_{kj}$



ASM equations of motion

EL, G. Düring, M. Wyart, PNAS **109**, 4798 (2012)

from model ingredients we derive the EOM

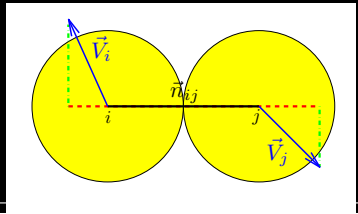
$$|\dot{R}\rangle = \left(\mathcal{I} - \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S}\right) |V^{\text{aff}}\rangle$$

encodes **deformation mode**

where $\mathcal{S}$ is the linear operator defined by:

$$|\dot{r}\rangle = \mathcal{S}|V\rangle \quad \Leftrightarrow \quad \dot{r}_{ij} = \left(\vec{V}_j - \vec{V}_i\right) \cdot \vec{n}_{ij}$$

and $\mathcal{N} \equiv \mathcal{S}\mathcal{S}^T \sim \vec{n}_{ij} \cdot \vec{n}_{kj}$



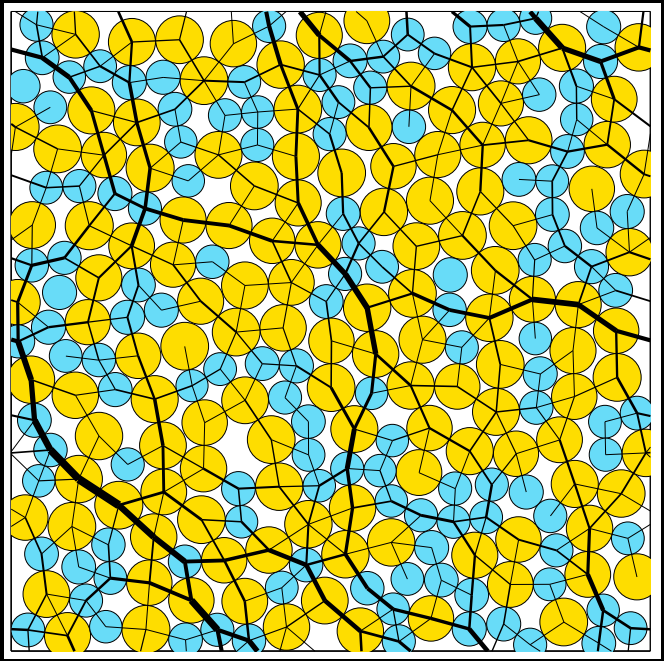
geometry fully determines mechanics:

contact forces $$|f\rangle = -\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle$$

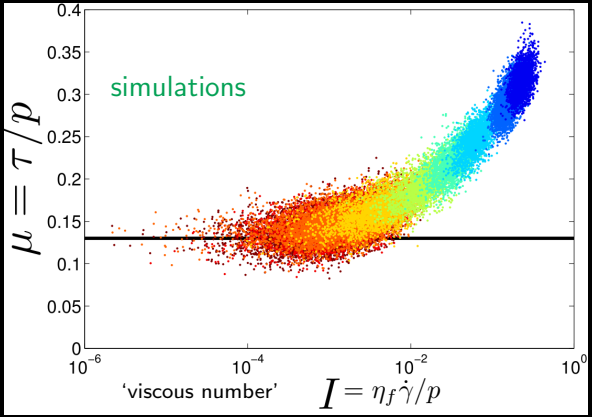
stress $$\tau = \frac{\langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle}{L^d}$$

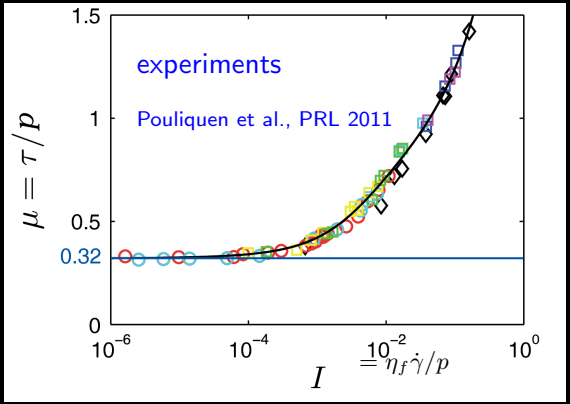
pressure $$p = -\frac{\langle r|\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle}{L^d}$$

motion picture



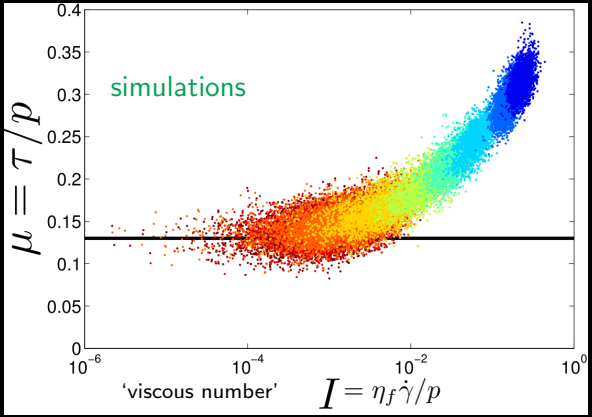
ASM rheology – 'friction law'

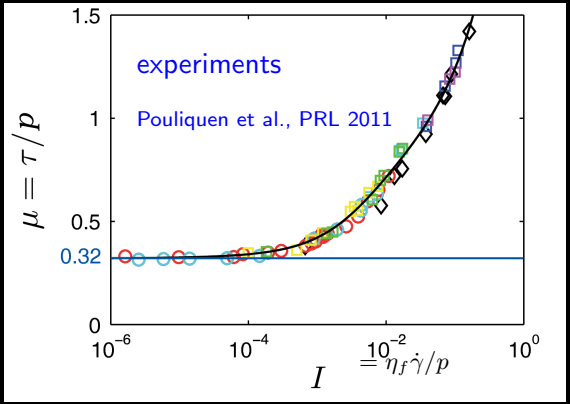




EL, G. Düring, M. Wyart, PNAS **109**, 4798 (2012)

ASM rheology – 'friction law'

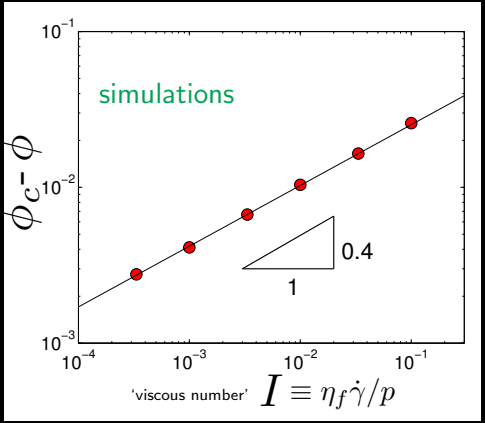




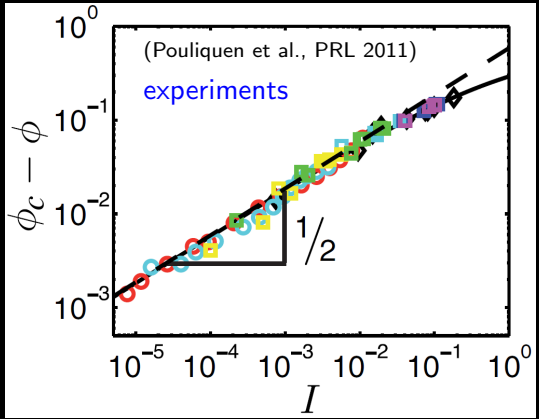
in the ASM: $I \sim 1/p$

EL, G. Düring, M. Wyart, PNAS **109**, 4798 (2012)

ASM rheology – 'dilatancy law'

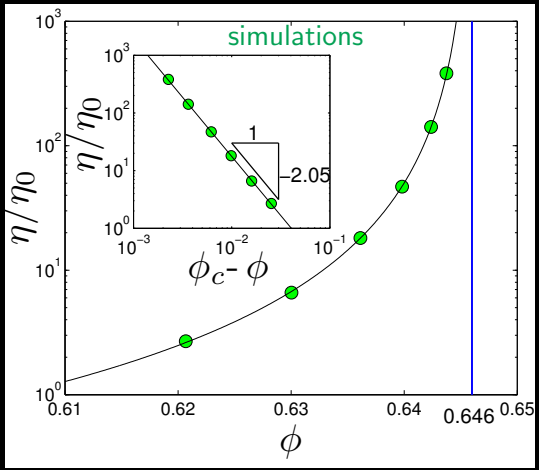


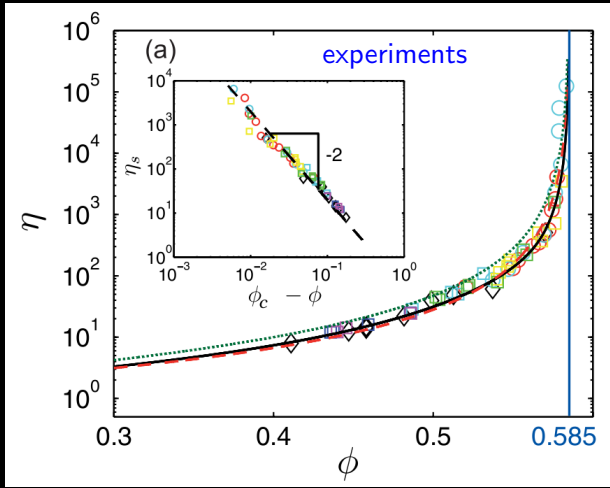
$\phi_c \approx 0.646$ (frictionless)



$\phi_c \approx 0.585$ (frictional)

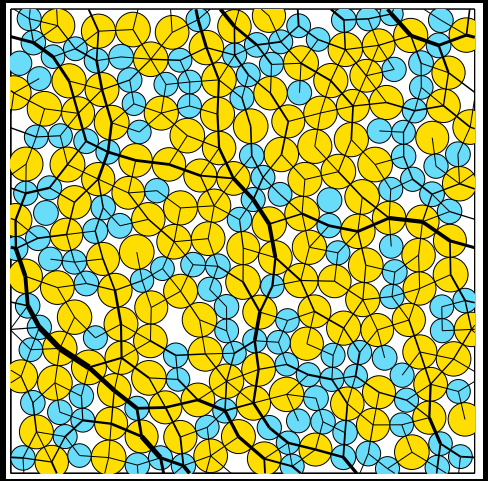
ASM rheology – viscosity





Pouliquen et al., PRL 2011

how do we understand the singular rheology from a **micromechanical** perspective?



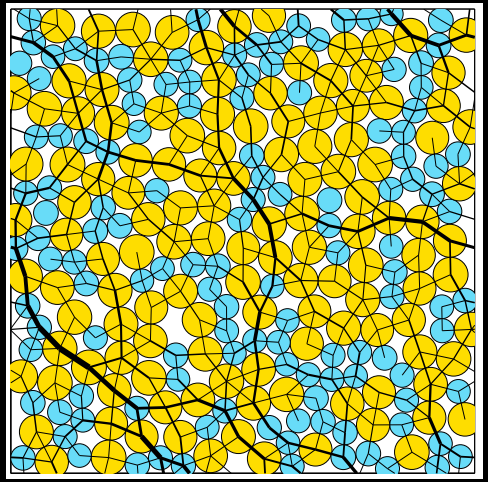
how do we understand the singular rheology from a **micromechanical** perspective?

geometry fully determines mechanics:

contact forces $$|f\rangle = -\mathcal{N}^{-1}\mathcal{S}|V^{\mathrm{aff}}\rangle$$

stress $$\tau = \frac{\langle V^{\mathrm{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\mathrm{aff}}\rangle}{L^d}$$

pressure $$p = -\frac{\langle r|\mathcal{N}^{-1}\mathcal{S}|V^{\mathrm{aff}}\rangle}{L^d}$$



how do we understand the singular rheology from a **micromechanical** perspective?

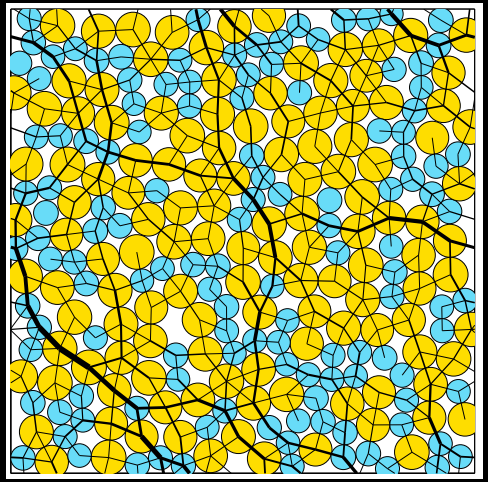
geometry fully determines mechanics:

contact forces $$|f\rangle = -\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle$$

stress $$\tau = \frac{\langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle}{L^d}$$

pressure $$p = -\frac{\langle r|\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle}{L^d}$$

$\mathcal{N}$ must have low-lying modes!!

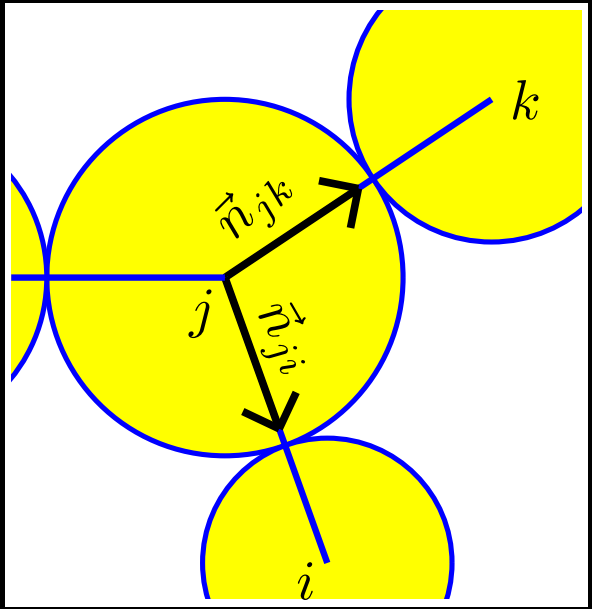


the $\mathcal{N}$ matrix

the $\mathcal{N}$ matrix

'lives' in the space of contacts

$$\mathcal{N}_{ji,jk} \sim \vec{n}_{ji} \cdot \vec{n}_{jk}$$

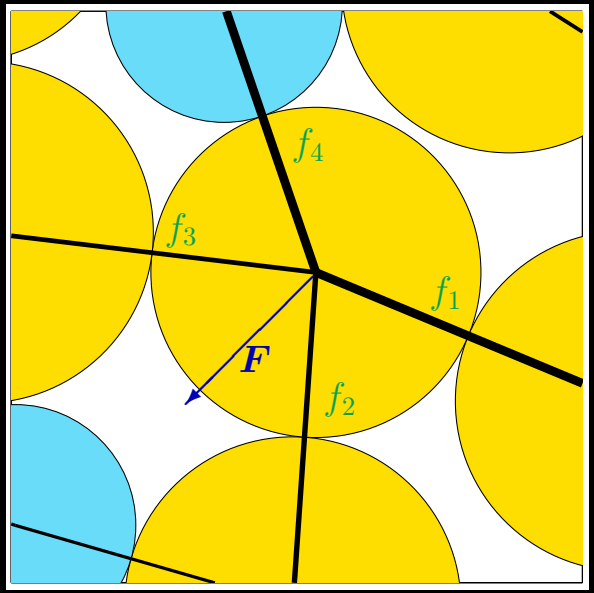


the $\mathcal{N}$ matrix

'lives' in the space of contacts

eigenvalues of $\mathcal{N} \sim \frac{\langle f|\mathcal{N}|f\rangle}{\langle f|f\rangle} \sim \frac{||F||^2}{||f||^2}$

(in jammed packings $\frac{||F||^2}{||f||^2} = 0$)

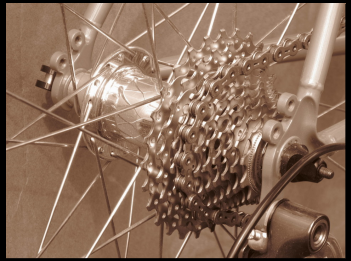


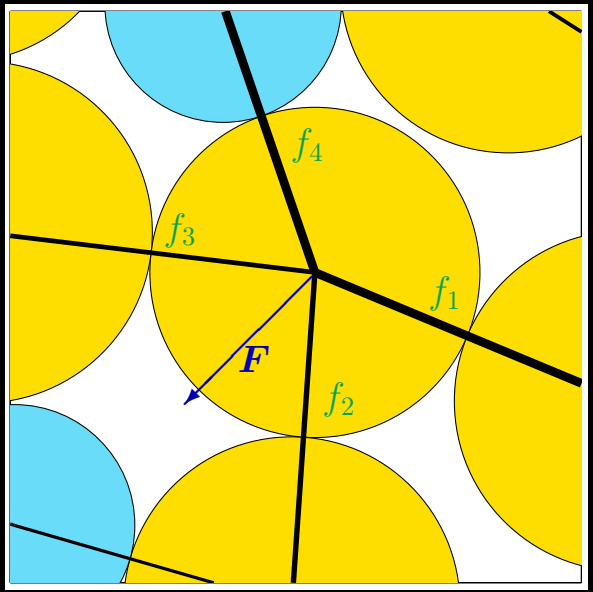
the $\mathcal{N}$ matrix

'lives' in the space of contacts

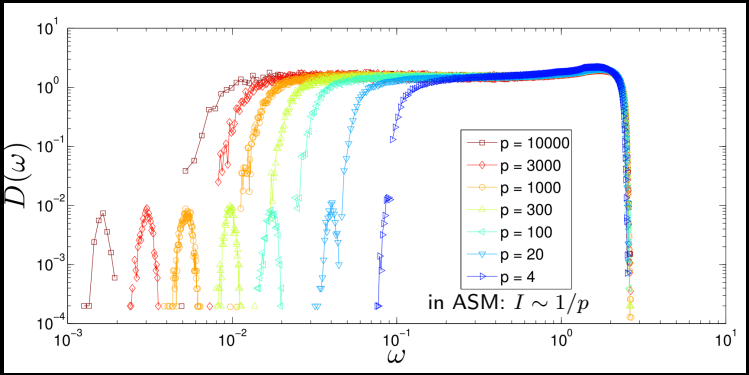
eigenvalues of $\mathcal{N} \sim \frac{\langle f|\mathcal{N}|f\rangle}{\langle f|f\rangle} \sim \frac{||F||^2}{||f||^2}$

(in jammed packings $\frac{||F||^2}{||f||^2} = 0$)





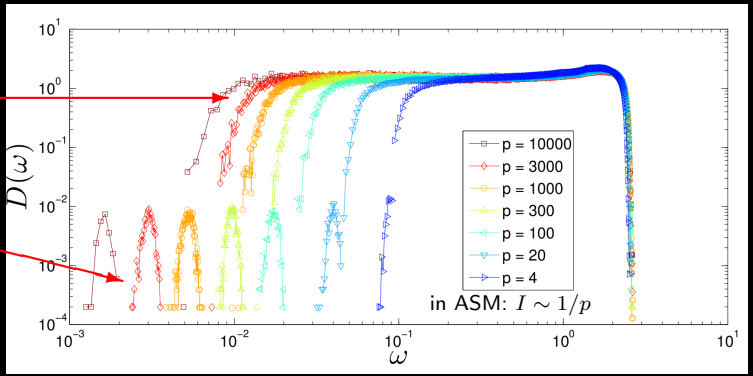
spectral analysis of $\mathcal{N}$



spectral analysis of $\mathcal{N}$

2 frequency scales:

- plateau emerging at $\omega_\star$
- SINGLE mode at $\omega_{\min}$

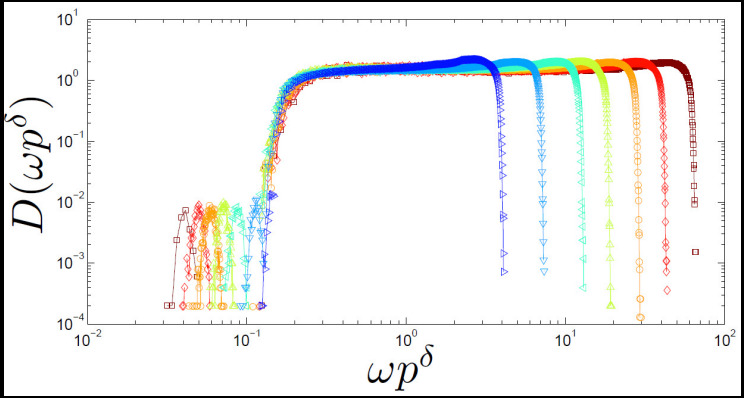


spectral analysis of $\mathcal{N}$

2 frequency scales:

- plateau emerging at $\omega_\star$

$$\omega_\star \sim p^{-\delta}, \quad \delta \approx 0.35$$



spectral analysis of $\mathcal{N}$

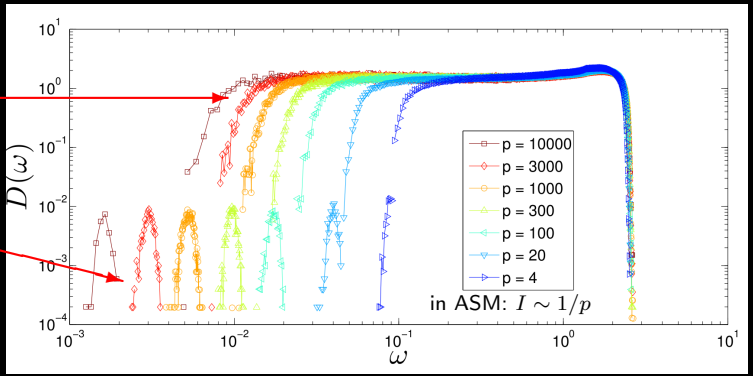
2 frequency scales:

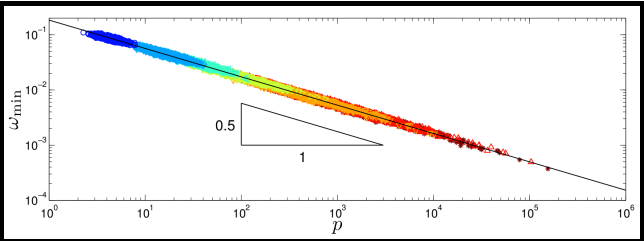
- plateau emerging at $\omega_\star$

  $\omega_\star \sim p^{-\delta}, \quad \delta \approx 0.35$

- SINGLE mode at $\omega_{\min}$

  $\omega_{\min} \sim 1/\sqrt{p}$





spectral analysis of $\mathcal{N}$

2 frequency scales:

- plateau emerging at $\omega_\star$

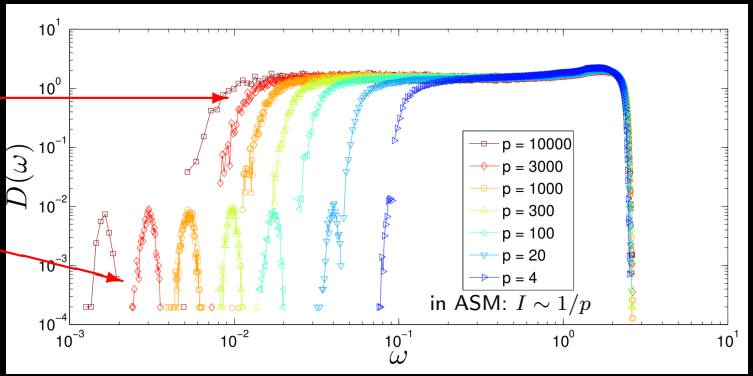
  $\omega_\star \sim p^{-\delta}, \quad \delta \approx 0.35$

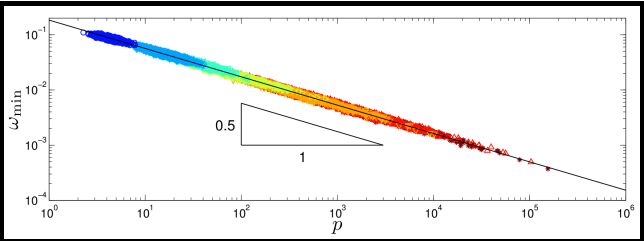
- SINGLE mode at $\omega_{\min}$

  $\omega_{\min} \sim 1/\sqrt{p}$

what is the origin of this bi-scaling?

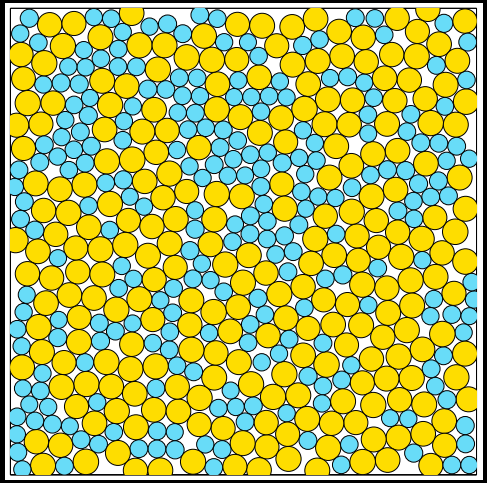
$\omega_\star \sim p^{-0.35}, \ \omega_{\min} \sim 1/\sqrt{p}$





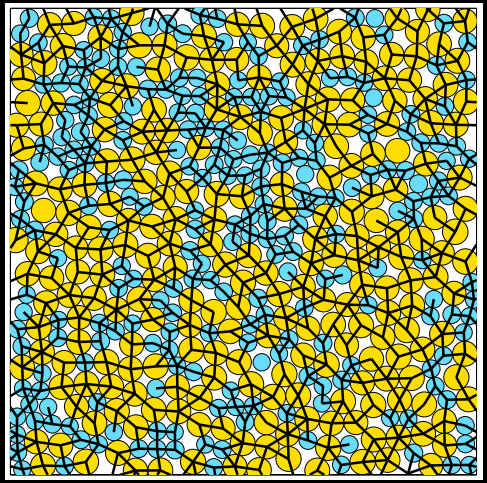
relation to floppy networks of **rigid** bars

instantaneous response of driven hard spheres



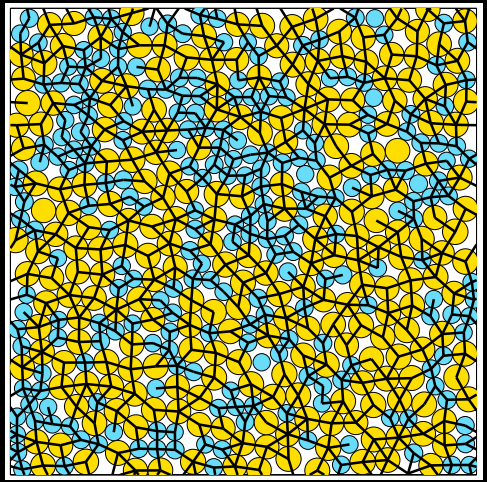
relation to floppy networks of **rigid** bars

instantaneous response of driven hard spheres = that of a network of **rigid bars**



relation to floppy networks of **rigid** bars

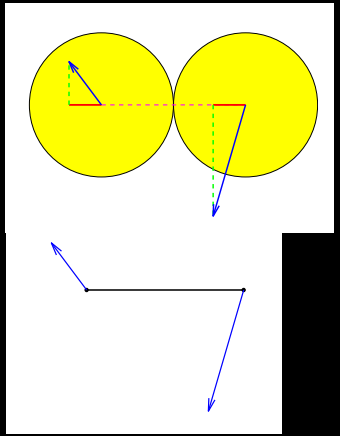
instantaneous response of driven hard spheres



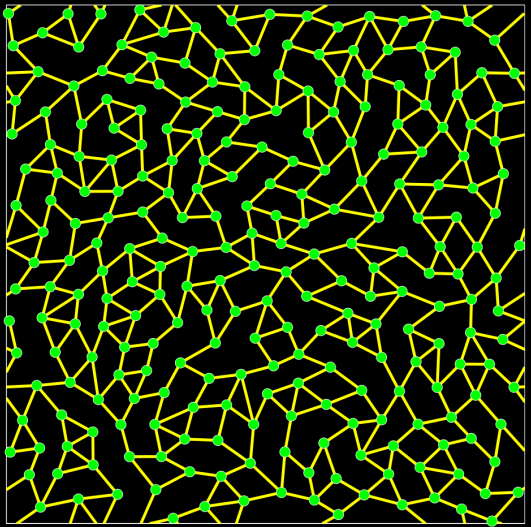
=

that of a network of **rigid bars**

displacements preserve contacts

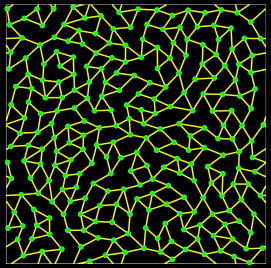


can we understand the spectrum of $\mathcal{N}$ by studying floppy networks of **rigid bars**?



mechanics of floppy networks of **rigid bars**

key control parameter $z \equiv$ mean connectivity

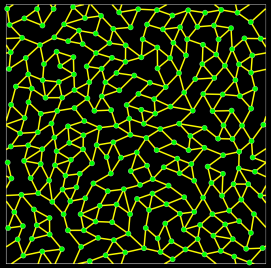


mechanics of floppy networks of **rigid bars**

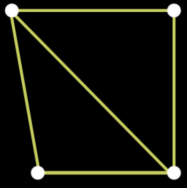
key control parameter $z \equiv$ mean connectivity

$\delta z \equiv z_c - z$ controls the degree of 'floppiness'

variables $\sim$ constraints $\Rightarrow$ $z_c = 2 \times \bar{d}$ with $\bar{d} \equiv$ space



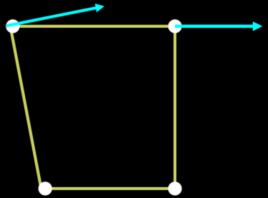
$z = z_c$



jammed

'isostatic'

$z < z_c$

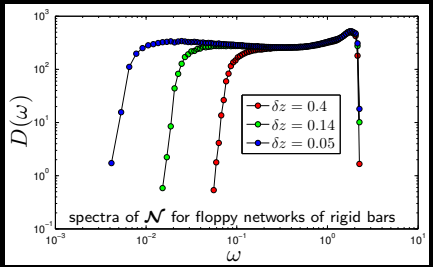


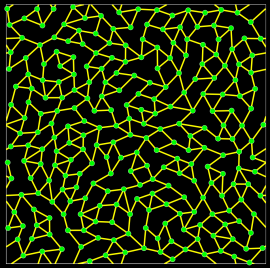
floppy

'hypostatic'

mechanics of floppy networks of **rigid bars**

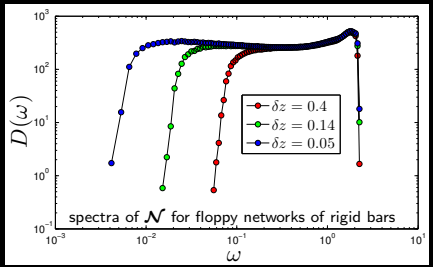
eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

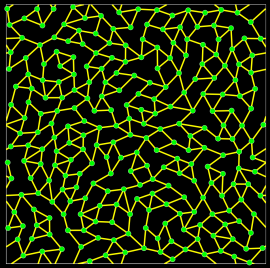


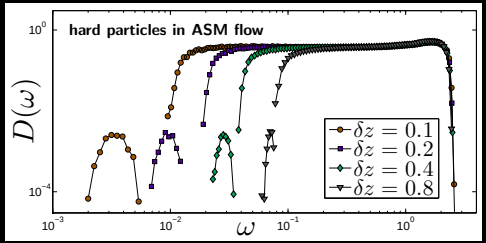


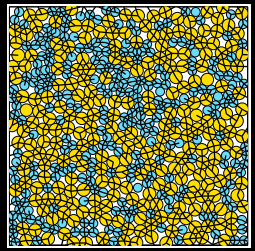
mechanics of floppy networks of **rigid bars**

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$



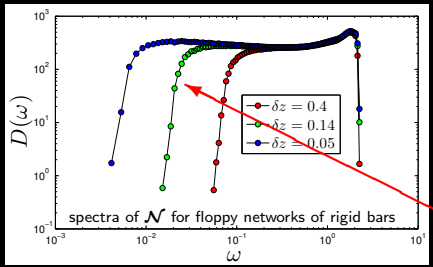


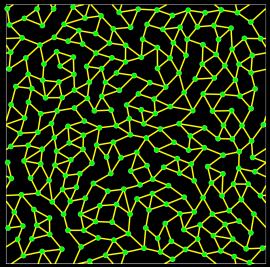


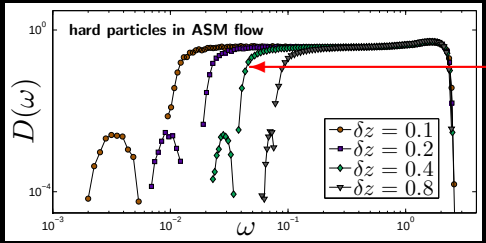


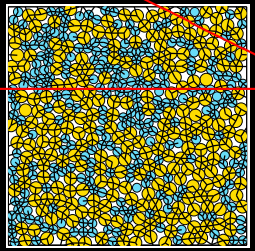
mechanics of floppy networks of **rigid bars**

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$





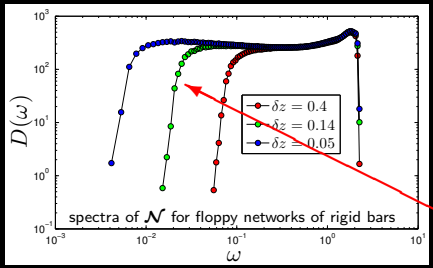


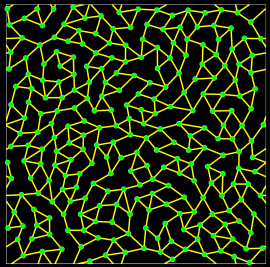


same dependence on connectivity!

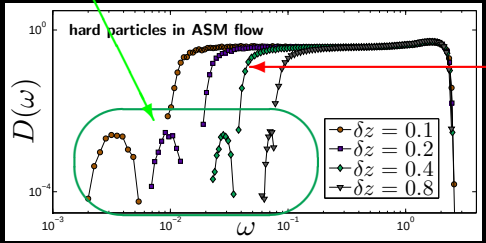
mechanics of floppy networks of **rigid bars**

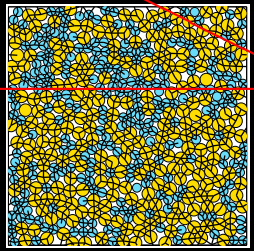
eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$





self-organization! more 'solid-like'





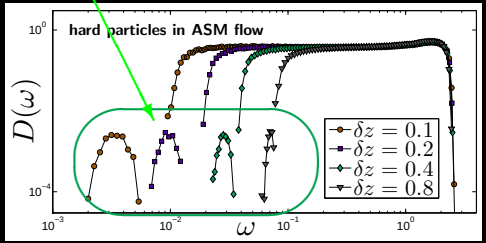
same dependence on connectivity!

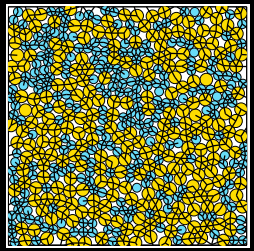
mechanics of floppy networks of **rigid bars**

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

how do lower frequency modes emerge??

self-organization! more 'solid-like'





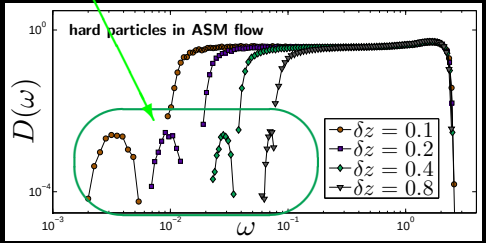
mechanics of floppy networks of **rigid bars**

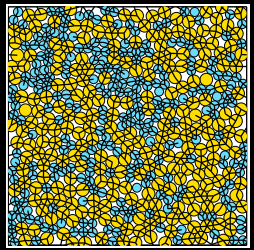
eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

how do lower frequency modes emerge??

how does one make a floppy system 'less floppy'??

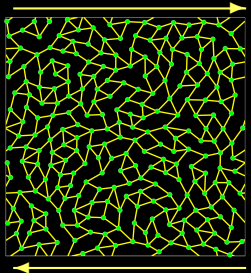
self-organization! more 'solid-like'





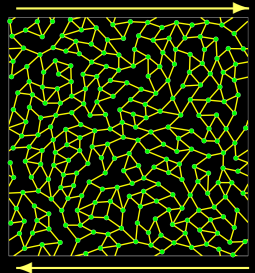
mechanics of floppy networks of **rigid bars**

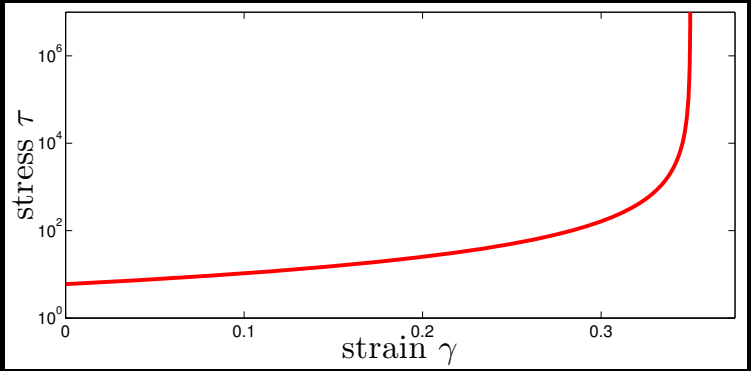
deform networks with overdamped dynamics



mechanics of floppy networks of **rigid bars**

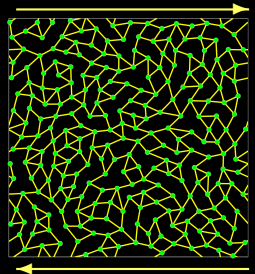
deform networks with overdamped dynamics

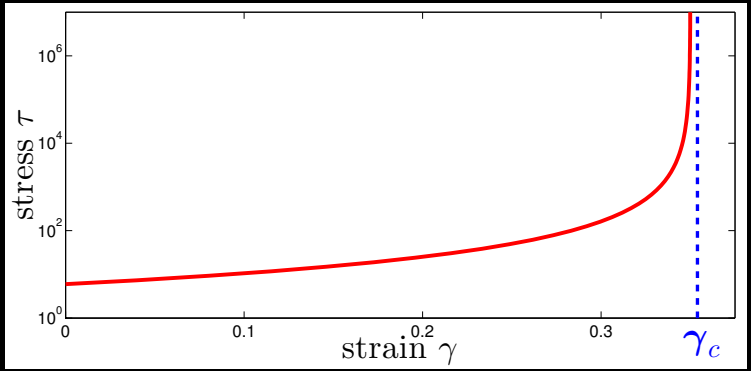




mechanics of floppy networks of **rigid bars**

deform networks with overdamped dynamics

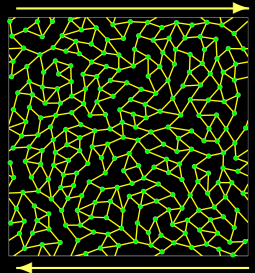


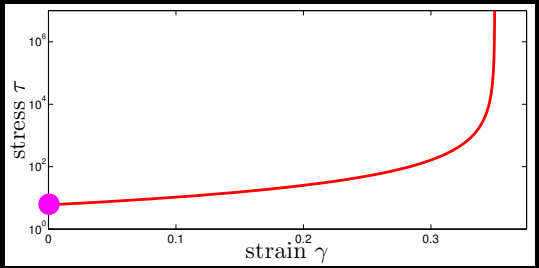


network stiffens (jams) at a strain $\gamma_c \sim \delta z$

mechanics of floppy networks of **rigid bars**

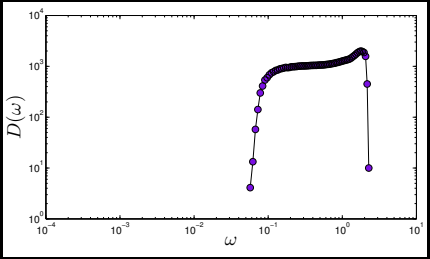
eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

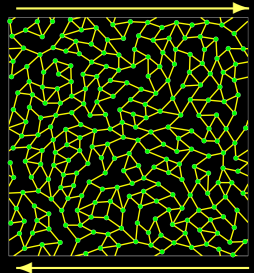


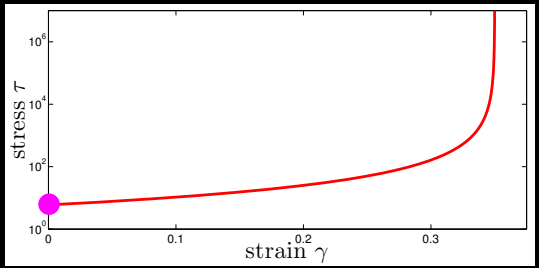


mechanics of floppy networks of **rigid bars**

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

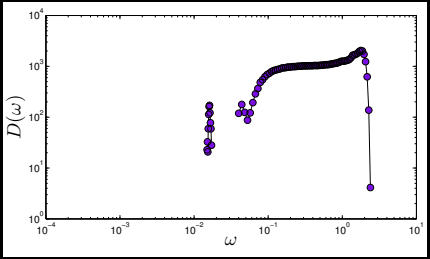


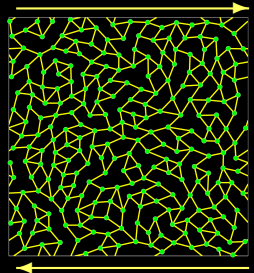


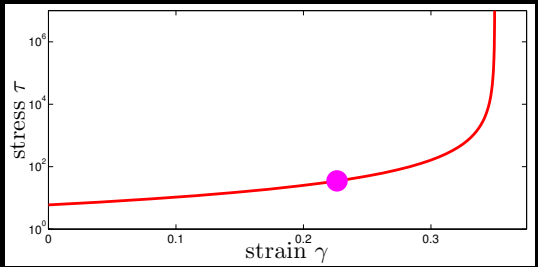


mechanics of floppy networks of **rigid bars**

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

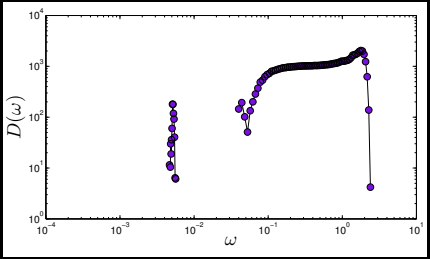


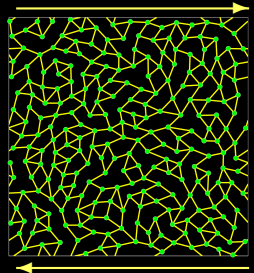


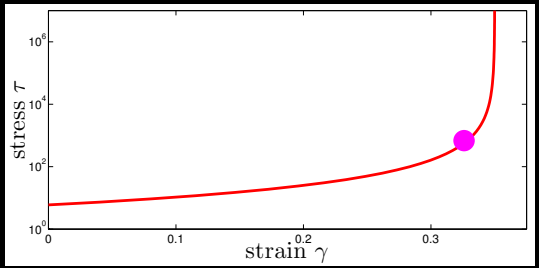


mechanics of floppy networks of **rigid bars**

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

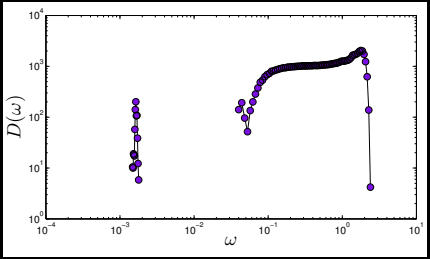


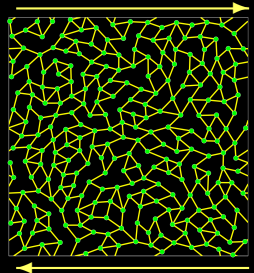


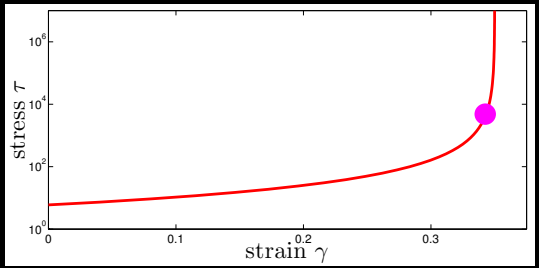


mechanics of floppy networks of **rigid bars**

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

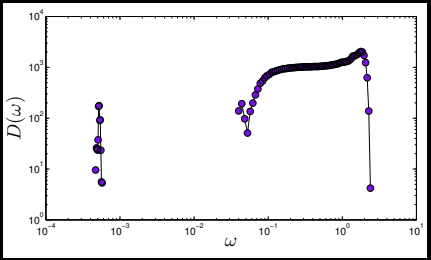


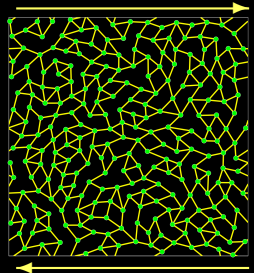


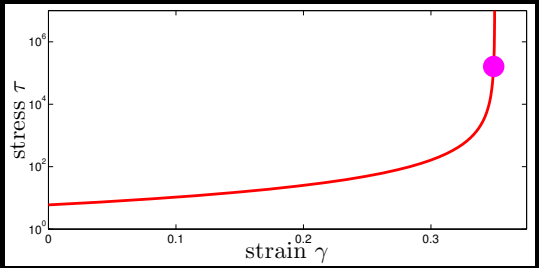


mechanics of floppy networks of **rigid bars**

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

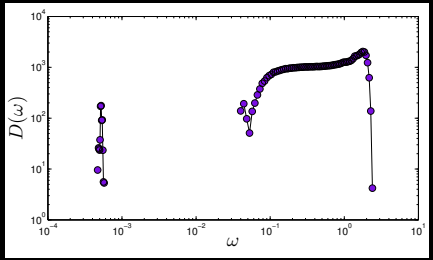


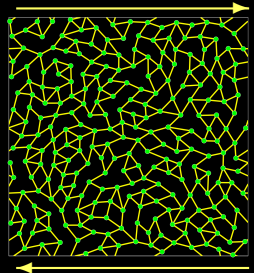


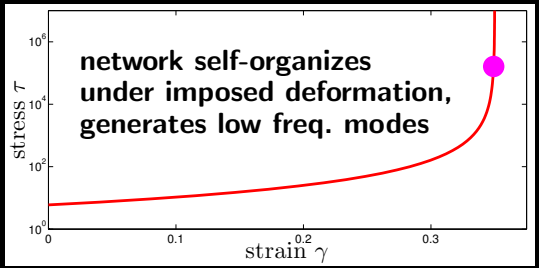


mechanics of floppy networks of **rigid bars**

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

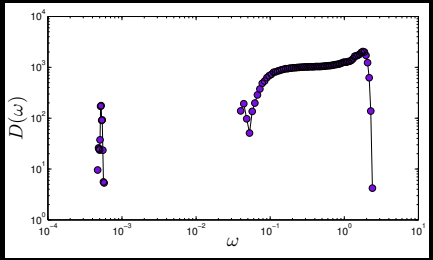


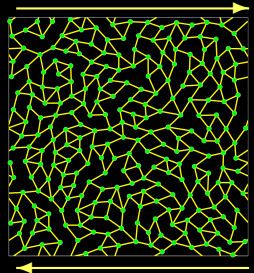


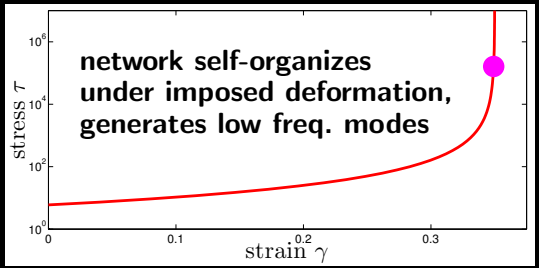


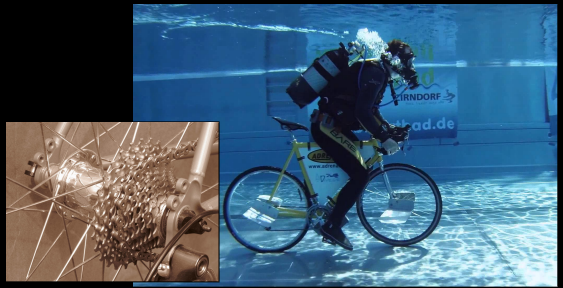
mechanics of floppy networks of **rigid bars**

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$



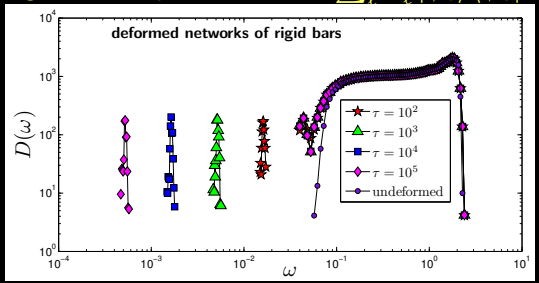


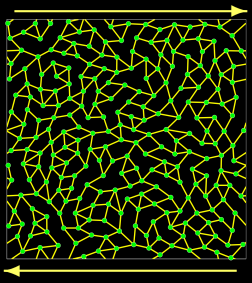


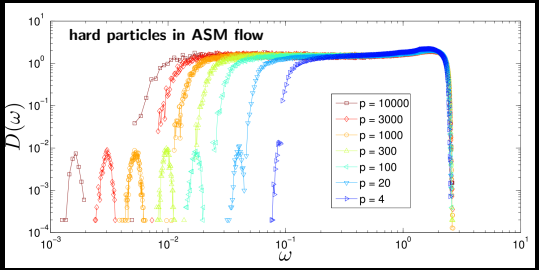


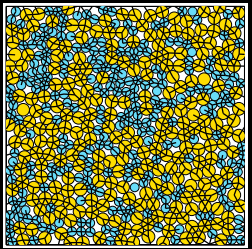
mechanics of floppy networks of **rigid bars**

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

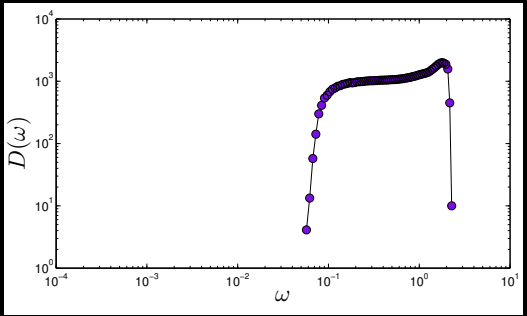


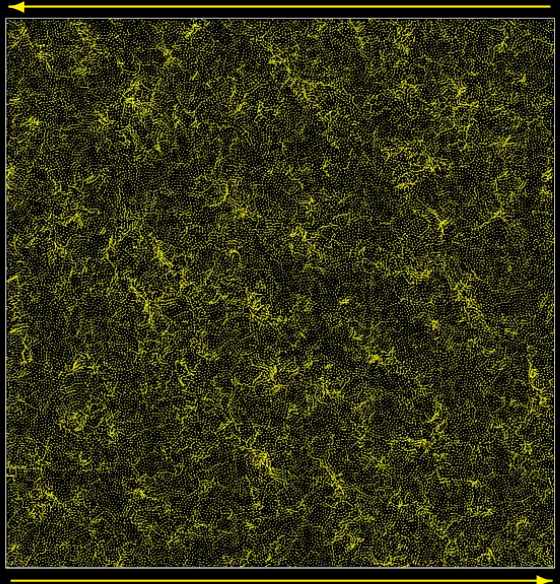




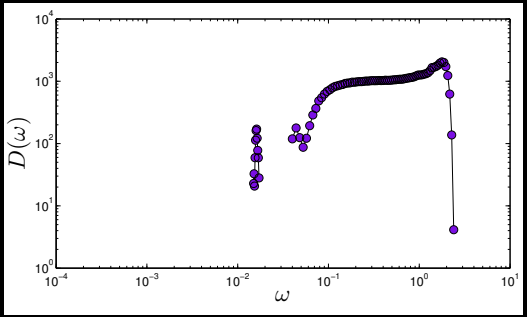


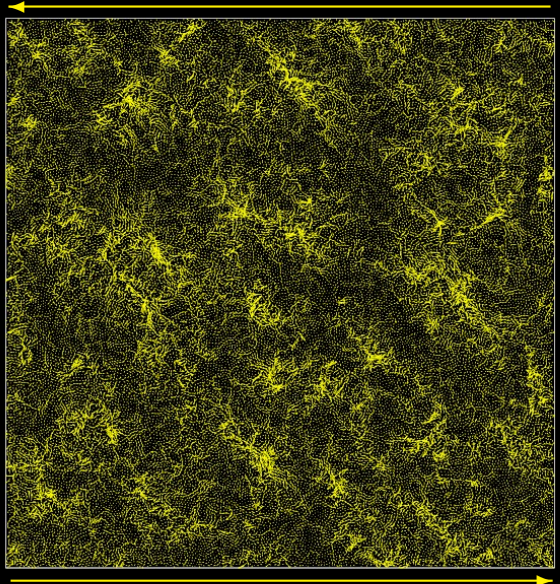
growing correlation length



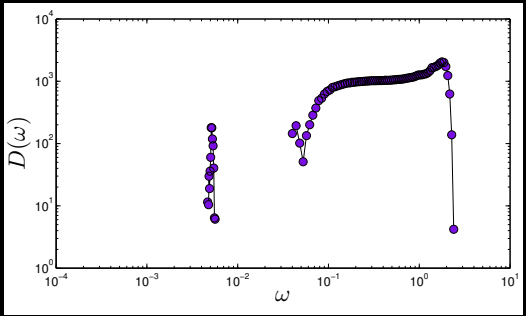


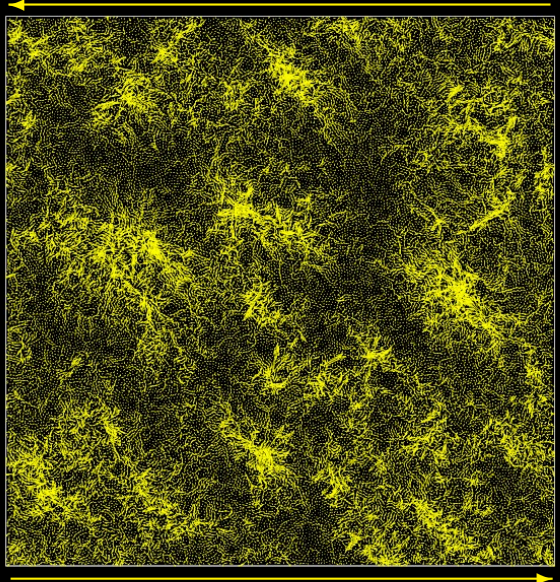
growing correlation length



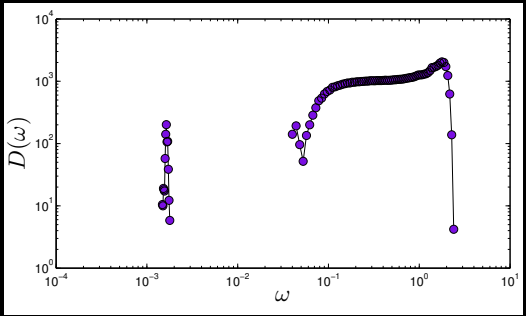


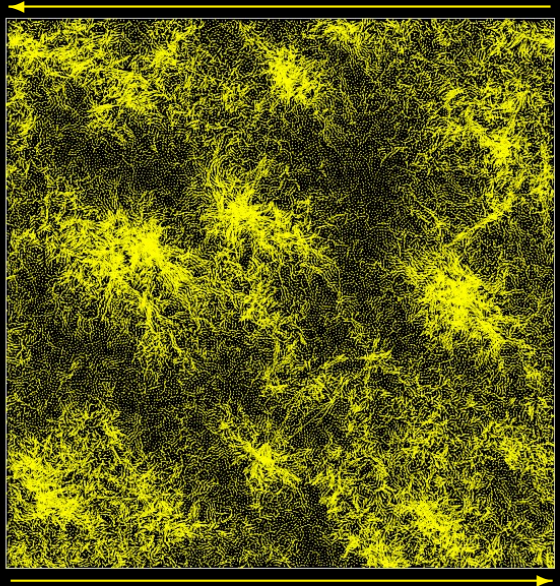
growing correlation length



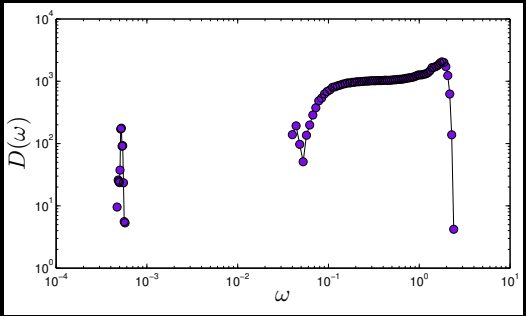


growing correlation length

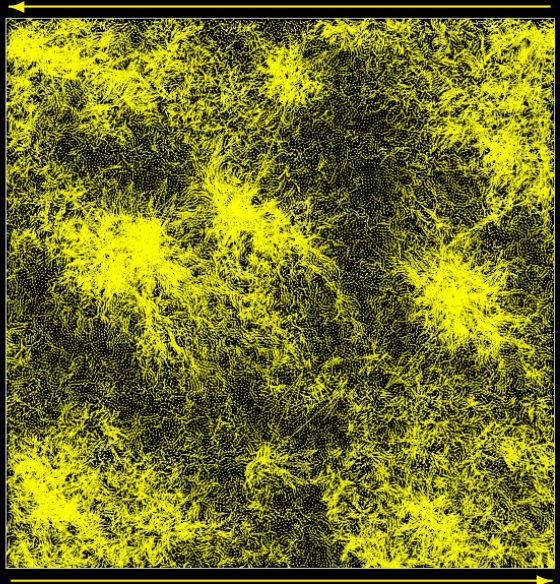




growing correlation length

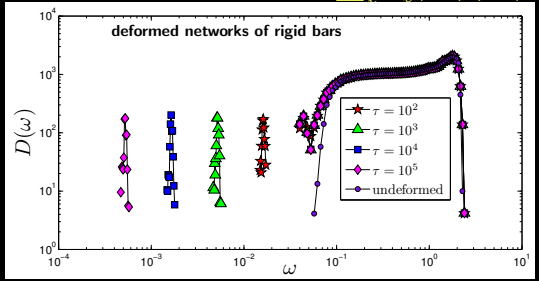


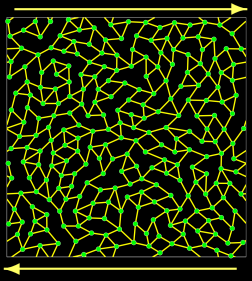
$$\ell \sim \frac{1}{\omega_{\min}}$$

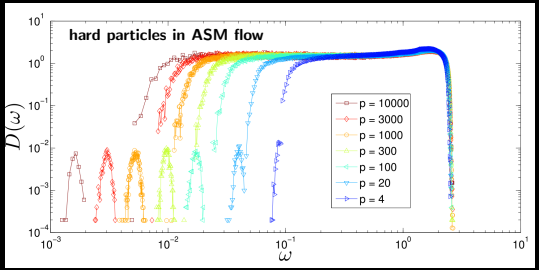


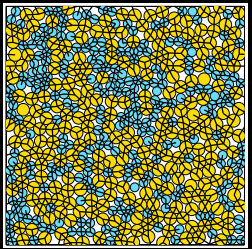
self-organization and rheology

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$



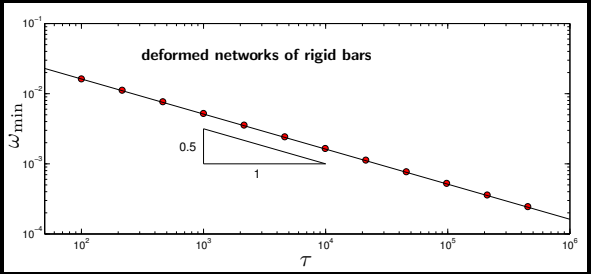


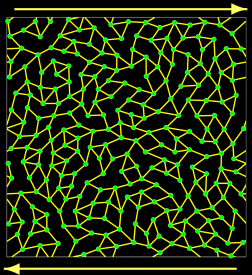


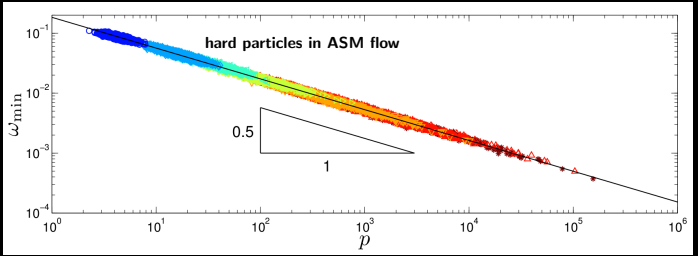


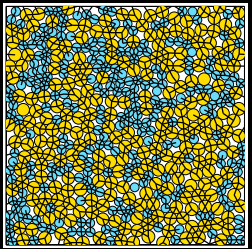
self-organization and rheology

eigenmode expansion: $\mathcal{N} = \sum_\ell \omega_\ell^2 |\psi_\ell\rangle\langle\psi_\ell|$

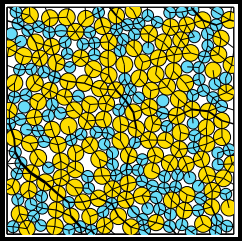








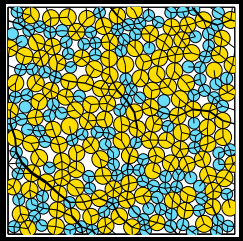
... back to particles: stress relaxation



stretched network: tensile forces

sheared hard particles: compressive forces

... back to particles: stress relaxation

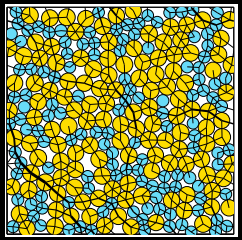


stretched network: tensile forces

sheared hard particles: compressive forces

stress relaxation: $\frac{d\tau}{d\gamma} \sim -\tau^2$

... back to particles: stress relaxation

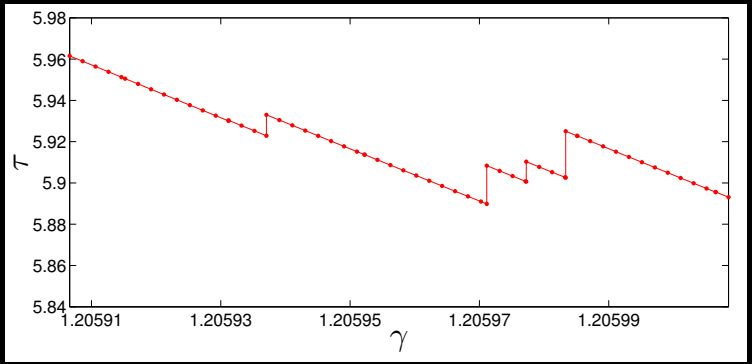


stretched network: tensile forces

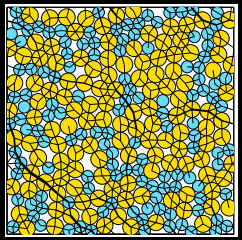
sheared hard particles: compressive forces

stress relaxation: $\frac{d\tau}{d\gamma} \sim -\tau^2$

hard particles in ASM flow



... back to particles: stress relaxation

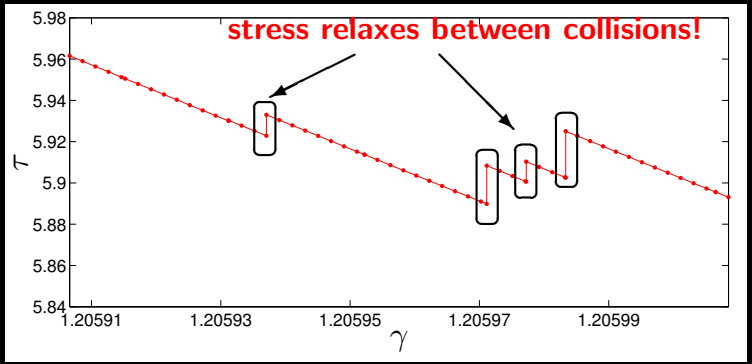


stretched network: tensile forces

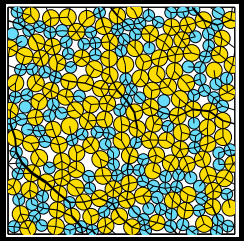
sheared hard particles: compressive forces

stress relaxation: $\frac{d\tau}{d\gamma} \sim -\tau^2$

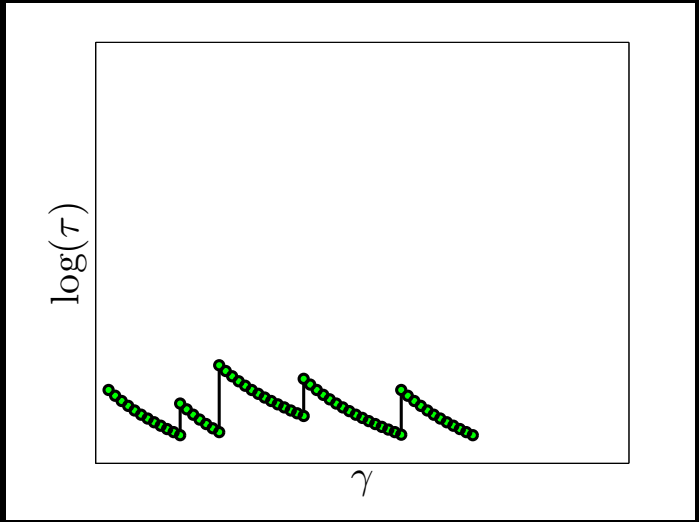
hard particles in ASM flow



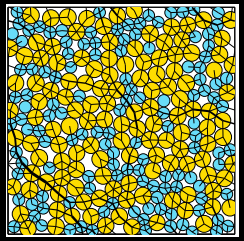
... back to particles: stress relaxation



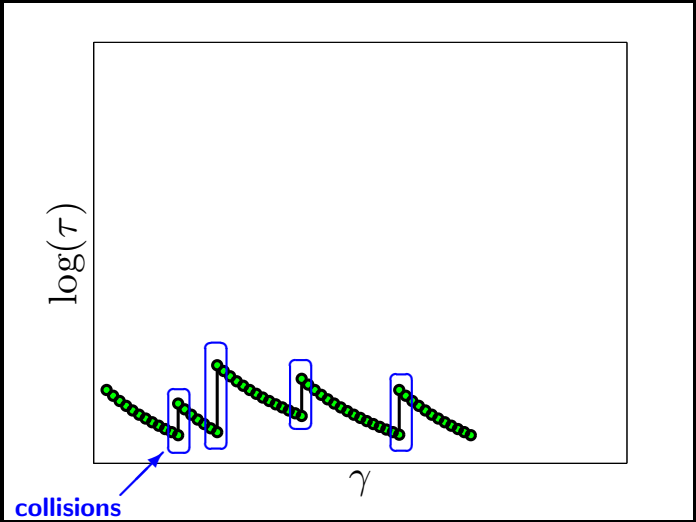
stress relaxes between collisions via $\frac{d\tau}{d\gamma} \sim -\tau^2$



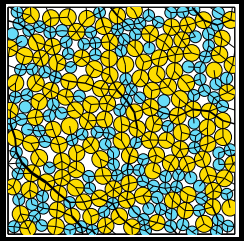
... back to particles: stress relaxation



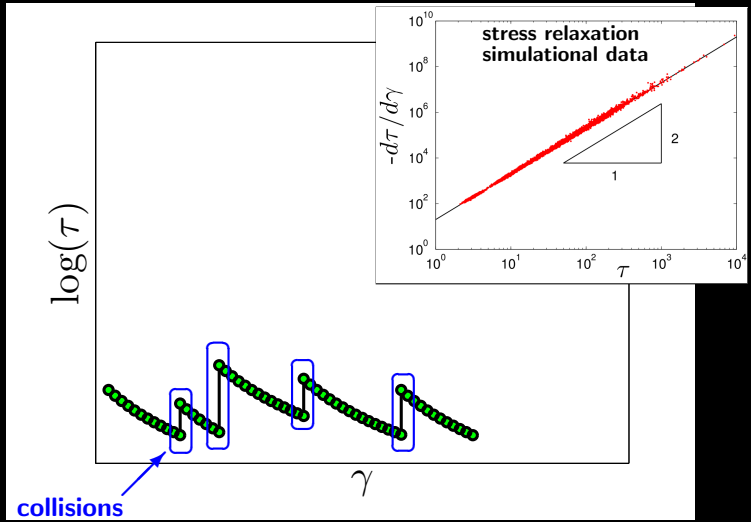
stress relaxes between collisions via $\frac{d\tau}{d\gamma} \sim -\tau^2$



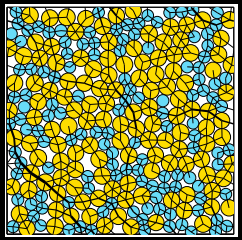
... back to particles: stress relaxation



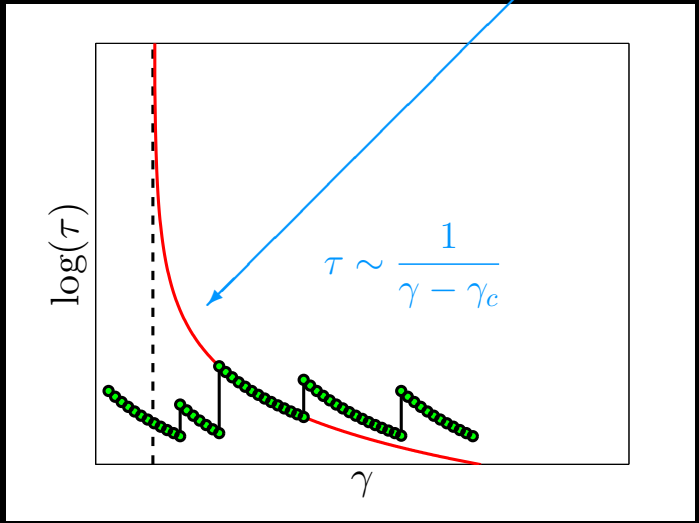
stress relaxes between collisions via $\frac{d\tau}{d\gamma} \sim -\tau^2$



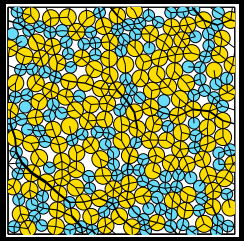
... back to particles: stress relaxation



stress relaxes between collisions via $\frac{d\tau}{d\gamma} \sim -\tau^2$

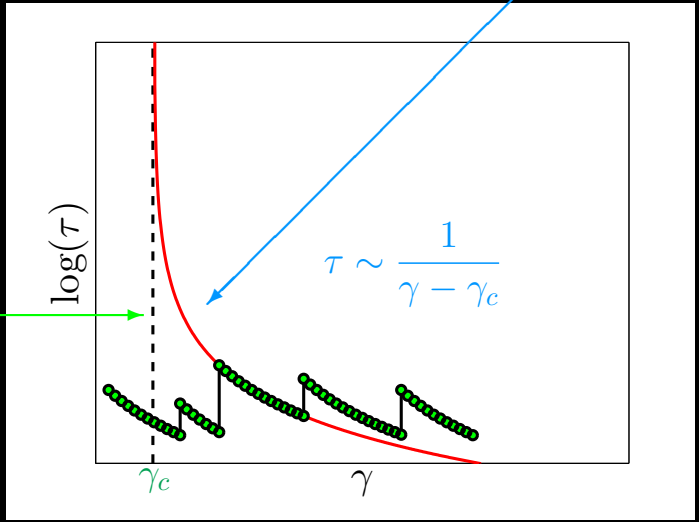


... back to particles: stress relaxation

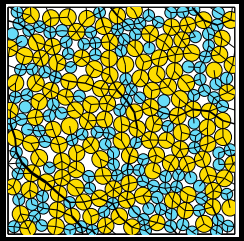


stress relaxes between collisions via $\frac{d\tau}{d\gamma} \sim -\tau^2$

system moves away from a 'fictitious' stiffening strain γ_c

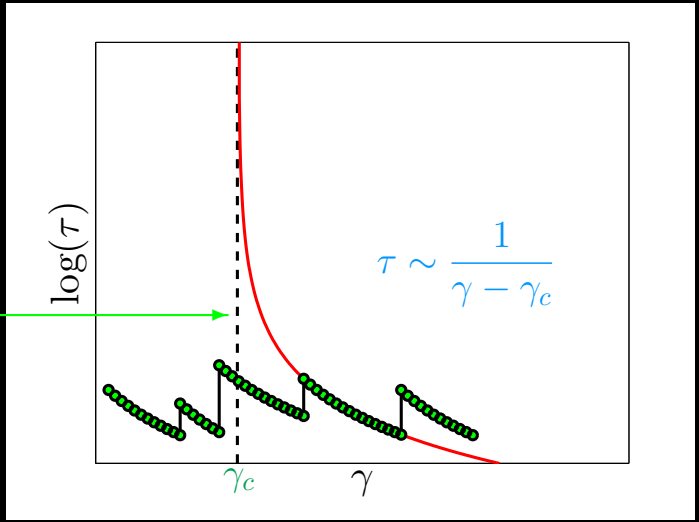


... back to particles: stress relaxation

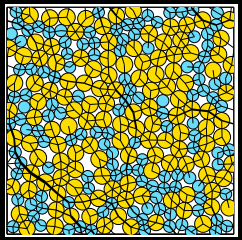


stress relaxes between collisions via $\frac{d\tau}{d\gamma} \sim -\tau^2$

system moves away from a 'fictitious' stiffening strain γ_c

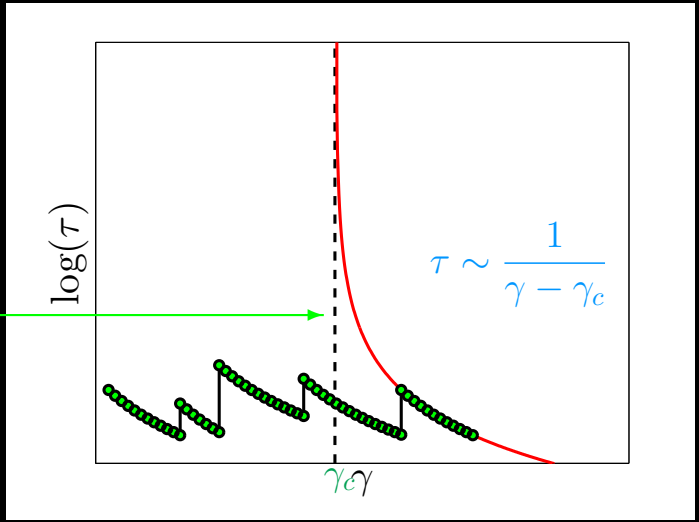


... back to particles: stress relaxation

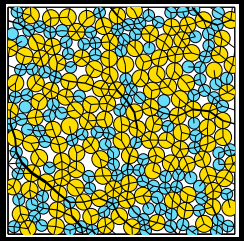


stress relaxes between collisions via $\frac{d\tau}{d\gamma} \sim -\tau^2$

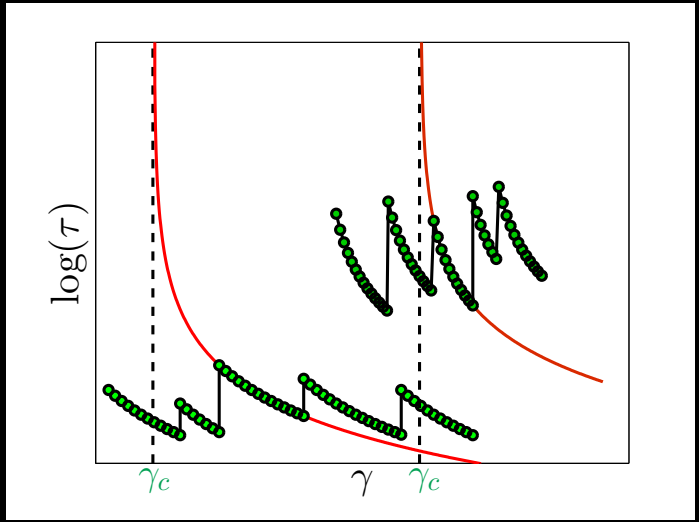
system moves away from a 'fictitious' stiffening strain γ_c



... back to particles: stress relaxation

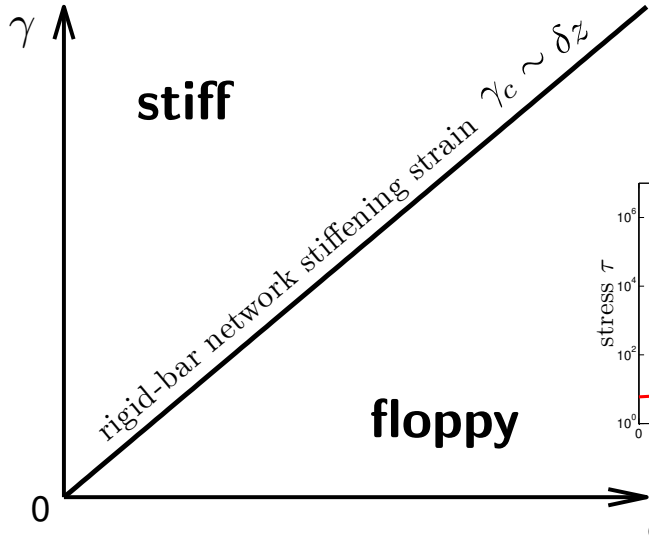


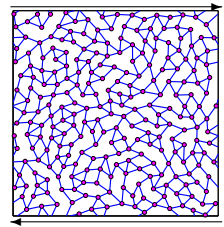
stress relaxes between collisions via $\frac{d\tau}{d\gamma} \sim -\tau^2$

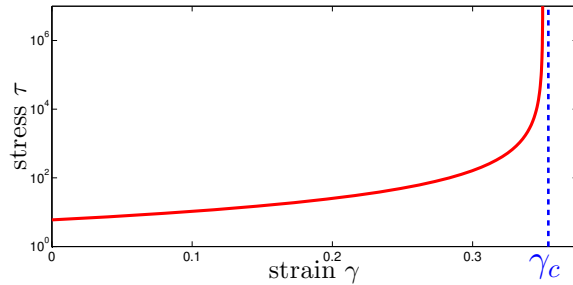


higher stress $\Rightarrow$
γ_c is 'closer behind',
system 'lives' closer to
a stiffening transition

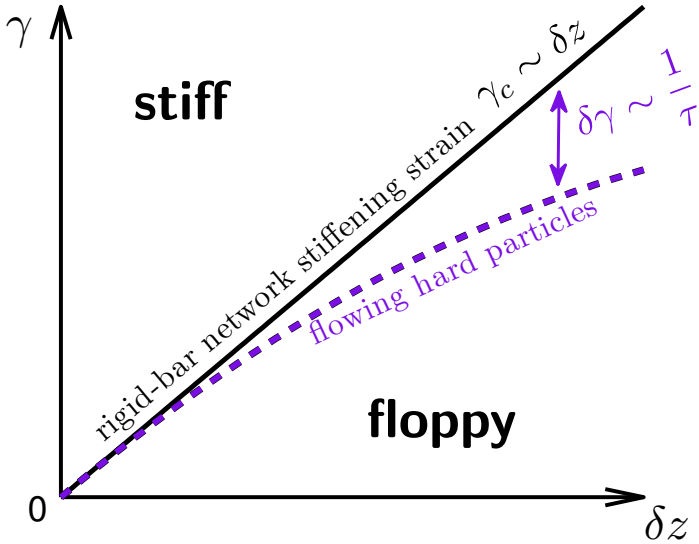
self-organization in flow – phase diagram

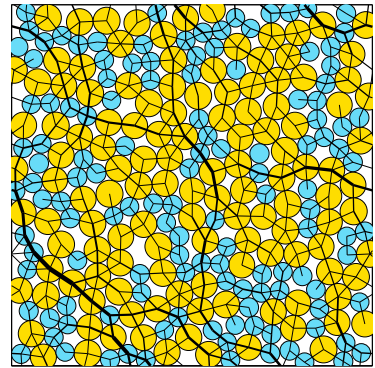




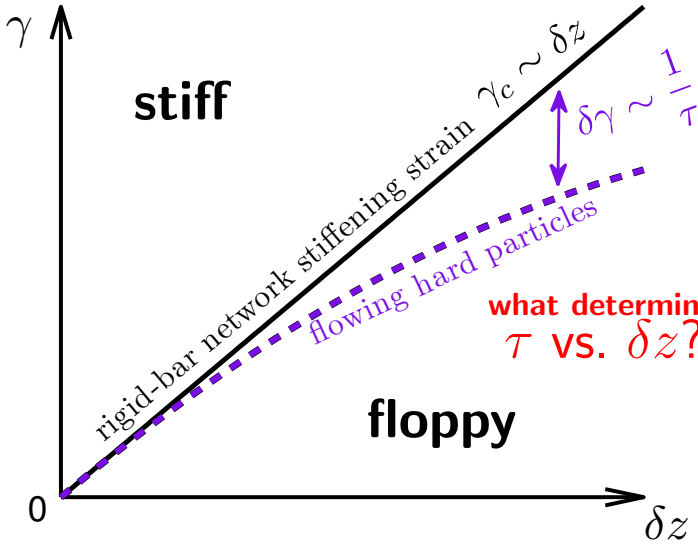


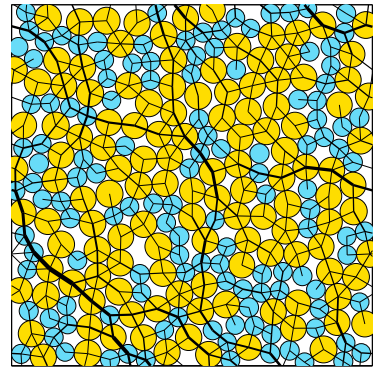
self-organization in flow – phase diagram



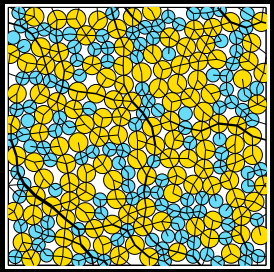


self-organization in flow – phase diagram



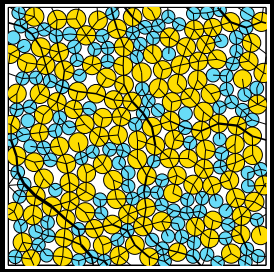


velocities vs. viscosity



velocities vs. viscosity

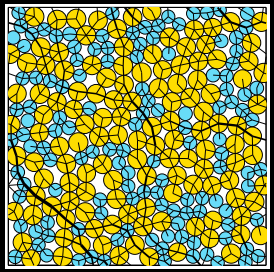
$\rightarrow$ velocities $|V\rangle \sim \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} |V^{\mathrm{aff}}\rangle$



velocities vs. viscosity

$\rightarrow$ velocities $|V\rangle \sim \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} |V^{\mathrm{aff}}\rangle$

$\Rightarrow \;\; V^2 \sim \langle V^{\mathrm{aff}} | \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} | V^{\mathrm{aff}} \rangle = \langle V^{\mathrm{aff}} | \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} | V^{\mathrm{aff}} \rangle \sim \tau$

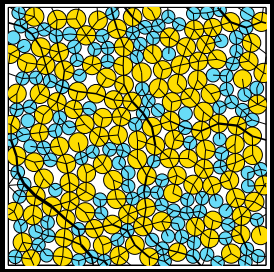


velocities vs. viscosity

$\rightarrow$ velocities $|V\rangle \sim \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} |V^{\text{aff}}\rangle$

$\Rightarrow \; V^2 \sim \langle V^{\text{aff}} | \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} | V^{\text{aff}} \rangle = \langle V^{\text{aff}} | \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} | V^{\text{aff}} \rangle \sim \tau$

$\rightarrow V \sim \dot{\gamma}$ and so we can consider displacements per shear $V/\dot{\gamma}$



velocities vs. viscosity

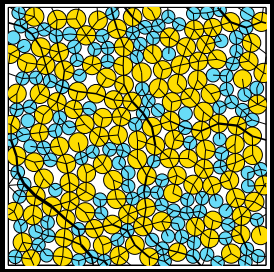
$\rightarrow$ velocities $|V\rangle \sim \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} |V^{\mathrm{aff}}\rangle$

$\Rightarrow \; V^2 \sim \langle V^{\mathrm{aff}} | \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} | V^{\mathrm{aff}} \rangle = \langle V^{\mathrm{aff}} | \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} | V^{\mathrm{aff}} \rangle \sim \tau$

$\rightarrow V \sim \dot{\gamma}$ and so we can consider displacements per shear $V/\dot{\gamma}$

---

next:



velocities vs. viscosity

$\rightarrow$ velocities $|V\rangle \sim \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} |V^{\text{aff}}\rangle$

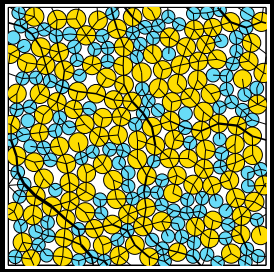
$\Rightarrow \; V^2 \sim \langle V^{\text{aff}} | \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} | V^{\text{aff}} \rangle = \langle V^{\text{aff}} | \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} | V^{\text{aff}} \rangle \sim \tau$

$\rightarrow V \sim \dot{\gamma}$ and so we can consider displacements per shear $V/\dot{\gamma}$

---

next:

$\rightarrow$ consider a **jammed state**



velocities vs. viscosity

$\rightarrow$ velocities $|V\rangle \sim \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} |V^{\text{aff}}\rangle$

$\Rightarrow \; V^2 \sim \langle V^{\text{aff}} | \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} | V^{\text{aff}} \rangle = \langle V^{\text{aff}} | \mathcal{S}^T \mathcal{N}^{-1} \mathcal{S} | V^{\text{aff}} \rangle \sim \tau$

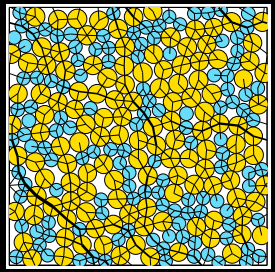
$\rightarrow V \sim \dot{\gamma}$ and so we can consider displacements per shear $V/\dot{\gamma}$

---

next:

$\rightarrow$ consider a **jammed state**

$\rightarrow$ select a few contacts such that removing them results in a state with some small δz



velocities vs. viscosity

$\rightarrow$ velocities $|V\rangle \sim \mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle$

$\Rightarrow \; V^2 \sim \langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle = \langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle \sim \tau$

$\rightarrow V \sim \dot{\gamma}$ and so we can consider displacements per shear $V/\dot{\gamma}$

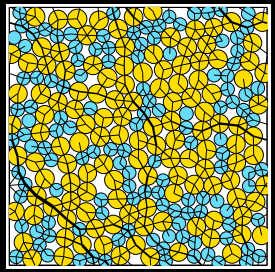
---

next:

$\rightarrow$ consider a **jammed state**

$\rightarrow$ select a few contacts such that removing them results in a state with some small δz

$\rightarrow$ what are the displacements per shear $V/\dot{\gamma}$ of the new state?



velocities vs. viscosity

$\rightarrow$ velocities $|V\rangle \sim \mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle$

$\Rightarrow \; V^2 \sim \langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle = \langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle \sim \tau$

$\rightarrow V \sim \dot{\gamma}$ and so we can consider displacements per shear $V/\dot{\gamma}$

---

next:

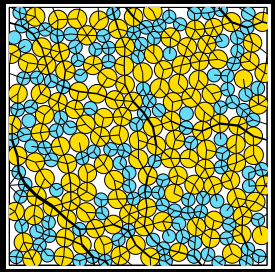
$\rightarrow$ consider a **jammed state**

$\rightarrow$ select a few contacts such that removing them results in a state with some small δz

$\rightarrow$ what are the displacements per shear $V/\dot{\gamma}$ of the new state?

$\rightarrow$ loading work = dissipated energy (+ 'virtual work theorem')

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\substack{\text{removed} \\ ij}} f_{ij}\delta r_{ij}$$



velocities vs. viscosity

$\rightarrow$ velocities $|V\rangle \sim \mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle$

$\Rightarrow \; V^2 \sim \langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle = \langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle \sim \tau$

$\rightarrow V \sim \dot{\gamma}$ and so we can consider displacements per shear $V/\dot{\gamma}$

---

next:

$\rightarrow$ consider a **jammed state**

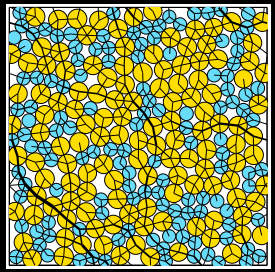
$\rightarrow$ select a few contacts such that removing them results in a state with some small δz

$\rightarrow$ what are the displacements per shear $V/\dot{\gamma}$ of the new state?

$\rightarrow$ loading work = dissipated energy (+ 'virtual work theorem')

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij}$$

sum runs over **removed** contacts
$\delta r_{ij} \equiv$ radial motion
$f_{ij} \equiv$ contact force



velocities vs. viscosity

$\rightarrow$ velocities $|V\rangle \sim \mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle$

$\Rightarrow \; V^2 \sim \langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle = \langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle \sim \tau$

$\rightarrow V \sim \dot{\gamma}$ and so we can consider displacements per shear $V/\dot{\gamma}$

---

next:

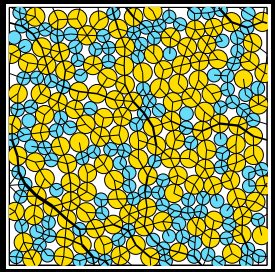
$\rightarrow$ consider a **jammed state**

$\rightarrow$ select a few contacts such that removing them results in a state with some small δz

$\rightarrow$ what are the displacements per shear $V/\dot{\gamma}$ of the new state?

$\rightarrow$ loading work = dissipated energy (+ 'virtual work theorem')

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\substack{\text{removed} \\ ij}} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$



velocities vs. viscosity

$\rightarrow$ velocities $|V\rangle \sim \mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle$

$\Rightarrow \; V^2 \sim \langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle = \langle V^{\text{aff}}|\mathcal{S}^T\mathcal{N}^{-1}\mathcal{S}|V^{\text{aff}}\rangle \sim \tau$

$\rightarrow V \sim \dot{\gamma}$ and so we can consider displacements per shear $V/\dot{\gamma}$

---

next:

$\rightarrow$ consider a **jammed state**

$\rightarrow$ select a few contacts such that removing them results in a state with some small δz

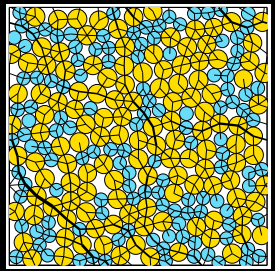
$\rightarrow$ what are the displacements per shear $V/\dot{\gamma}$ of the new state?

$\rightarrow$ loading work = dissipated energy (+ 'virtual work theorem')

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$

$\bar{f}$ ≡typical force scale

$\bar{\delta r}$ ≡typical radial displacement



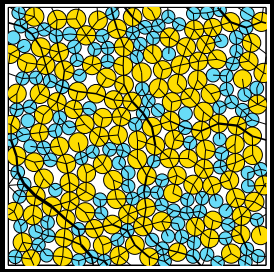
displacements per shear vs. δz

$\longrightarrow$ select a few contacts such that removing them results in a state with some small δz

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$

$\bar{f}$ ≡typical force scale

$\bar{\delta r}$ ≡typical radial displacement



displacements per shear vs. δz

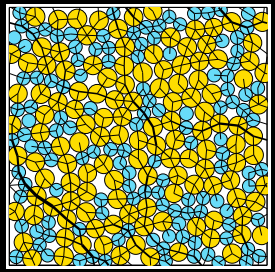
which contacts should be removed?

$\rightarrow$ select a few contacts such that removing them results in a state with some small δz

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$

$\bar{f}$ $\equiv$typical force scale

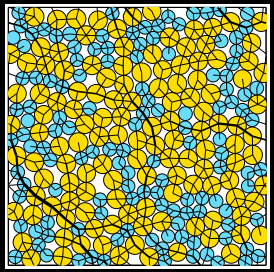
$\bar{\delta r}$ $\equiv$typical radial displacement



displacements per shear vs. δz

$\bar{f} \equiv$typical force scale
$\delta r \equiv$typical radial displacement

$$\underbrace{\Omega \tau \delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij} \delta r_{ij} \sim N \delta z \bar{f} \bar{\delta r}$$

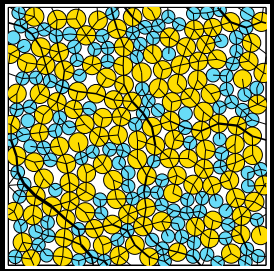


displacements per shear vs. δz

$\bar{f}$ ≡typical force scale
δr ≡typical radial displacement

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$

$\rightarrow$ if random contacts are removed: $\bar{f} \sim p \quad \Rightarrow \quad \dfrac{\bar{\delta r}}{\delta\gamma} \sim \dfrac{1}{\delta z}$



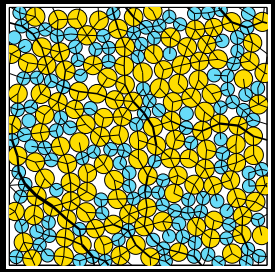
displacements per shear vs. δz

$\bar{f} \equiv$typical force scale
$\delta r \equiv$typical radial displacement

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$

$\rightarrow$ if random contacts are removed: $\bar{f} \sim p \quad \Rightarrow \quad \dfrac{\bar{\delta r}}{\delta\gamma} \sim \dfrac{1}{\delta z}$

$\rightarrow$ recall that stress $\sim$ viscosity $\sim$ (displacments per shear)^2



displacements per shear vs. δz

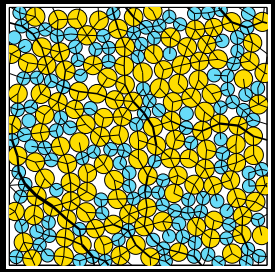
$\bar{f}$ ≡typical force scale
δr ≡typical radial displacement

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$

$\rightarrow$ if random contacts are removed: $\bar{f} \sim p \quad \Rightarrow \quad \dfrac{\bar{\delta r}}{\delta\gamma} \sim \dfrac{1}{\delta z}$

$\rightarrow$ recall that stress $\sim$ viscosity $\sim$ (displacments per shear)^2

$$\tau \sim \delta z^{-2}$$



displacements per shear vs. δz

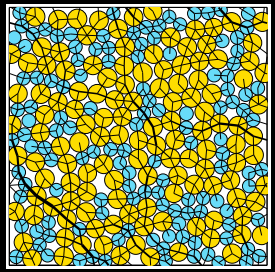
$\bar{f}$ ≡typical force scale
δr ≡typical radial displacement

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$

$\rightarrow$ if random contacts are removed: $\bar{f} \sim p \quad \Rightarrow \quad \dfrac{\bar{\delta r}}{\delta\gamma} \sim \dfrac{1}{\delta z}$

$\rightarrow$ recall that stress $\sim$ viscosity $\sim$ (displacments per shear)^2

$$\tau \sim \delta z^{-2}$$ (inconsistent with observations)



displacements per shear vs. δz

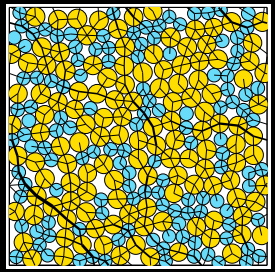
$\bar{f}$ ≡typical force scale
δr ≡typical radial displacement

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$

$\rightarrow$ if random contacts are removed: $\bar{f} \sim p \quad \Rightarrow \quad \dfrac{\bar{\delta r}}{\delta\gamma} \sim \dfrac{1}{\delta z}$

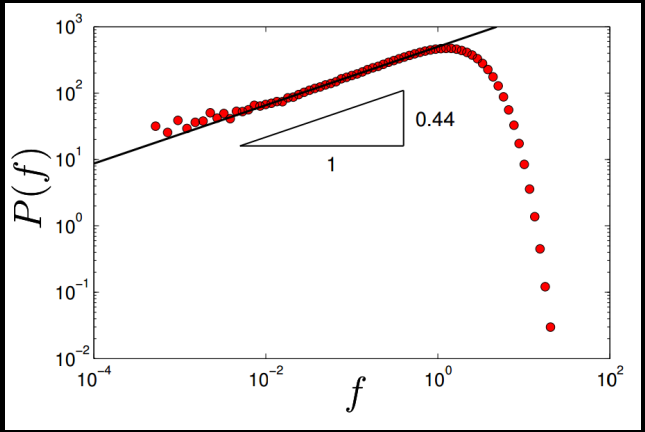
$\rightarrow$ recall that stress $\sim$ viscosity $\sim$ (displacments per shear)^2

$$\tau \sim \delta z^{-2}$$ (inconsistent with observations)



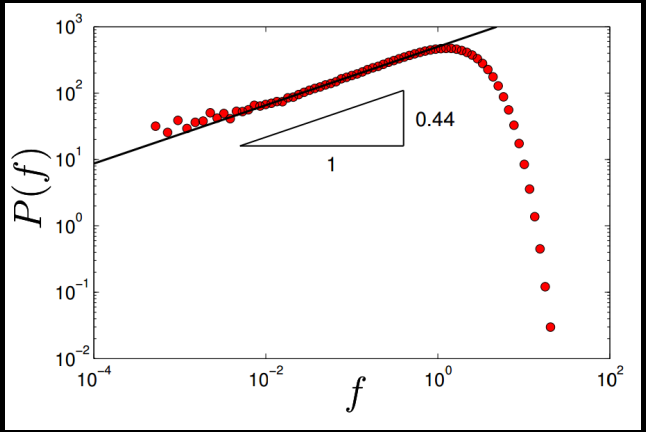
what if we removed the **weakest forces**?

contact force distribution in jammed states



$$P(f) \sim f^{\theta}$$

contact force distribution in jammed states



$$P(f) \sim f^{\theta}$$

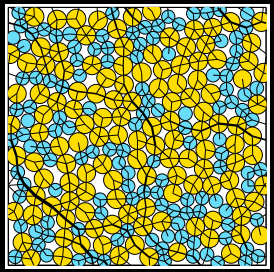
mean-field gives:
$\theta \approx 0.423\ldots$

Charbonneau, Kurchan, Parisi, Urbani, & Zamponi
Nat. Comm. 2014

displacements per shear vs. δz

$\bar{f} \equiv$typical force scale
$\delta r \equiv$typical radial displacement

$$\underbrace{\Omega \tau \delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij} \delta r_{ij} \sim N \delta z \bar{f} \bar{\delta r}$$



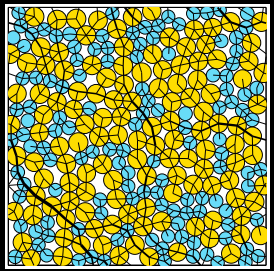
displacements per shear vs. δz

$\bar{f}$ ≡typical force scale
δr ≡typical radial displacement

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$

$\rightarrow$ if weakest contacts are removed:

$$\bar{f} \sim p\,\delta z^{1/(1+\theta)} \quad \Rightarrow \quad \frac{\bar{\delta r}}{\delta\gamma} \sim \delta z^{-\frac{2+\theta}{1+\theta}} \sim \delta z^{-1.7}$$



displacements per shear vs. δz

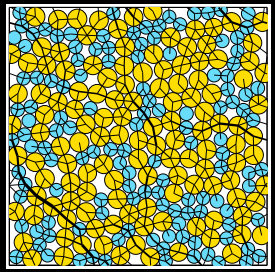
$\bar{f} \equiv$typical force scale
$\delta r \equiv$typical radial displacement

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$

$\longrightarrow$ if weakest contacts are removed:

$$\bar{f} \sim p\,\delta z^{1/(1+\theta)} \quad \Rightarrow \quad \frac{\bar{\delta r}}{\delta\gamma} \sim \delta z^{-\frac{2+\theta}{1+\theta}} \sim \delta z^{-1.7}$$

$\longrightarrow$ recall that stress $\sim$ viscosity $\sim$ (displacments per shear)^2



displacements per shear vs. δz

$\bar{f}$ ≡typical force scale
δr ≡typical radial displacement

$$\underbrace{\Omega\tau\delta\gamma}_{\text{work invested}} = \sum_{\text{removed } ij} f_{ij}\delta r_{ij} \sim N\delta z\bar{f}\bar{\delta r}$$

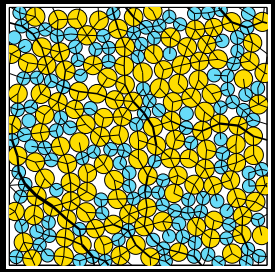
$\rightarrow$ if weakest contacts are removed:

$$\bar{f} \sim p\,\delta z^{1/(1+\theta)} \quad \Rightarrow \quad \frac{\bar{\delta r}}{\delta\gamma} \sim \delta z^{-\frac{2+\theta}{1+\theta}} \sim \delta z^{-1.7}$$

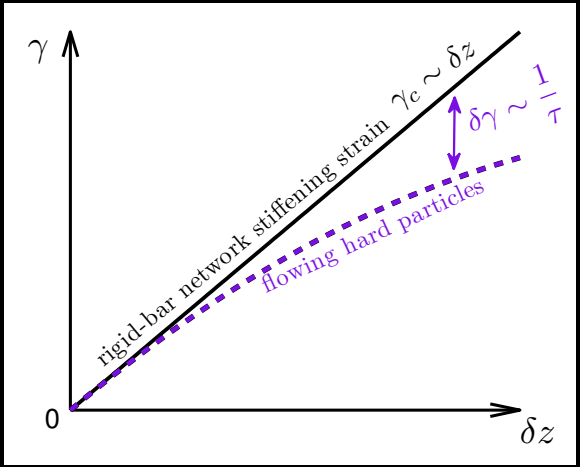
$\rightarrow$ recall that stress $\sim$ viscosity $\sim$ (displacments per shear)^2

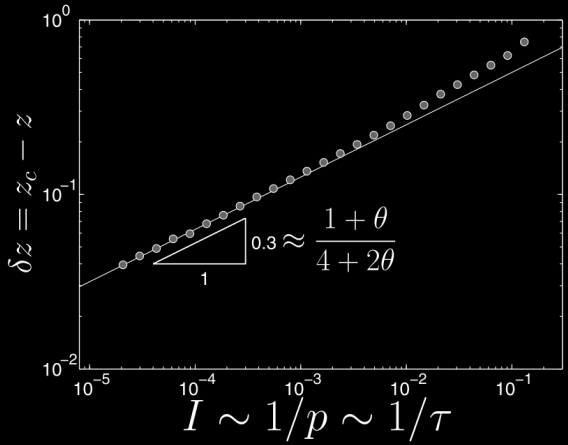
$$\tau \sim \delta z^{-\frac{4+2\theta}{1+\theta}}$$

if **weakest** contacts are removed



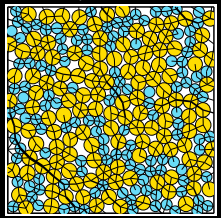
stress vs. coordination δz



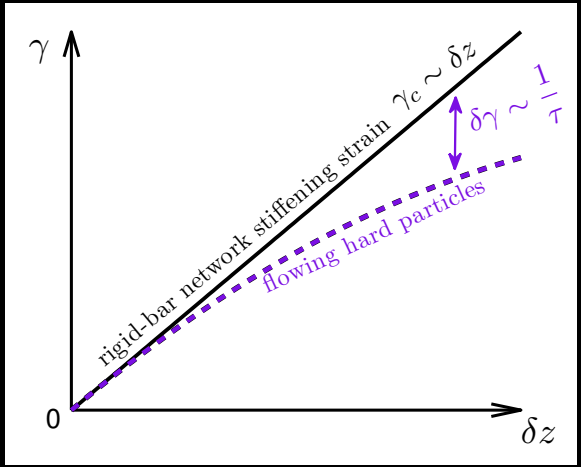


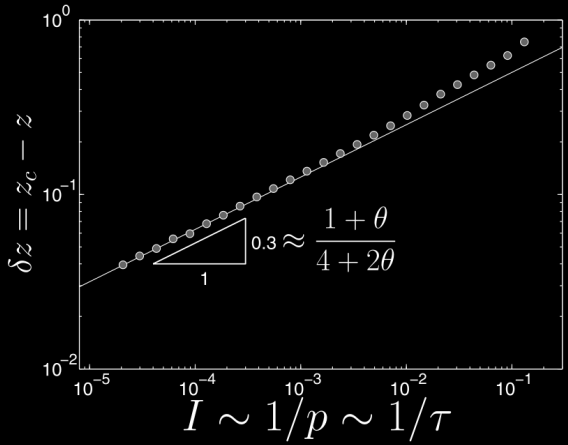
$$\tau \sim \delta z^{-\frac{4+2\theta}{1+\theta}}$$

if **weakest** contacts are removed



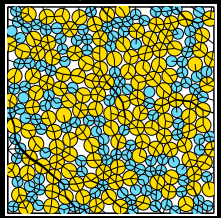
stress vs. coordination δz



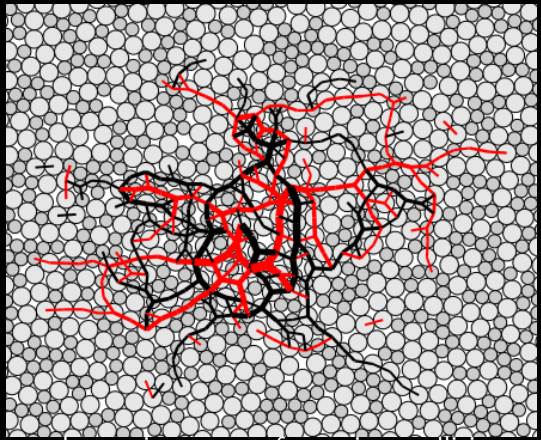


$$\tau \sim \delta z^{-\frac{4+2\theta}{1+\theta}}$$

if **weakest** contacts are removed ← why?



collisional mechanics – why remove the **weakest** forces?

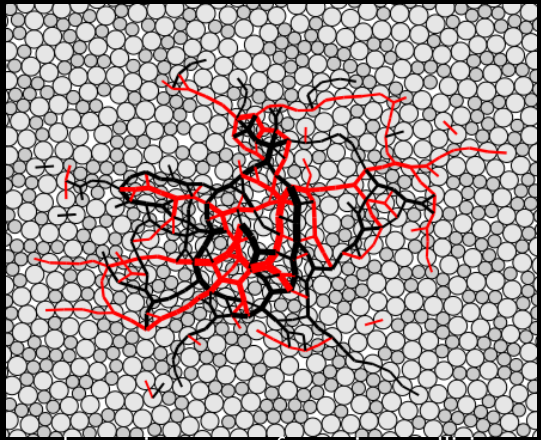


change in contact forces in a collision

collisional mechanics – why remove the **weakest** forces?

one can show:

$$f_{\text{collision}} \sim p^{\frac{3+2\theta}{4+2\theta}} \ll p \sim f_{\text{typical}}$$



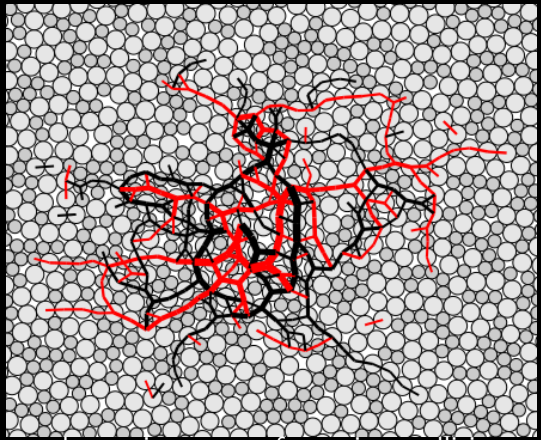
change in contact forces in a collision

collisional mechanics – why remove the **weakest** forces?

one can show:

$$f_{\text{collision}} \sim p^{\frac{3+2\theta}{4+2\theta}} \ll p \sim f_{\text{typical}}$$

newly-formed contact forces are
much weaker than the average!



change in contact forces in a collision

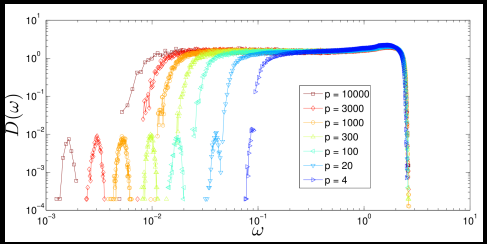
collisional mechanics – why remove the **weakest** forces?

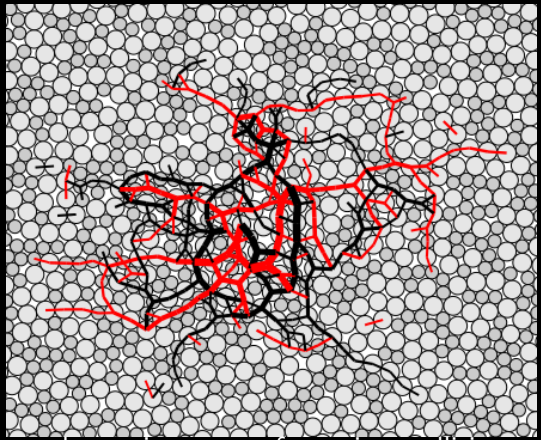
one can show:

$$f_{\text{collision}} \sim p^{\frac{3+2\theta}{4+2\theta}} \ll p \sim f_{\text{typical}}$$

newly-formed contact forces are
much weaker than the average!

(has to do with bi-scaling & plateau)



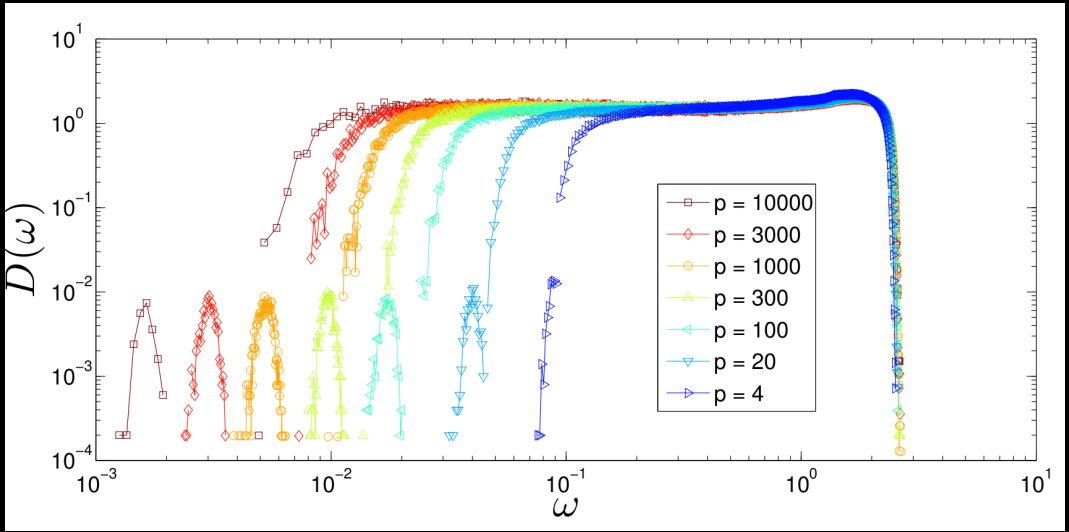


change in contact forces in a collision

non-Brownian suspension viscosity – summary

non-Brownian suspension viscosity – summary

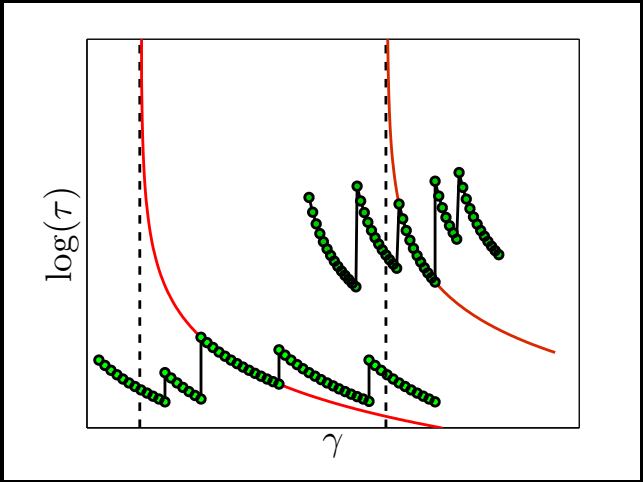
$\rightarrow$ self-organization of the geometry produces more 'solid-like' states



non-Brownian suspension viscosity – summary

$\longrightarrow$ self-organization of the geometry produces more 'solid-like' states

$\longrightarrow$ system 'lives' near a fictitious strain-stiffening transition

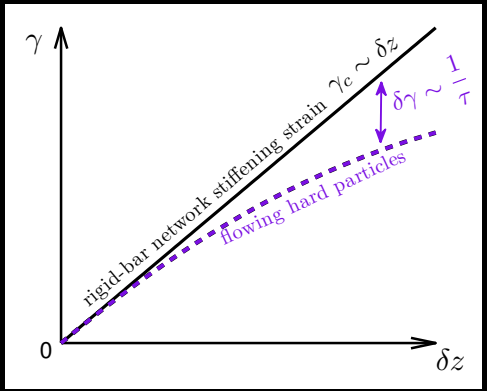


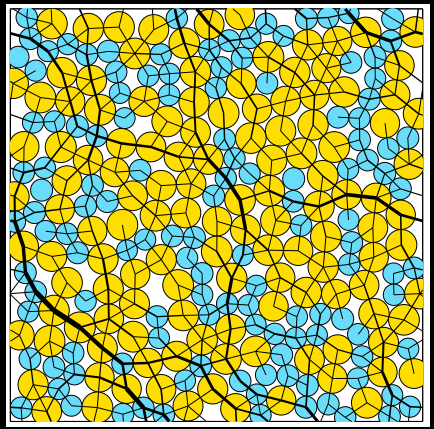
non-Brownian suspension viscosity – summary

$\longrightarrow$ self-organization of the geometry produces more 'solid-like' states

$\longrightarrow$ system 'lives' near a fictitious strain-stiffening transition

$\longrightarrow$ by perturbing the critical state — obtain relation between stress and coordination

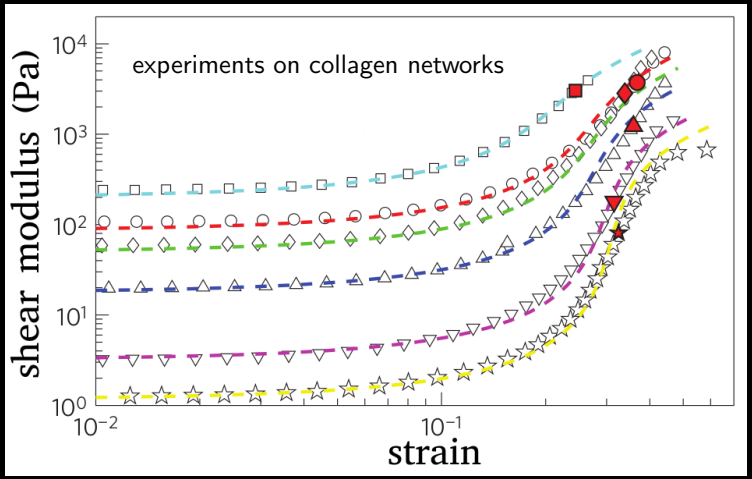




strain stiffening

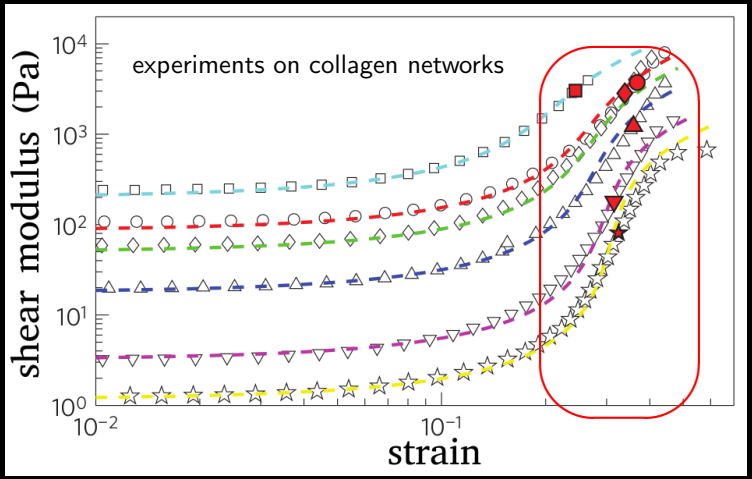
strain stiffening is the deformation-induced increase in a material's elastic modulus

strain stiffening is the deformation-induced increase in a material's elastic modulus



Sharma et al., Nature Physics **12**, 584 (2016)

strain stiffening is the deformation-induced increase in a material's elastic modulus



today:

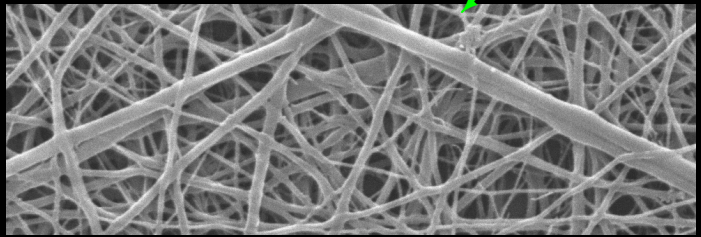
why?
how?

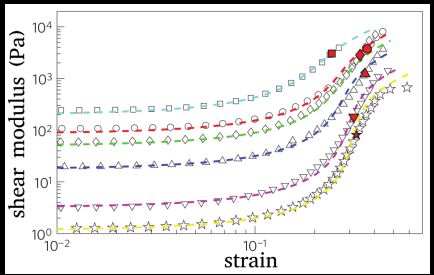
Sharma et al., Nature Physics **12**, 584 (2016)

strain stiffening is the deformation-induced increase in a material's elastic modulus

fibers that are **easy to bend** but **hard to stretch**

collagen networks





Sharma et al., Nature Physics **12**, 584 (2016)

strain-stiffening occurs in different ‘stretchy’ materials, but has different origin/scaling laws

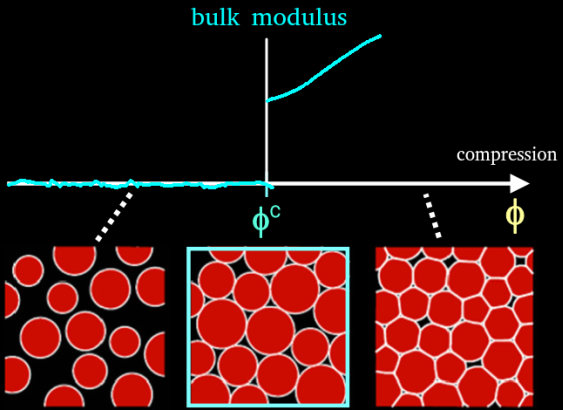
$E \sim \sigma^3$

$\neq$

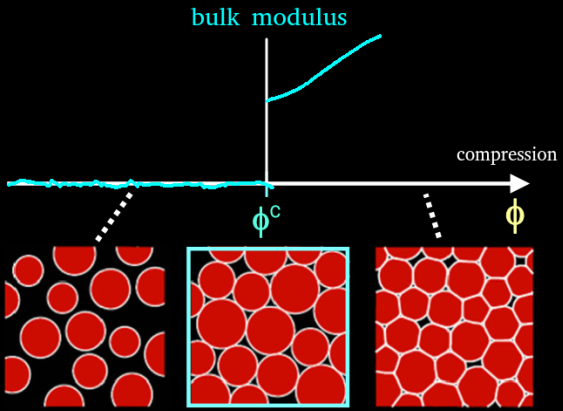
$E \sim \sigma^2$

strain stiffening is a member of a set of ‘jamming’ problems:

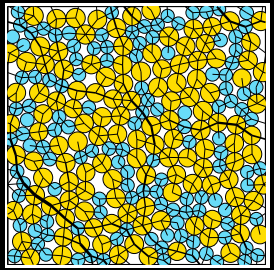
strain stiffening is a member of a set of ‘jamming’ problems:

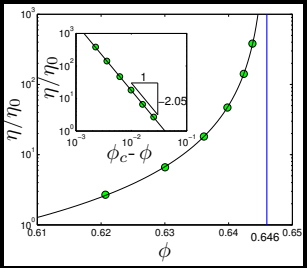


strain stiffening is a member of a set of ‘jamming’ problems:

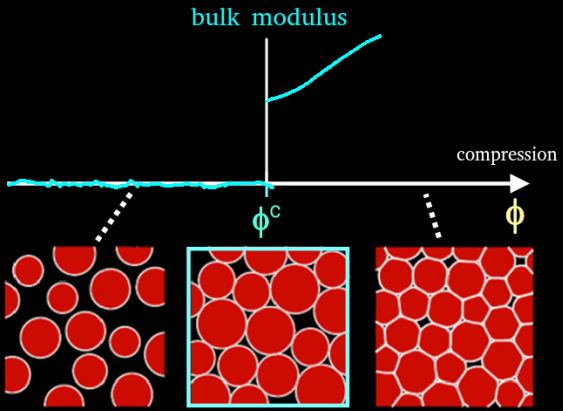


non-Brownian suspension viscosity

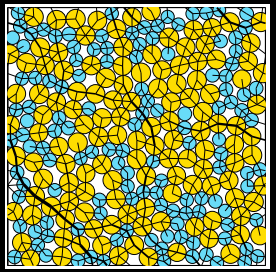


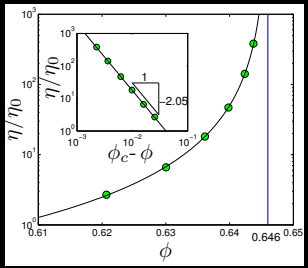


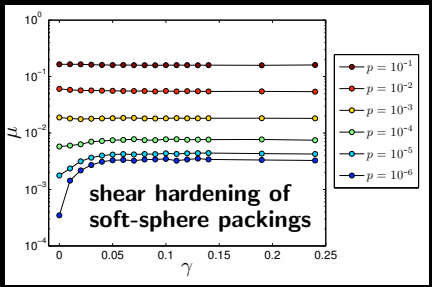
strain stiffening is a member of a set of 'jamming' problems:



non-Brownian suspension viscosity







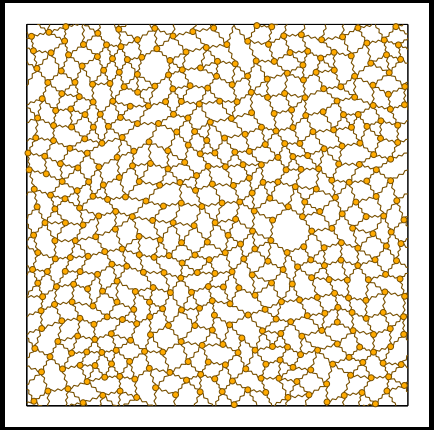
today's plan:

today's plan:

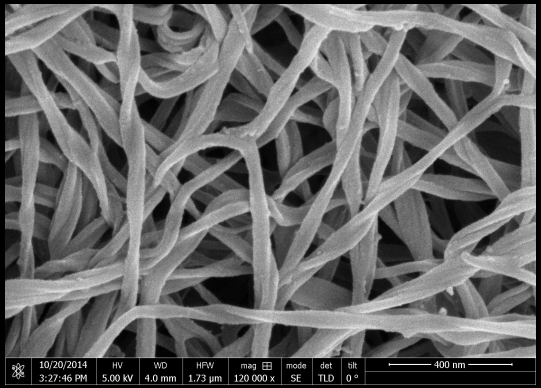
- model (for any disordered material): unit masses connected by Hookean springs

today's plan:

- model (for any disordered material): unit masses connected by Hookean springs



=



today's plan:

- model (for any disordered material): unit masses connected by Hookean springs
- what are **floppy networks**?

today's plan:

- model (for any disordered material): unit masses connected by Hookean springs
- what are **floppy networks**?
- geometric analysis of strain-stiffening networks – **states of self-stress**

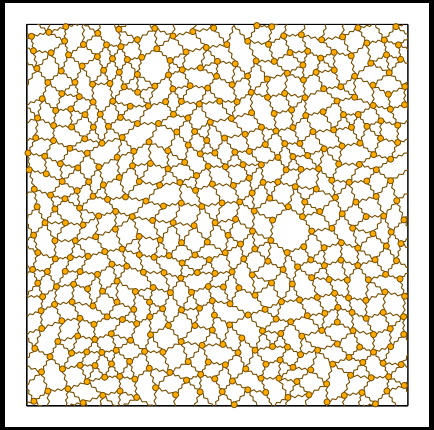
today's plan:

- model (for any disordered material): unit masses connected by Hookean springs
- what are **floppy networks**?
- geometric analysis of strain-stiffening networks – **states of self-stress**
- adding **bending forces** into the picture

today's plan:

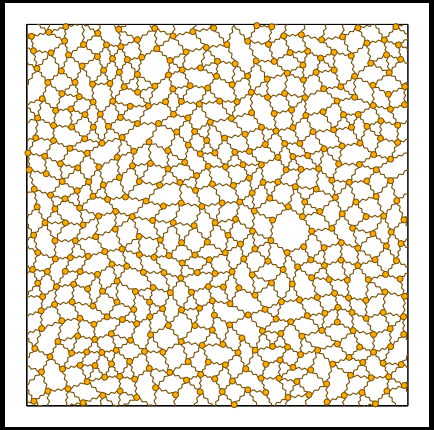
- model (for any disordered material): unit masses connected by Hookean springs
- what are **floppy networks**?
- geometric analysis of strain-stiffening networks – **states of self-stress**
- adding **bending forces** into the picture
- scaling theory of strain stiffening

disordered networks of masses connected by **relaxed** Hookean springs



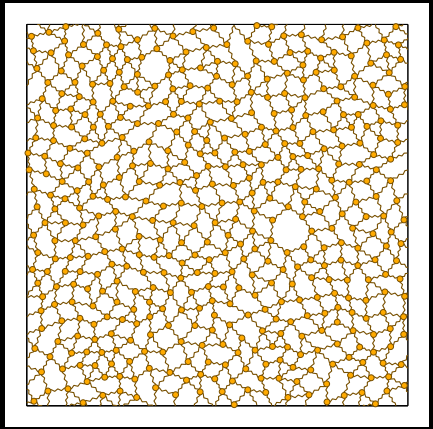
disordered networks of masses connected by **relaxed** Hookean springs

key control parameter: coordination z



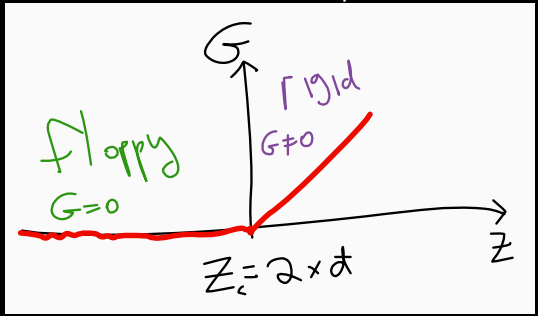
disordered networks of masses connected by **relaxed** Hookean springs

key control parameter: coordination z



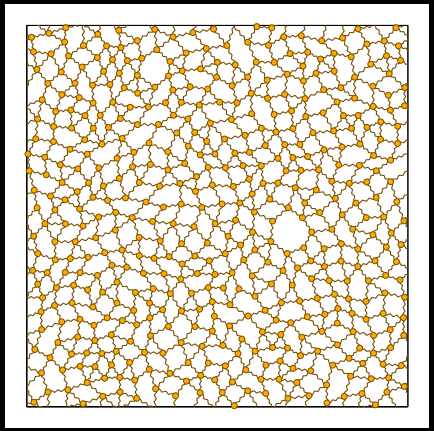
$G \equiv$ shear modulus

$d \equiv$ dimension of space



disordered networks of masses connected by **relaxed** Hookean springs

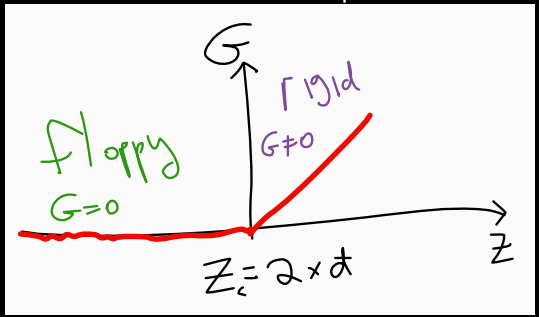
key control parameter: coordination $z < z_c \equiv 2 \times \mathit{đ}$



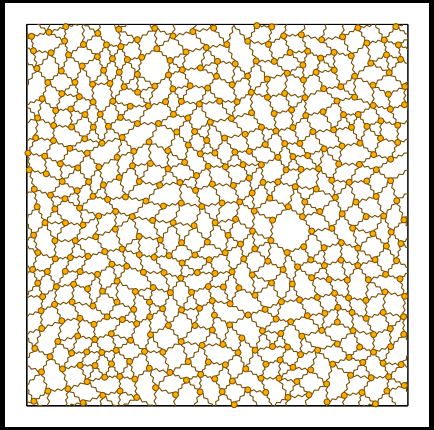
'floppy' networks

$G \equiv$ shear modulus

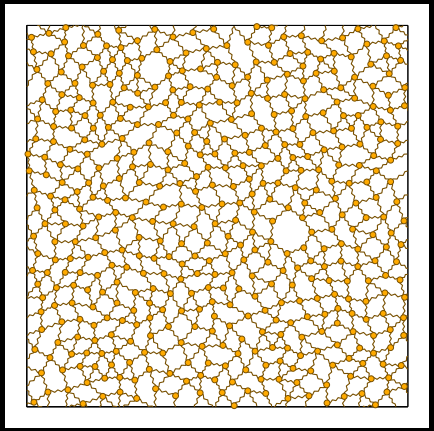
$\mathit{đ} \equiv$ dimension of space



‘floppy’ networks feature ‘floppy modes’



‘floppy’ networks feature ‘floppy modes’

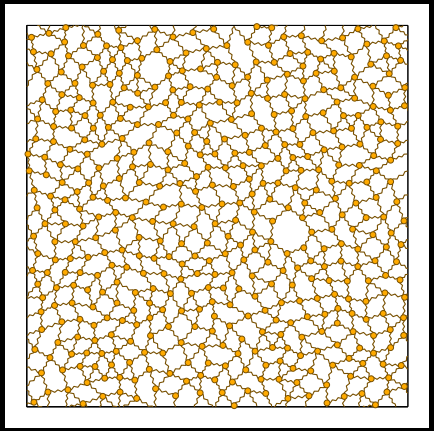


‘floppy modes’ are zero-energy modes.

They are displacements $\boldsymbol{u}$ that

do **not** stretch **nor** compress any spring

‘floppy’ networks feature ‘floppy modes’



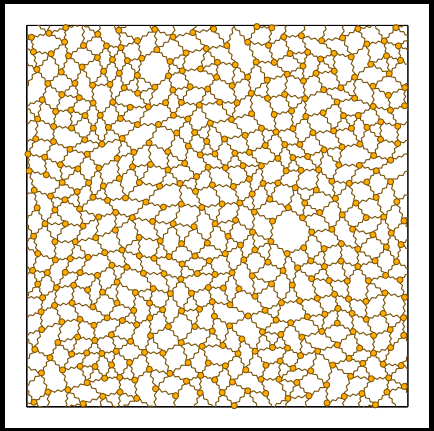
‘floppy modes’ are zero-energy modes.

They are displacements $\boldsymbol{u}$ that
do **not** stretch **nor** compress any spring

$$\text{if } \hat{\boldsymbol{n}}_{ij} \cdot (\boldsymbol{u}_j - \boldsymbol{u}_i) = 0 \quad \text{for all springs } i,j$$

$$\Rightarrow \boldsymbol{u} \text{ is a } \textbf{floppy mode}$$

‘floppy’ networks feature ‘floppy modes’



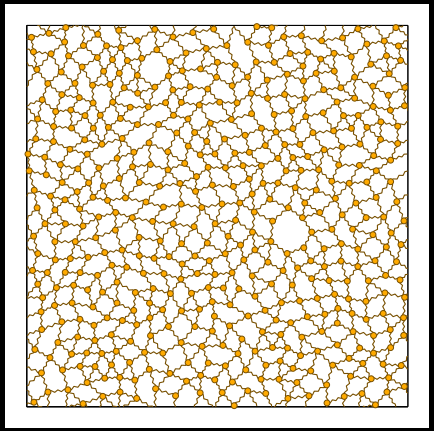
‘floppy modes’ are zero-energy modes.

They are displacements $\boldsymbol{u}$ that
do **not** stretch **nor** compress any spring

$$\text{if } \mathcal{S}|u\rangle = 0$$

$$\Rightarrow \boldsymbol{u} \text{ is a } \textbf{floppy mode}$$

‘floppy’ networks feature ‘floppy modes’



‘floppy modes’ are zero-energy modes.

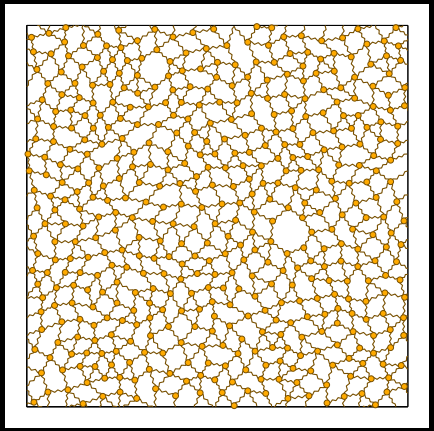
They are displacements $\boldsymbol{u}$ that
do **not** stretch **nor** compress any spring

$$\text{if } \mathcal{S}|u\rangle = 0$$

$$\Rightarrow \boldsymbol{u} \text{ is a } \textbf{floppy mode}$$

$\mathcal{S}$ is known as the ‘*compatibility matrix*’

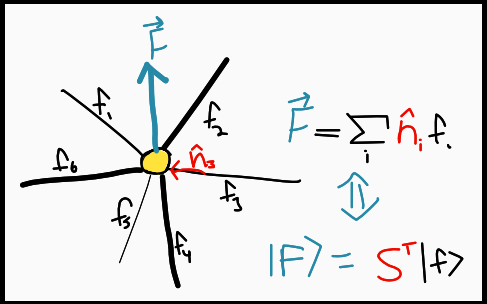
floppy networks **do not** feature ‘states of self-stress’



‘states of self-stress’ are assignments of spring-forces

that are **vectorically self-balanced**

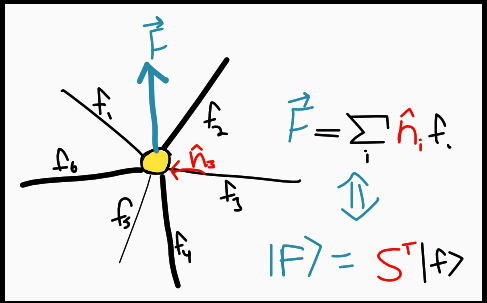
floppy networks **do not** feature ‘states of self-stress’



‘states of self-stress’ are assignments of spring-forces

that are **vectorically self-balanced**

floppy networks **do not** feature 'states of self-stress'



'states of self-stress' are assignments of spring-forces that are **vectorically self-balanced**

if $\mathcal{S}^T |f\rangle = \mathbf{0}$

$\Rightarrow$ $|f\rangle$ is a **state of self-stress**

floppy networks **do not** feature ‘states of self-stress’

– why do we care about this ?

floppy networks **do not** feature 'states of self-stress'

– why do we care about this ?

Wyart (phd thesis, 2005) showed that (for relaxed spring networks)

$$G = \frac{1}{V} \sum_{\substack{\text{states of} \\ \text{self-stress } \varphi_\ell}} \langle \varphi_\ell | \partial r / \partial \gamma \rangle^2$$

floppy networks **do not** feature 'states of self-stress'

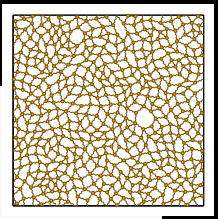
– why do we care about this 👆?

Wyart (phd thesis, 2005) showed that (for relaxed spring networks)

$$G = \frac{1}{V} \sum_{\substack{\text{states of} \\ \text{self-stress } \varphi_\ell}} \langle \varphi_\ell | \partial r / \partial \gamma \rangle^2$$

no states-of-self-stress? then $G = 0$.

floppy networks – summary



floppy

$G=0$

NO states of self-stress

G

rigid

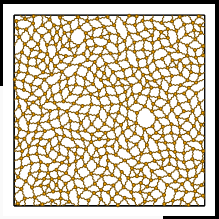
$G\neq 0$

$N(z-z_c)$ states of self-stress

z

$z_c = 2\times d$

floppy networks – summary



floppy

$G=0$

NO states of self-stress

G

rigid

$G\neq 0$

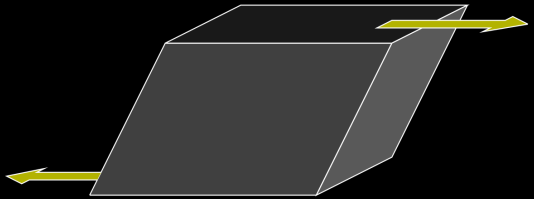
$N(z-z_c)$ states of self-stress

z

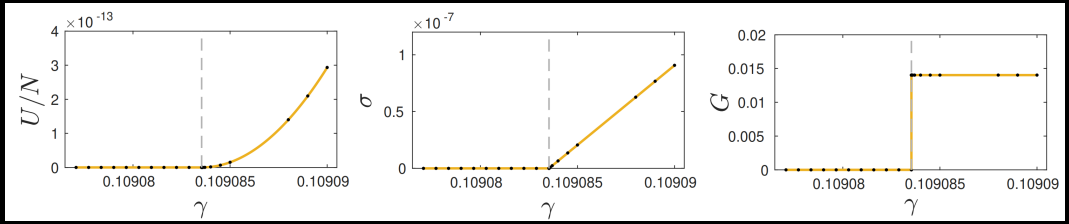
$z_c = 2\times d$

for the rest of this session

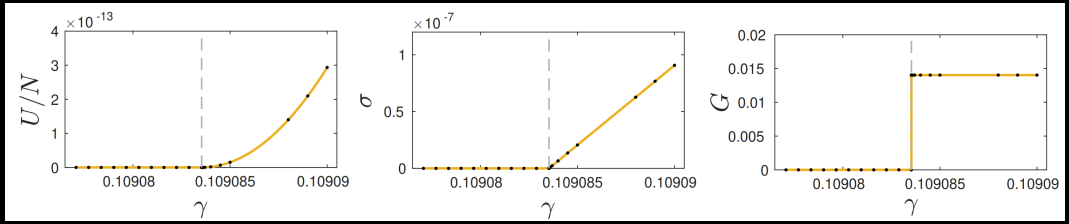
what happens when a floppy network is sheared?



what happens when a floppy network is sheared?

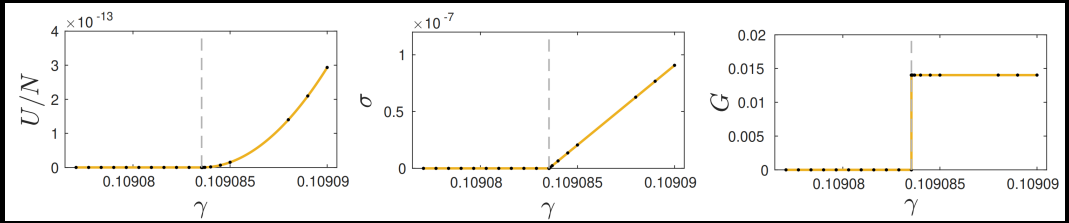


what happens when a floppy network is sheared?



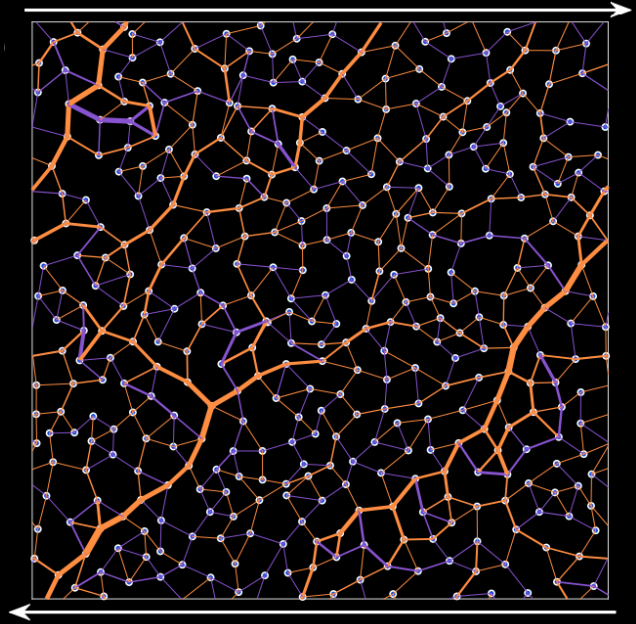
at a **critical** strain γ_c the shear modulus **jumps**

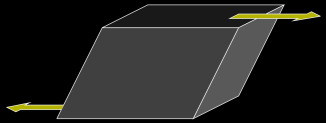
what happens when a floppy network is sheared?



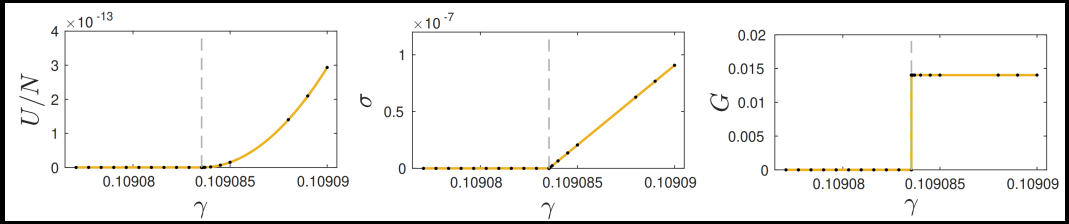
at a **critical** strain γ_c the shear modulus **jumps**

$\Rightarrow$ a **state of self-stress developed**





what happens when a floppy network is sheared?

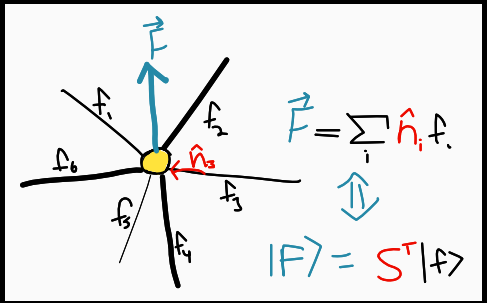


at a **critical** strain γ_c the shear modulus **jumps**

$\Rightarrow$ a **state of self-stress developed**

how can this 👆 be quantified?

recall: states of self-stress



'states of self-stress' are assignments of spring-forces that are **vectorically self-balanced**

if $\mathcal{S}^T |f\rangle = \mathbf{0}$

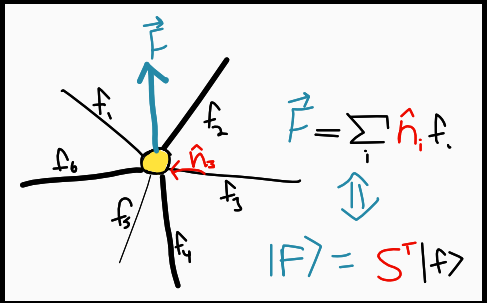
$\Rightarrow \quad |f\rangle$ is a **state of self-stress**

recall: states of self-stress

we construct the operator: $\mathcal{S}\mathcal{S}^T$

and consider its **spectrum**

$$\mathcal{S}\mathcal{S}^T|f\rangle = \omega^2|f\rangle$$

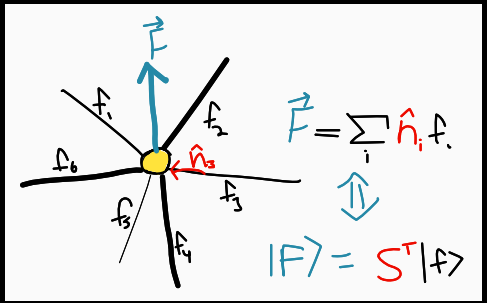


recall: states of self-stress

we construct the operator: $\mathcal{S}\mathcal{S}^T$

and consider its **spectrum**

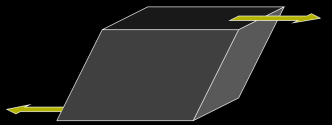
$$\mathcal{S}\mathcal{S}^T|f\rangle = \omega^2|f\rangle$$

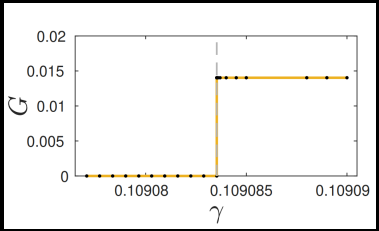


eigenvectors $|f\rangle$: sets of **spring-forces**,

eigenvalues ω^2: **dimensionless force unballance**:

$$\omega^2 = \frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle} = \frac{\langle F|F\rangle}{\langle f|f\rangle}$$



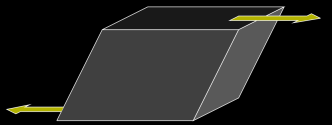


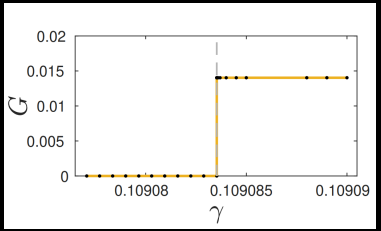
eigenvalues ω^2: **dimensionless force unballance**:

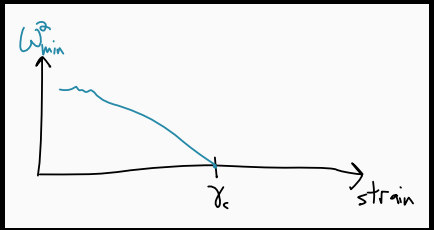
$$\omega^2 = \frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle} = \frac{\langle F|F\rangle}{\langle f|f\rangle}$$

we expect:

at γ_c, $\omega^2 \to 0$





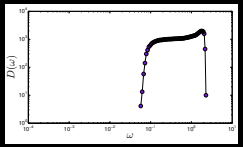


eigenvalues ω^2: **dimensionless force unballance**:

$$\omega^2 = \frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle} = \frac{\langle F|F\rangle}{\langle f|f\rangle}$$

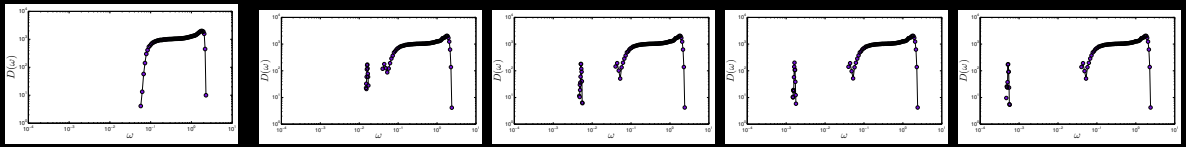
spectrum of $\mathcal{S}\mathcal{S}^T$ in sheared floppy networks $\left(\mathcal{S}\mathcal{S}^T|f\rangle = \omega^2|f\rangle\right)$

spectrum of $\mathcal{S}\mathcal{S}^T$ in sheared floppy networks $\left(\mathcal{S}\mathcal{S}^T|f\rangle = \omega^2|f\rangle\right)$



isotropic

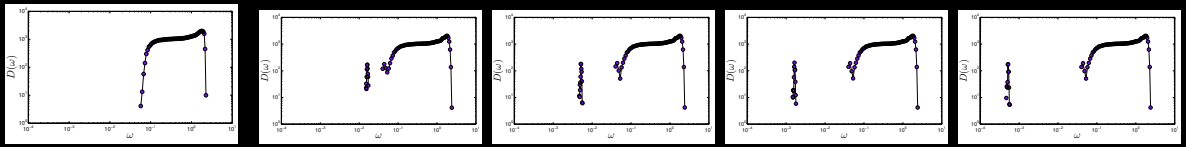
spectrum of $\mathcal{S}\mathcal{S}^T$ in sheared floppy networks $\left(\mathcal{S}\mathcal{S}^T|f\rangle = \omega^2|f\rangle\right)$



isotropic

sheared

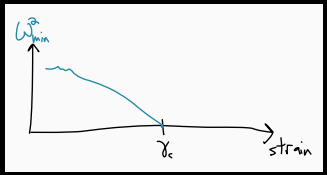
spectrum of $\mathcal{S}\mathcal{S}^T$ in sheared floppy networks $\left(\mathcal{S}\mathcal{S}^T|f\rangle = \omega^2|f\rangle\right)$



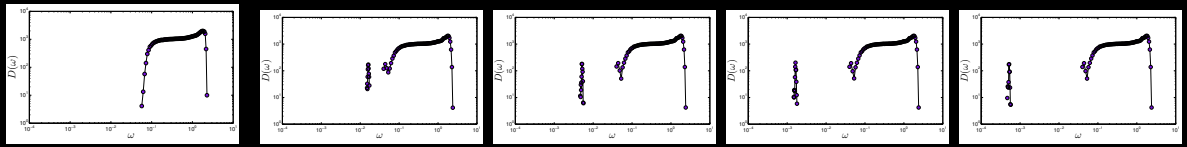
isotropic

sheared

development of a **state of self-stress** $\Leftrightarrow$ $\omega^2_{\min} \to 0$



spectrum of $\mathcal{S}\mathcal{S}^T$ in sheared floppy networks $\left(\mathcal{S}\mathcal{S}^T|f\rangle = \omega^2|f\rangle\right)$

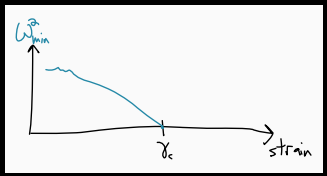


isotropic

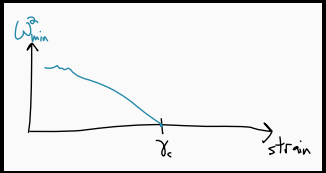
sheared

development of a **state of self-stress** $\Leftrightarrow$ $\omega^2_{\min} \to 0$

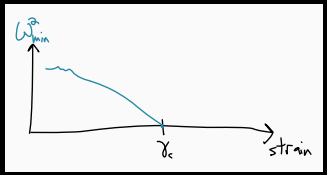
how does $\omega^2_{\min}$ vanish?



how does $\omega^2_{\rm min}$ vanish?

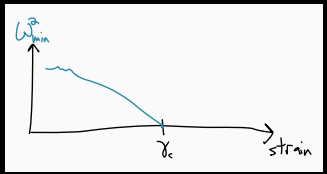


how does $\omega^2_{\min}$ vanish?



the (quasistatic) dynamics of a **floppy network** under shear is **ill-defined**

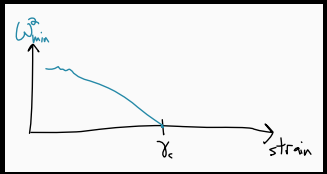
how does $\omega^2_{\min}$ vanish?



the (quasistatic) dynamics of a **floppy network** under shear is **ill-defined**

In generic elastic solids: $\boldsymbol{\mathcal{H}} \cdot \dfrac{\mathcal{D}\boldsymbol{x}}{\mathcal{D}\gamma} = \dfrac{\partial^2 U}{\partial\gamma\partial\boldsymbol{x}}$

how does $\omega^2_{\min}$ vanish?

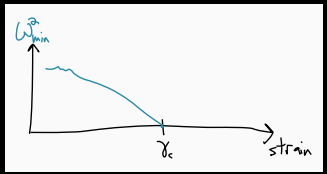


the (quasistatic) dynamics of a **floppy network** under shear is **ill-defined**

In generic elastic solids: $\boldsymbol{\mathcal{H}} \cdot \dfrac{\mathcal{D}\boldsymbol{x}}{\mathcal{D}\gamma} = \dfrac{\partial^2 U}{\partial\gamma\partial\boldsymbol{x}}$

$\Rightarrow$ one can add any zero mode $\boldsymbol{\psi}$ $(\boldsymbol{\mathcal{H}} \cdot \boldsymbol{\psi} = \mathbf{0})$ to the (under-determined) solution for $\dfrac{\mathcal{D}\boldsymbol{x}}{\mathcal{D}\gamma}$

how does $\omega^2_{\rm min}$ vanish?



the (quasistatic) dynamics of a **floppy network** under shear is **ill-defined**

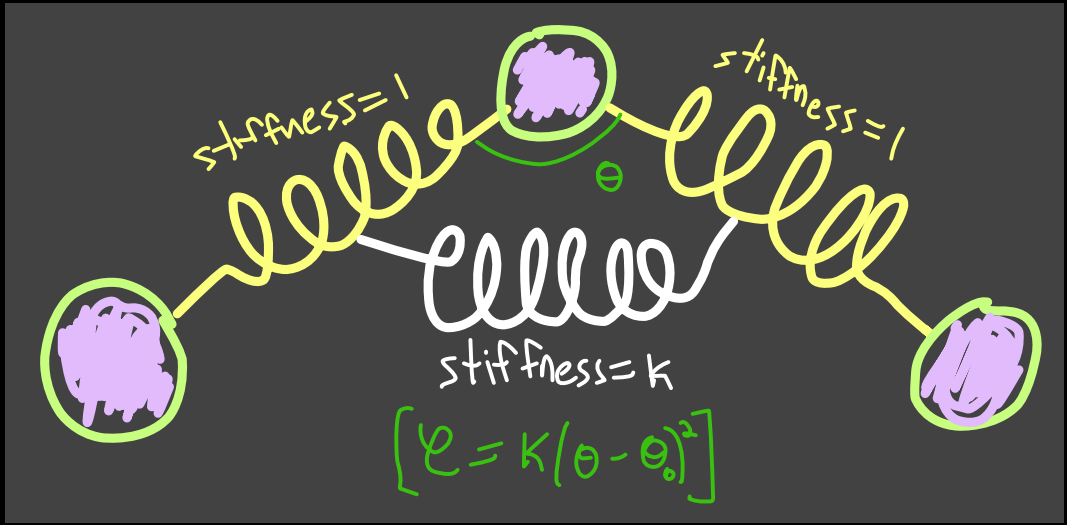
In generic elastic solids: $\boldsymbol{\mathcal{H}} \cdot \dfrac{\mathcal{D}\boldsymbol{x}}{\mathcal{D}\gamma} = \dfrac{\partial^2 U}{\partial\gamma\partial\boldsymbol{x}}$

$\Rightarrow$ one can add any zero mode $\boldsymbol{\psi}$ $(\boldsymbol{\mathcal{H}} \cdot \boldsymbol{\psi} = \mathbf{0})$ to the (under-determined) solution for $\dfrac{\mathcal{D}\boldsymbol{x}}{\mathcal{D}\gamma}$

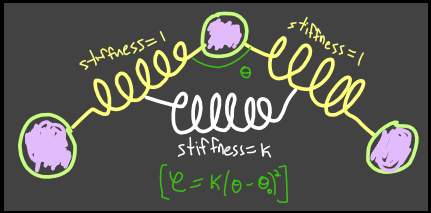
to proceed, we introduce a **weak interaction** of typical stiffness κ, that **eliminates** the indeterminacy of dynamics/mechanics

introducing weak interactions

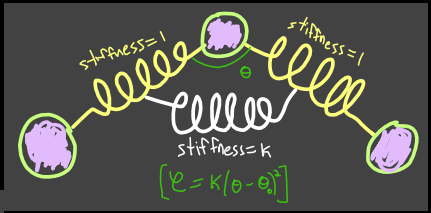
introducing weak interactions

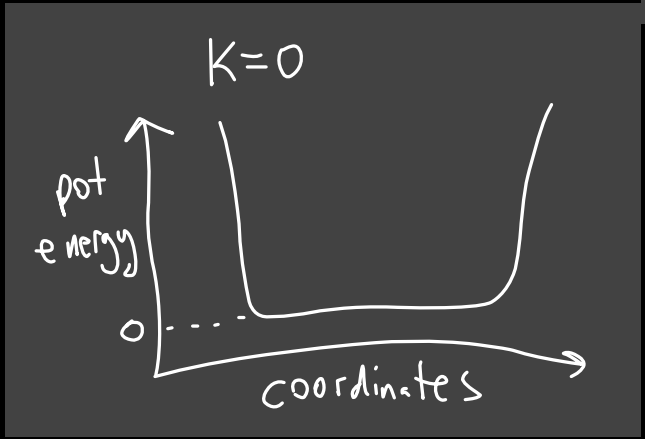


after introducing weak interactions of stiffness κ,
indeterminacy is **removed**.

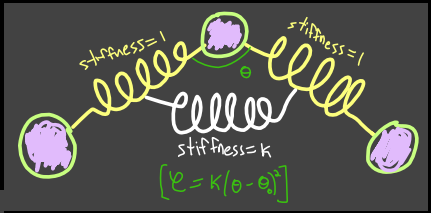


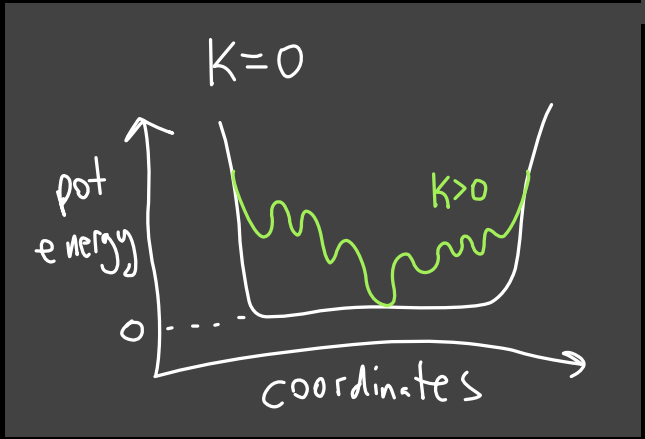
after introducing weak interactions of stiffness κ, **indeterminacy** is **removed**.



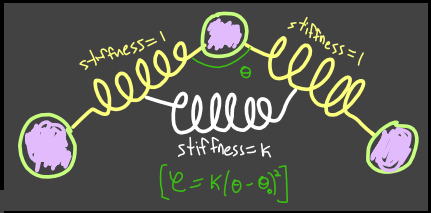


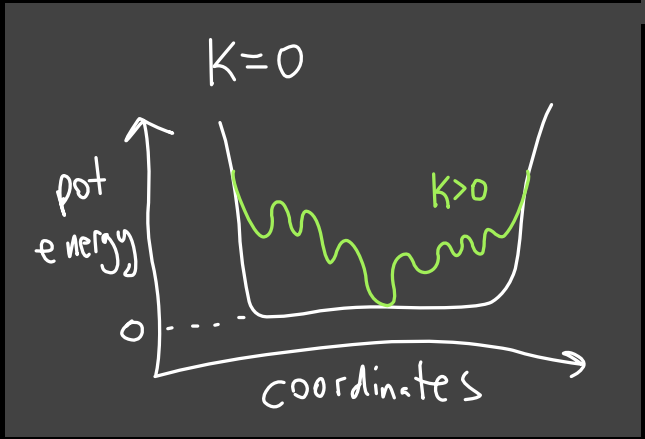
after introducing weak interactions of stiffness κ, **indeterminacy** is **removed**.





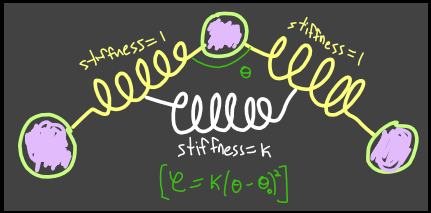
after introducing weak interactions of stiffness κ, **indeterminacy** is **removed**.





($\kappa > 0$ is a singular perturbation)

after introducing weak interactions of stiffness κ, **indeterminacy** is **removed**.



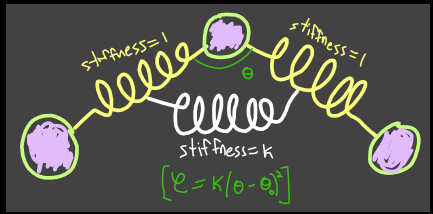
one useful limit is $\kappa \to 0^+$, then one finds:

$$\omega^2_{\min} \sim \gamma_c - \gamma$$

(recall $\mathcal{S}\mathcal{S}^T|f\rangle = \omega^2|f\rangle$)

R. Rens, C. Villarroel, G. Düring, and EL, PRE 2018

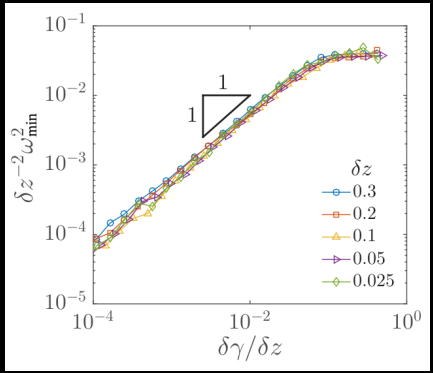
after introducing weak interactions of stiffness κ, **indeterminacy** is **removed**.



one useful limit is $\kappa \to 0^+$, then one finds:

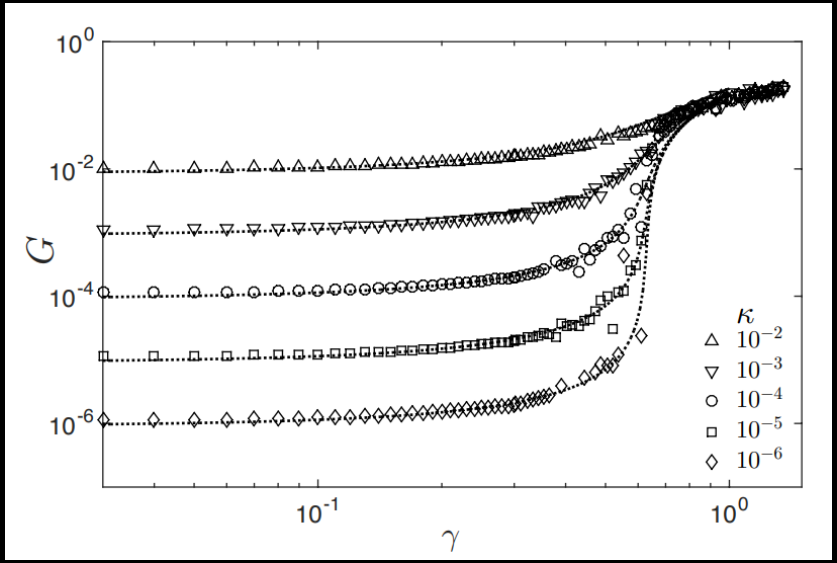
$$\omega_{\min}^2 \sim \gamma_c - \gamma$$

(recall $\mathcal{S}\mathcal{S}^T|f\rangle = \omega^2|f\rangle$)



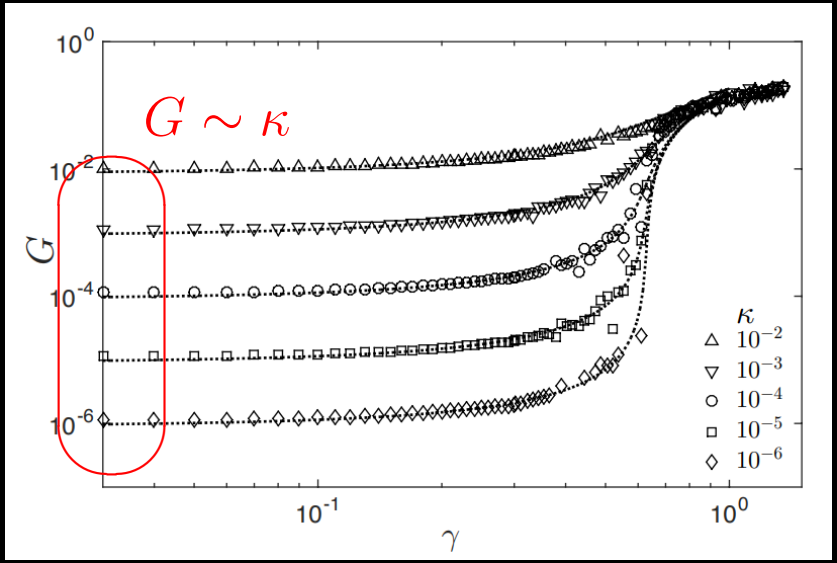
R. Rens, C. Villarroel, G. Düring, and EL, PRE 2018

back to the shear modulus G in the presence of (weak) **bending** interactions:



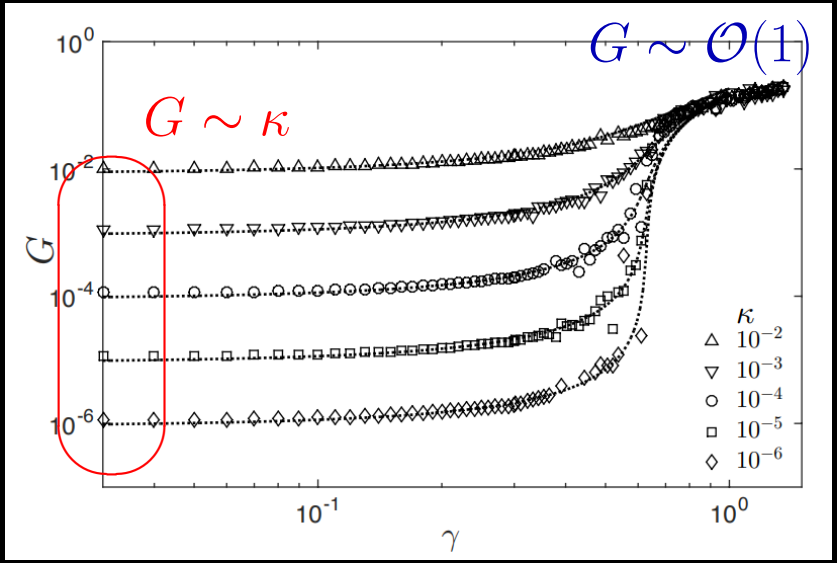
R. Rens et al. J. Phys. Chem. B, 2016

back to the shear modulus G in the presence of (weak) **bending** interactions:



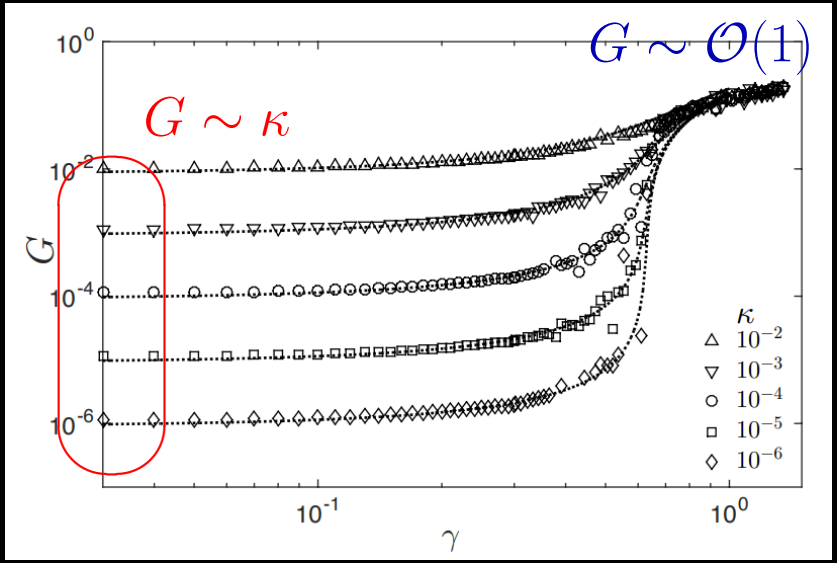
R. Rens et al. J. Phys. Chem. B, 2016

back to the shear modulus G in the presence of (weak) **bending** interactions:

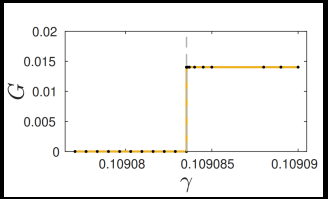


R. Rens et al. J. Phys. Chem. B, 2016

back to the shear modulus G in the presence of (weak) **bending** interactions:

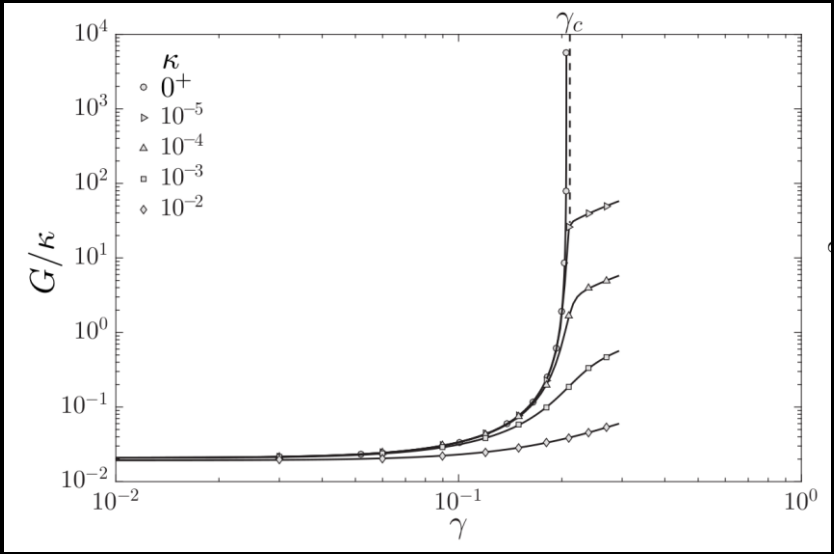


recall that at $\kappa = 0$:



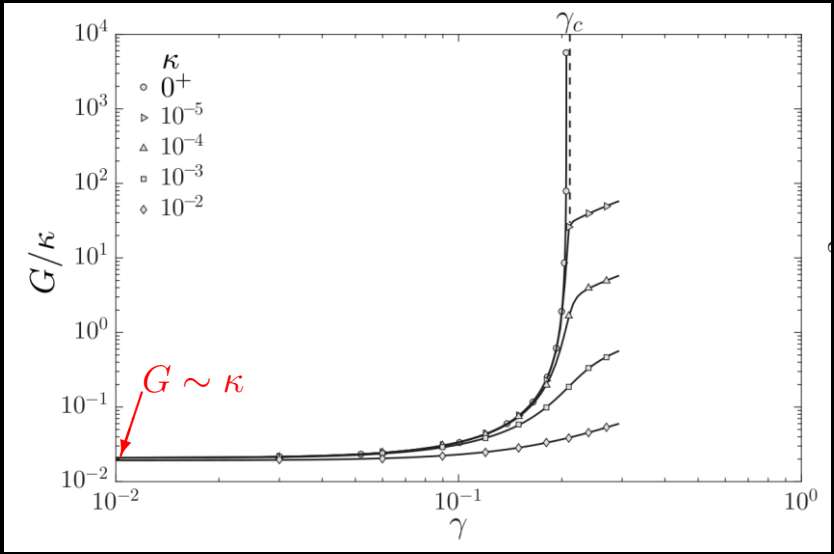
R. Rens et al. J. Phys. Chem. B, 2016

shear modulus G in the presence of (weak) **bending** interactions:



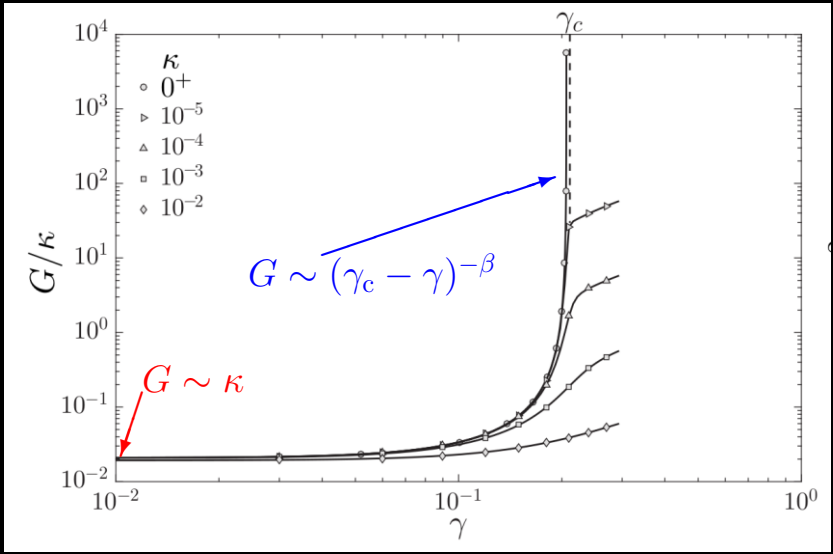
R. Rens, C. Villarroel, G. Düring, and EL, PRE 2018

shear modulus G in the presence of (weak) **bending** interactions:



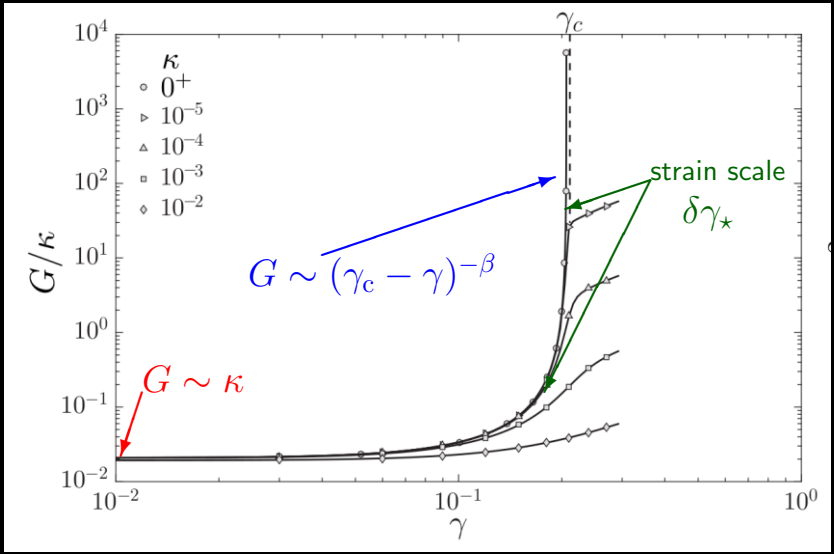
R. Rens, C. Villarroel, G. Düring, and EL, PRE 2018

shear modulus G in the presence of (weak) **bending** interactions:



R. Rens, C. Villarroel, G. Düring, and EL, PRE 2018

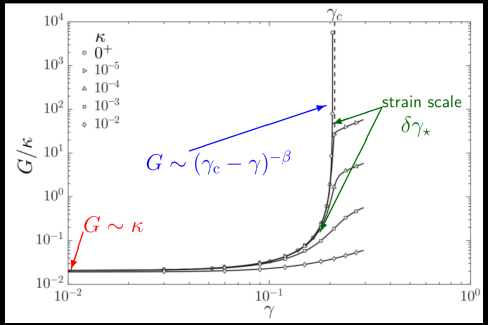
shear modulus G in the presence of (weak) **bending** interactions:



R. Rens, C. Villarroel, G. Düring, and EL, PRE 2018

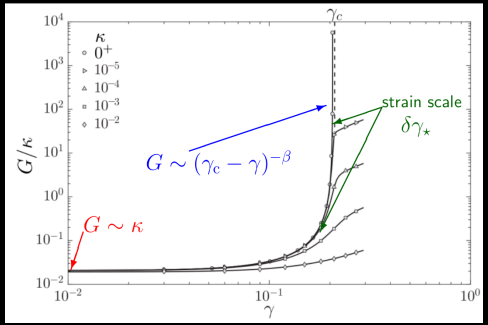
shear modulus G in the presence of (weak) **bending** interactions:

- in **isotropic** states $G \sim \kappa$
- if $\delta\gamma > \delta\gamma_\star(\kappa)$, $G \sim (\gamma_c - \gamma)^{-\beta}$



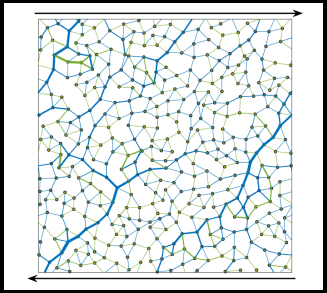
shear modulus G in the presence of (weak) **bending** interactions:

- in **isotropic** states $G \sim \kappa$
- if $\delta\gamma > \delta\gamma_\star(\kappa)$, $G \sim (\gamma_c - \gamma)^{-\beta}$

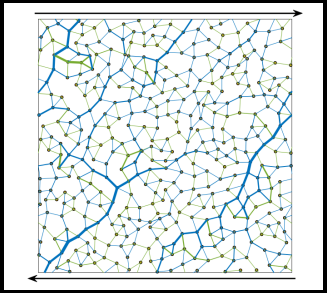


how can these observations be understood?

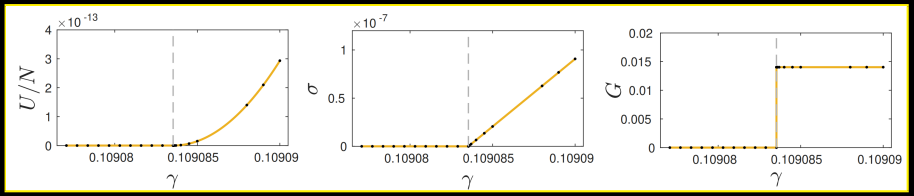
consider a shear-stiffened network with $\kappa = 0$;



consider a shear-stiffened network with $\kappa = 0$;

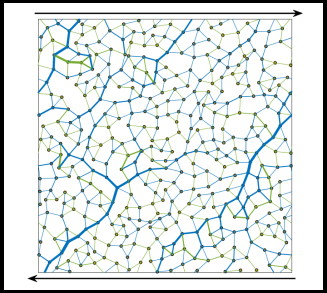


recall:



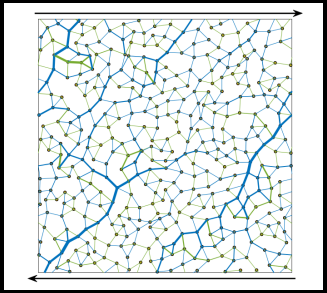
consider a shear-stiffened network with $\kappa = 0$;

counting DOF vs. interactions, $\sim N$ floppy modes exist



consider a shear-stiffened network with $\kappa = 0$;

counting DOF vs. interactions, $\sim N$ floppy modes exist

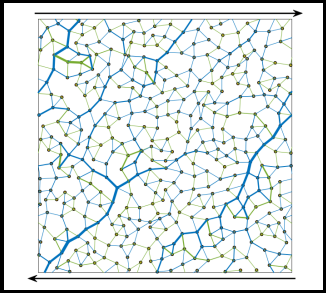


1) expand the energy in the floppy-mode space:

$$U(u) \simeq \frac{1}{2}\frac{\partial^2 U}{\partial x^2}u^2 + \frac{1}{6}\frac{\partial^3 U}{\partial x^3}u^3 + \frac{1}{24}\frac{\partial^4 U}{\partial x^4}u^4$$

consider a shear-stiffened network with $\kappa = 0$;

counting DOF vs. interactions, $\sim N$ floppy modes exist



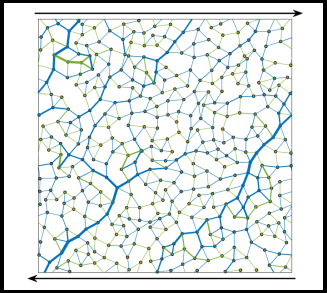
1) expand the energy in the floppy-mode space:

$$U(u) \simeq \cancel{\frac{1}{2}\frac{\partial^2 U}{\partial x^2}u^2} + \frac{1}{6}\frac{\partial^3 U}{\partial x^3}u^3 + \frac{1}{24}\frac{\partial^4 U}{\partial x^4}u^4$$

floppy modes

consider a shear-stiffened network with $\kappa = 0$;

counting DOF vs. interactions, $\sim N$ floppy modes exist



1) expand the energy in the floppy-mode space:

$$U(u) \simeq \cancel{\frac{1}{2}\frac{\partial^2 U}{\partial x^2}u^2} + \cancel{\frac{1}{6}\frac{\partial^3 U}{\partial x^3}u^3} + \frac{1}{24}\frac{\partial^4 U}{\partial x^4}u^4$$

floppy modes stability

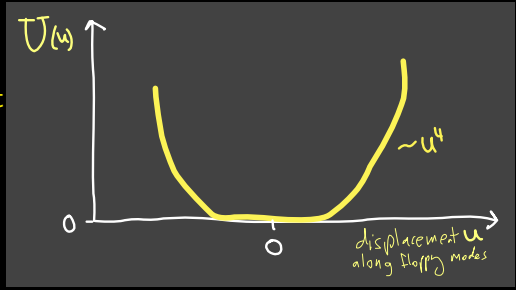
consider a shear-stiffened network with $\kappa = 0$;

counting DOF vs. interactions, $\sim N$ floppy modes exist

1) expand the energy in the floppy-mode space:

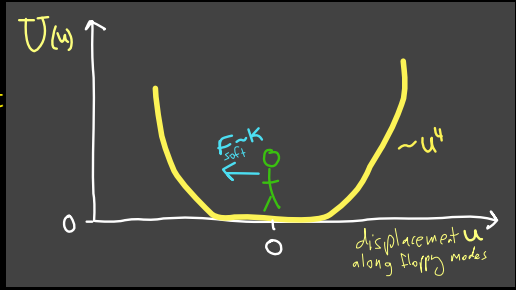
$$U(u) \simeq \cancel{\frac{1}{2}\frac{\partial^2 U}{\partial x^2}u^2} + \cancel{\frac{1}{6}\frac{\partial^3 U}{\partial x^3}u^3} + \frac{1}{24}\frac{\partial^4 U}{\partial x^4}u^4$$

floppy modes — stability



consider a shear-stiffened network with $\kappa = 0$;

counting DOF vs. interactions, $\sim N$ floppy modes exist



1) expand the energy in the floppy-mode space:

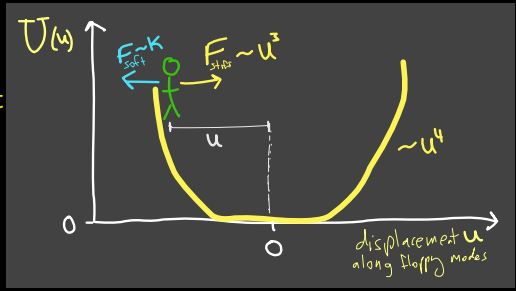
$$U(u) \simeq \cancel{\frac{1}{2}\frac{\partial^2 U}{\partial x^2}u^2} + \cancel{\frac{1}{6}\frac{\partial^3 U}{\partial x^3}u^3} + \frac{1}{24}\frac{\partial^4 U}{\partial x^4}u^4$$

floppy modes stability

2) turn on weak interactions. Nodes are now unballanced by a force $F_{\text{soft}} \sim \kappa$

consider a shear-stiffened network with $\kappa = 0$;

counting DOF vs. interactions, $\sim N$ floppy modes exist



1) expand the energy in the floppy-mode space:

$$U(u) \simeq \cancel{\frac{1}{2}\frac{\partial^2 U}{\partial x^2}u^2} + \cancel{\frac{1}{6}\frac{\partial^3 U}{\partial x^3}u^3} + \frac{1}{24}\frac{\partial^4 U}{\partial x^4}u^4$$

floppy modes stability

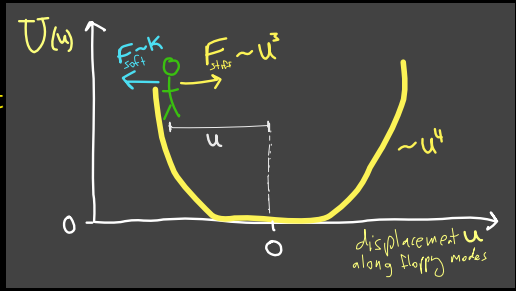
2) turn on weak interactions. Nodes are now unballanced by a force $F_{\text{soft}} \sim \kappa$

3) Nodes move a displacement $u_\star$ & recover mechanical equilibrium

when **anharmonic** force balances weak force: $F_{\text{soft}} \sim \kappa \sim F_{\text{stiff}} \sim u_\star^3$

consider a shear-stiffened network with $\kappa = 0$;

counting DOF vs. interactions, $\sim N$ floppy modes exist



1) expand the energy in the floppy-mode space:

$$U(u) \simeq \cancel{\frac{1}{2}\frac{\partial^2 U}{\partial x^2}u^2} + \cancel{\frac{1}{6}\frac{\partial^3 U}{\partial x^3}u^3} + \frac{1}{24}\frac{\partial^4 U}{\partial x^4}u^4$$

floppy modes — stability

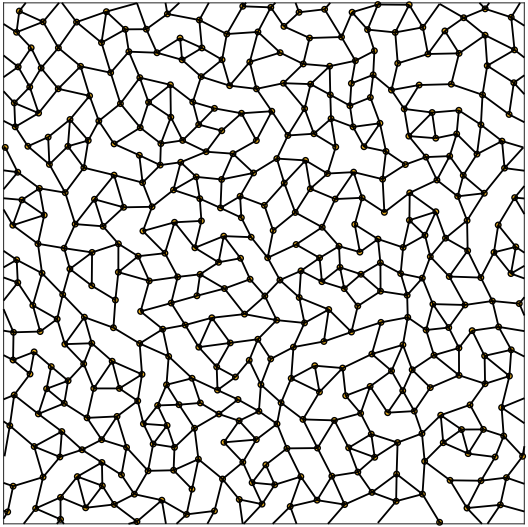
2) turn on weak interactions. Nodes are now unballanced by a force $F_{\rm soft} \sim \kappa$

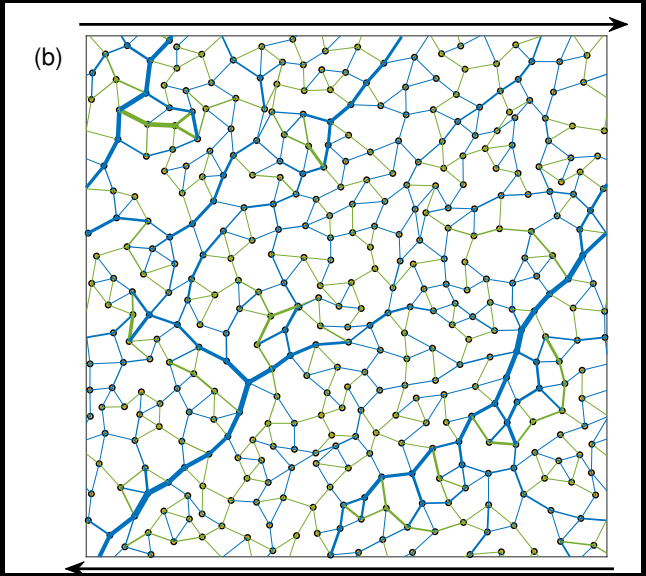
3) Nodes move a displacement $u_\star$ & recover mechanical equilibrium

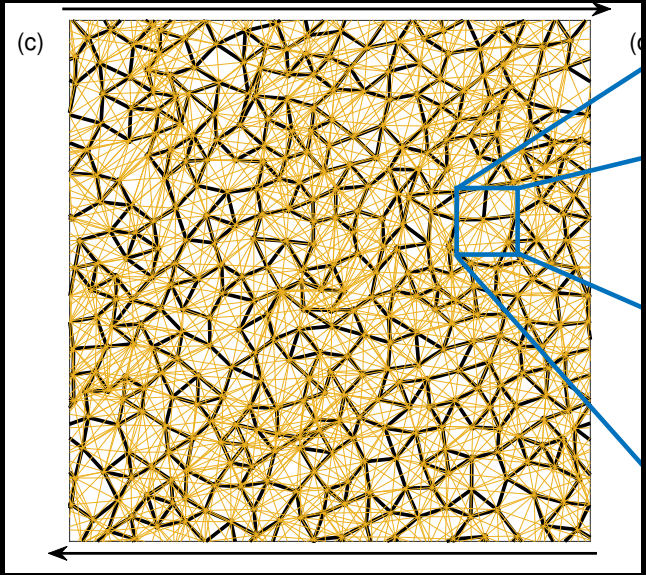
when **anharmonic** force balances weak force: $F_{\rm soft} \sim \kappa \sim F_{\rm stiff} \sim u_\star^3$

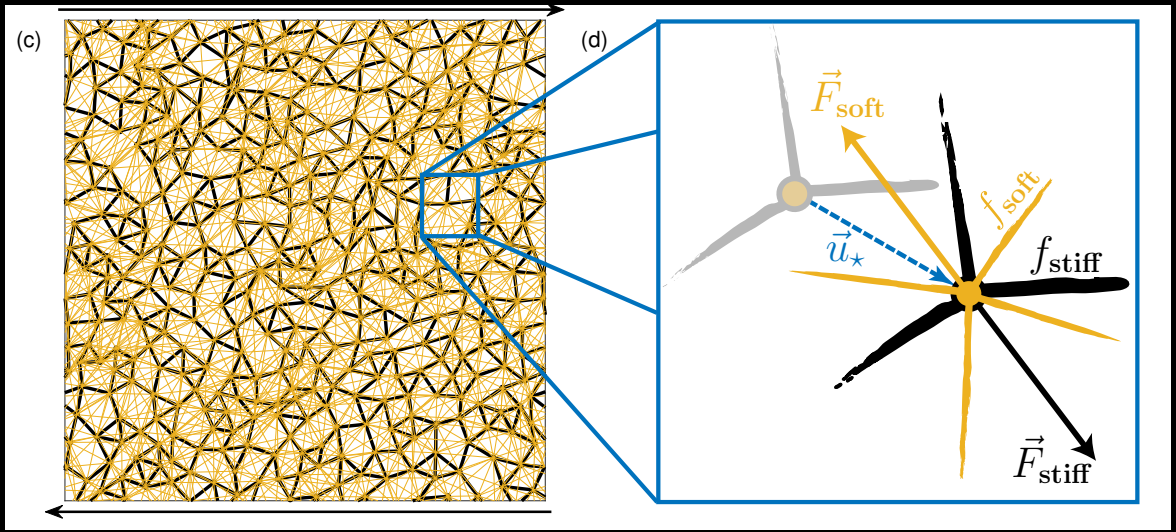
$$u_\star \sim \kappa^{1/3}$$

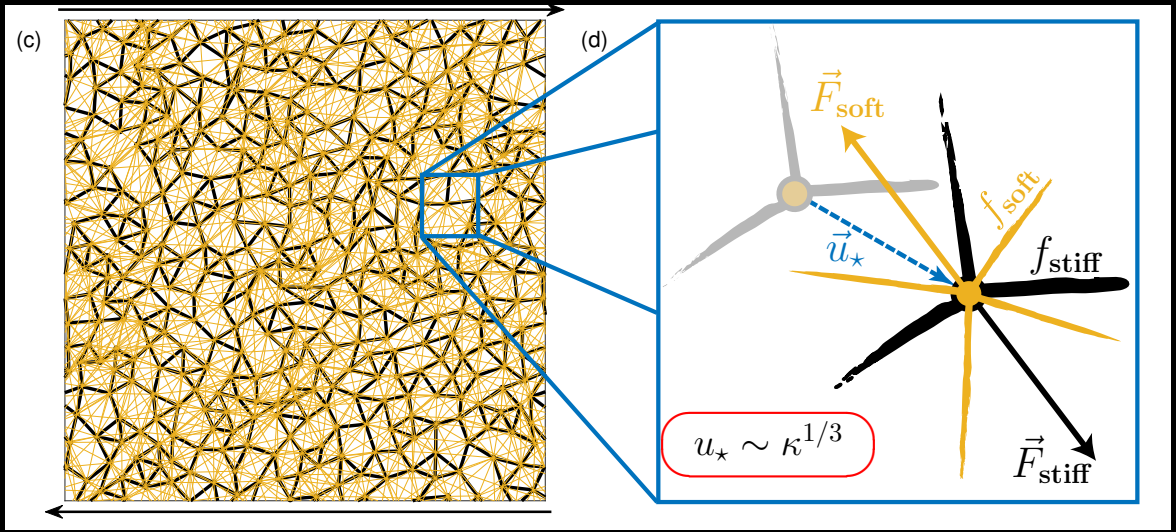
(a)



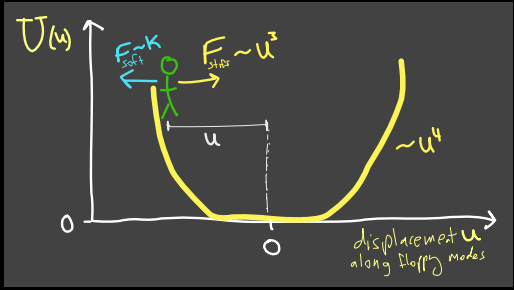

(b)


(c)


(c)
(d)
$\vec{F}_{\mathrm{soft}}$
f_{soft}
$\vec{u}_{\star}$
f_{stiff}
$\vec{F}_{\mathrm{stiff}}$

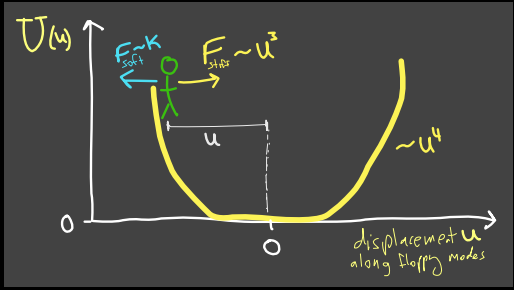

(c)
(d)
$\vec{F}_{\mathrm{soft}}$
f_{soft}
$\vec{u}_\star$
f_{stiff}
$u_\star \sim \kappa^{1/3}$
$\vec{F}_{\mathrm{stiff}}$

why do we need this 2-step perturbation approach?

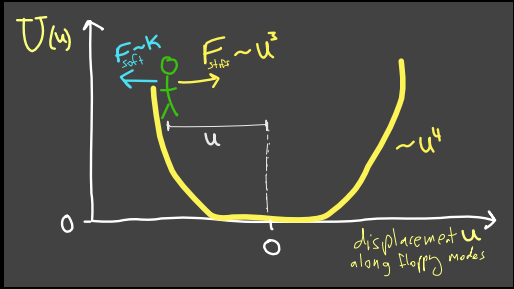


why do we need this 2-step perturbation approach?

1) theoretical handle (to be explained hereafter)

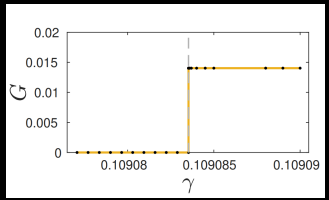


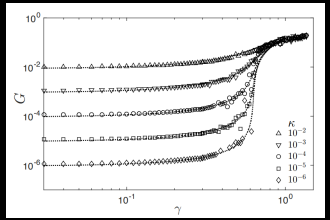
why do we need this 2-step perturbation approach?



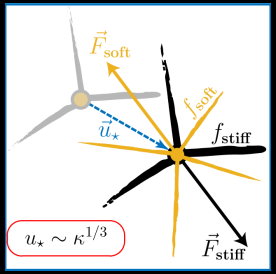
1) theoretical handle (to be explained hereafter)

2) allows to *simulate* systems **at the critical strain** (unfeasible otherwise)



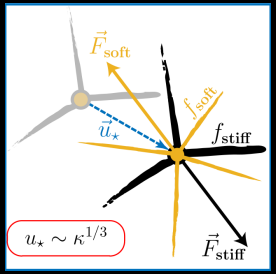


properties of perturbed ($\kappa > 0$), strain-stiffened states



properties of perturbed ($\kappa > 0$), strain-stiffened states

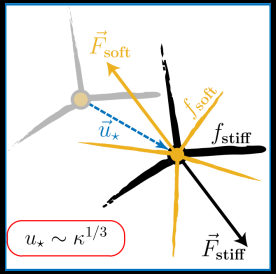
1) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **state-of-self-stress**



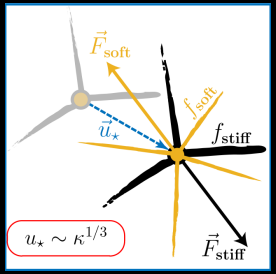
properties of perturbed ($\kappa > 0$), strain-stiffened states

1) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **state-of-self-stress**

$$\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle} \sim u_\star^2 \sim \kappa^{2/3}$$



properties of perturbed ($\kappa > 0$), strain-stiffened states



1) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **state-of-self-stress**

$$\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle} \sim u_\star^2 \sim \kappa^{2/3}$$

since the stiff network needs to balance the soft ($\sim \kappa$) force,
one expects an **amplification** over $\sim \kappa$:

shear stress $\quad \sigma \sim \kappa \Big/ \sqrt{\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle}} \sim \kappa^{2/3}$

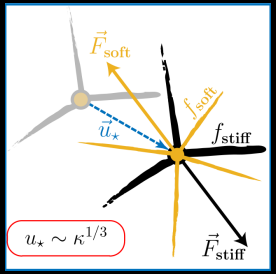
properties of perturbed ($\kappa > 0$), strain-stiffened states

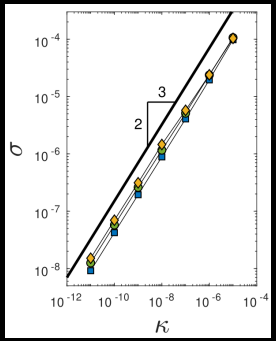
1) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **state-of-self-stress**

$$\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle} \sim u_\star^2 \sim \kappa^{2/3}$$

since the stiff network needs to balance the soft ($\sim \kappa$) force, one expects an **amplification** over $\sim \kappa$:

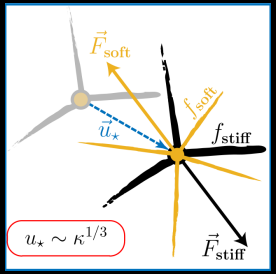
shear stress $\sigma \sim \kappa \Big/ \sqrt{\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle}} \sim \kappa^{2/3}$





properties of perturbed ($\kappa > 0$), strain-stiffened states

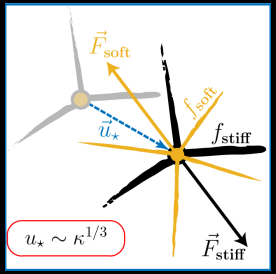
2) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **zero modes**:



properties of perturbed ($\kappa > 0$), strain-stiffened states

2) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **zero modes**:

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2 + \mathcal{H}_{\text{soft}}$$

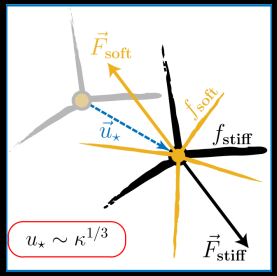


properties of perturbed ($\kappa > 0$), strain-stiffened states

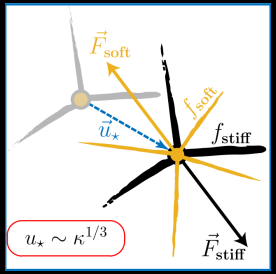
2) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **zero modes**:

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2 + \mathcal{H}_{\rm soft}$$

stiffness — $\mathcal{H}_1$ — force — $\mathcal{H}_2$ — bending — $\mathcal{H}_{\rm soft}$



properties of perturbed ($\kappa > 0$), strain-stiffened states



2) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **zero modes**:

$$\boldsymbol{\mathcal{H}} = \boldsymbol{\mathcal{H}}_1 + \boldsymbol{\mathcal{H}}_2 + \boldsymbol{\mathcal{H}}_{\rm soft}$$

stiffness force bending

stiffness term:

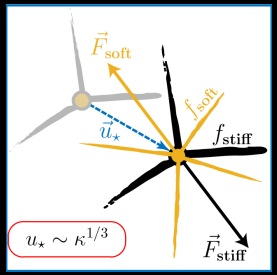
$$\boldsymbol{\mathcal{H}}_1 = \sum_{\langle i,j \rangle} \boldsymbol{n}_{ij} \bigotimes \boldsymbol{n}_{ij} \quad \Rightarrow \quad \delta\boldsymbol{\mathcal{H}}_1 \sim \delta\boldsymbol{n} \bigotimes \delta\boldsymbol{n} \sim u_\star^2 \sim \kappa^{2/3}$$

properties of perturbed ($\kappa > 0$), strain-stiffened states

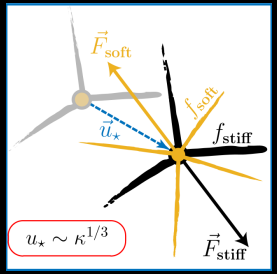
2) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **zero modes**:

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2 + \mathcal{H}_{\rm soft}$$

stiffness $\sim \kappa^{2/3}$ force bending



properties of perturbed ($\kappa > 0$), strain-stiffened states



2) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **zero modes**:

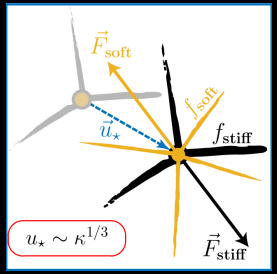
$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2 + \mathcal{H}_{\rm soft}$$

stiffness $\sim \kappa^{2/3}$ — force — bending

force term:

$$\mathcal{H}_2 \sim f \quad \Rightarrow \quad \delta\mathcal{H}_2 \sim f \sim \kappa \Big/ \sqrt{\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle}} \sim \kappa^{2/3}$$

properties of perturbed ($\kappa > 0$), strain-stiffened states



2) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **zero modes**:

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2 + \mathcal{H}_{\rm soft}$$

stiffness $\sim \kappa^{2/3}$ force $\sim \kappa^{2/3}$ bending

force term:

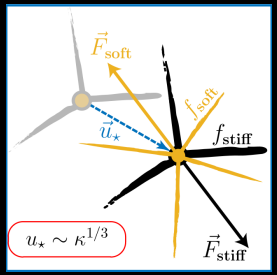
$$\mathcal{H}_2 \sim f \quad \Rightarrow \quad \delta\mathcal{H}_2 \sim f \sim \kappa \Big/ \sqrt{\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle}} \sim \kappa^{2/3}$$

properties of perturbed ($\kappa > 0$), strain-stiffened states

2) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **zero modes**:

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2 + \mathcal{H}_{\text{soft}}$$

stiffness $\sim \kappa^{2/3}$ force $\sim \kappa^{2/3}$ bending $\sim \kappa$

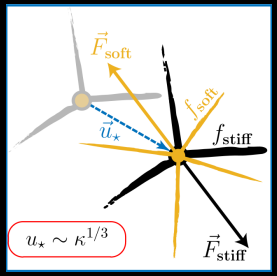


properties of perturbed ($\kappa > 0$), strain-stiffened states

2) displacements $u_\star$ distort (and ruin) the $\kappa = 0$ **zero modes**:

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2 + \mathcal{H}_{\text{soft}}$$

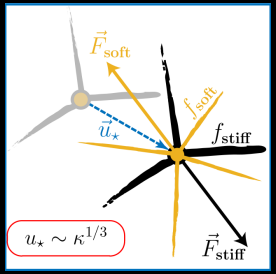
stiffness $\sim \kappa^{2/3}$ force $\sim \kappa^{2/3}$ bending $\sim \kappa$



new frequency of previously-zero-modes $\omega(\kappa) \sim \kappa^{1/3}$

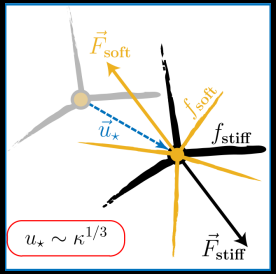
properties of perturbed ($\kappa > 0$), strain-stiffened states

3) shear modulus $G = G_{\text{affine}} + G_{\text{nonaffine}}$



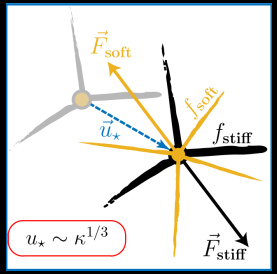
properties of perturbed ($\kappa > 0$), strain-stiffened states

3) shear modulus $G = \overbrace{G_{\text{affine}}}^{\text{regular}} + G_{\text{nonaffine}}$



properties of perturbed ($\kappa > 0$), strain-stiffened states

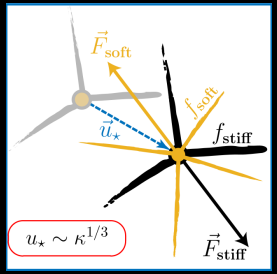
3) shear modulus $G =$ regular $G_{\text{affine}} +$?? $G_{\text{nonaffine}}$



properties of perturbed ($\kappa > 0$), strain-stiffened states

3) shear modulus $G =$ regular $G_{\text{affine}} +$?? $G_{\text{nonaffine}}$

why worry?

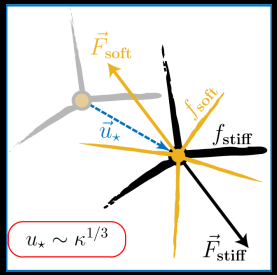


properties of perturbed ($\kappa > 0$), strain-stiffened states

3) shear modulus $G =$ regular $G_{\text{affine}} +$?? $G_{\text{nonaffine}}$

why worry?

$$G_{\text{nonaffine}} = \sum_{\ell} \frac{\left(\boldsymbol{F}_{\gamma} \cdot \boldsymbol{\psi}_{\ell}\right)^2}{\omega_{\ell}^2}$$



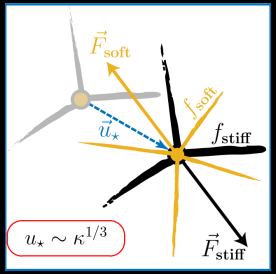
properties of perturbed ($\kappa > 0$), strain-stiffened states

3) shear modulus $G =$ regular $G_{\text{affine}} +$?? $G_{\text{nonaffine}}$

why worry?

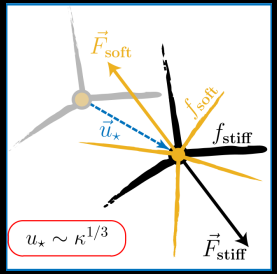
$$G_{\text{nonaffine}} = \sum_{\ell} \frac{\left(\boldsymbol{F}_{\gamma} \cdot \boldsymbol{\psi}_{\ell}\right)^2}{\omega_{\ell}^2}$$

$$\omega_{\text{soft}} \sim \kappa^{1/3}$$



properties of perturbed ($\kappa > 0$), strain-stiffened states

3) shear modulus $G =$ regular $G_{\text{affine}} +$?? $G_{\text{nonaffine}}$



why worry?

$$G_{\text{nonaffine}} = \sum_\ell \frac{\left(\boldsymbol{F}_\gamma \cdot \boldsymbol{\psi}_\ell\right)^2}{\omega_\ell^2}$$

$\omega_{\text{soft}} \sim \kappa^{1/3}$ but $\boldsymbol{F}_\gamma \equiv \frac{\partial^2 U}{\partial \gamma \partial \boldsymbol{x}} \simeq \mathcal{S}^T |\partial r / \partial \gamma\rangle$ (in the $\kappa \to 0$ limit)

and $\mathcal{S}|\boldsymbol{\psi}\rangle \sim \omega \sim \kappa^{1/3}$

properties of perturbed ($\kappa > 0$), strain-stiffened states

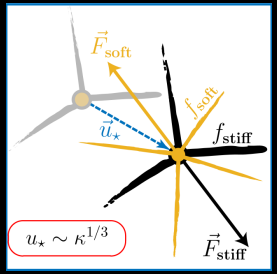
3) shear modulus $G =$ regular $G_{\text{affine}} +$?? $G_{\text{nonaffine}}$

why worry?

$G_{\text{nonaffine}} = \sum_\ell \frac{\left(\boldsymbol{F}_\gamma \cdot \boldsymbol{\psi}_\ell\right)^2}{\omega_\ell^2} \sim \frac{\kappa^{2/3}}{\kappa^{2/3}} \sim \kappa^0$ is regular too!

$\omega_{\text{soft}} \sim \kappa^{1/3}$ but $\boldsymbol{F}_\gamma \equiv \frac{\partial^2 U}{\partial\gamma\partial\boldsymbol{x}} \simeq \mathcal{S}^T|\partial r/\partial\gamma\rangle$ (in the $\kappa \to 0$ limit)

and $\mathcal{S}|\boldsymbol{\psi}\rangle \sim \omega \sim \kappa^{1/3}$

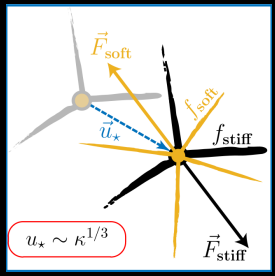


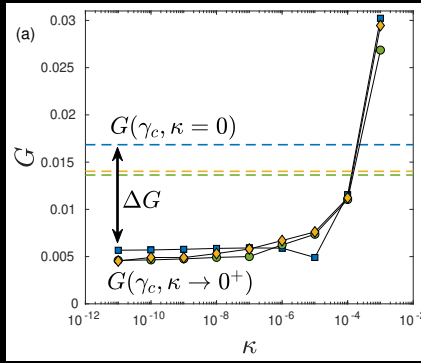
properties of perturbed ($\kappa > 0$), strain-stiffened states

3) shear modulus $G =$ regular $G_{\text{affine}} +$?? $G_{\text{nonaffine}}$

why worry?

$$G_{\text{nonaffine}} = \sum_{\ell} \frac{\left(\boldsymbol{F}_\gamma \cdot \boldsymbol{\psi}_\ell\right)^2}{\omega_\ell^2} \sim \frac{\kappa^{2/3}}{\kappa^{2/3}} \sim \kappa^0$$ is regular too!





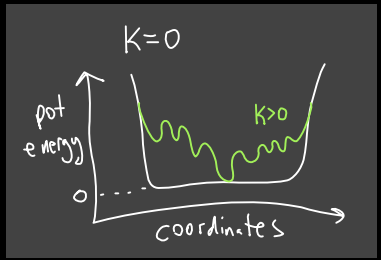
properties of perturbed ($\kappa > 0$), strain-stiffened states

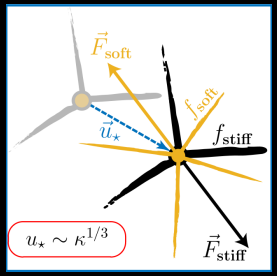
3) shear modulus $G =$ regular $G_{\text{affine}} +$?? $G_{\text{nonaffine}}$

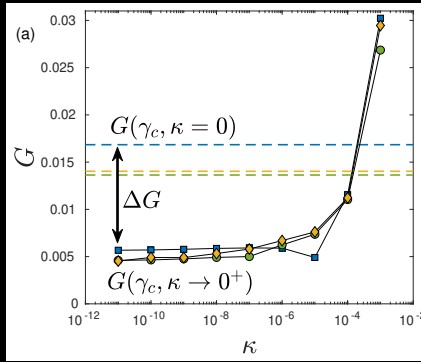
why worry?

$$G_{\text{nonaffine}} = \sum_{\ell} \frac{\left(\boldsymbol{F}_\gamma \cdot \boldsymbol{\psi}_\ell\right)^2}{\omega_\ell^2} \sim \frac{\kappa^{2/3}}{\kappa^{2/3}} \sim \kappa^0$$ is regular too!

again, $\kappa > 0$ is a **singular perturbation**



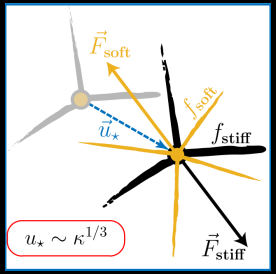




properties of perturbed ($\kappa > 0$), strain-stiffened states

4) nonlinear shear modulus $dG/d\gamma$

$$\frac{dG}{d\gamma} \simeq \frac{1}{V}\sum_{\ell mn} \frac{(\boldsymbol{\psi}_\ell \cdot \boldsymbol{F}_\gamma)(\boldsymbol{\psi}_m \cdot \boldsymbol{F}_\gamma)(\boldsymbol{\psi}_n \cdot \boldsymbol{F}_\gamma)\,(\boldsymbol{\mathcal{U}}''' \vdots \boldsymbol{\psi}_\ell \boldsymbol{\psi}_m \boldsymbol{\psi}_n)}{\omega_\ell^2 \omega_m^2 \omega_n^2} + \mathcal{O}(\boldsymbol{\mathcal{H}}^{-2})$$



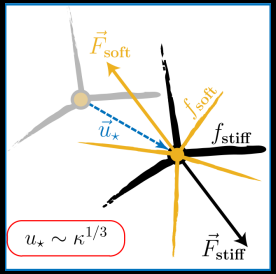
properties of perturbed ($\kappa > 0$), strain-stiffened states

4) nonlinear shear modulus $dG/d\gamma$

$$\frac{dG}{d\gamma} \simeq \frac{1}{V}\sum_{\ell mn} \frac{(\boldsymbol{\psi}_\ell \cdot \boldsymbol{F}_\gamma)(\boldsymbol{\psi}_m \cdot \boldsymbol{F}_\gamma)(\boldsymbol{\psi}_n \cdot \boldsymbol{F}_\gamma)\,(\boldsymbol{\mathcal{U}}''' \therefore \boldsymbol{\psi}_\ell \boldsymbol{\psi}_m \boldsymbol{\psi}_n)}{\omega_\ell^2 \omega_m^2 \omega_n^2} + \mathcal{O}(\boldsymbol{\mathcal{H}}^{-2})$$

for soft modes $\boldsymbol{\psi}$: $(i)\ \boldsymbol{\psi} \cdot \boldsymbol{F}_\gamma \sim \omega_{\text{soft}} \sim \kappa^{1/3}$

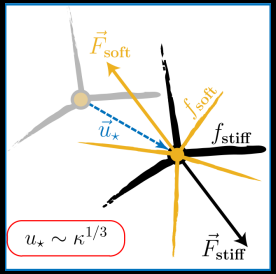
$(ii)\ \boldsymbol{\mathcal{U}}''' \therefore \boldsymbol{\psi}\boldsymbol{\psi}\boldsymbol{\psi} \sim u_\star \sim \kappa^{1/3}$



properties of perturbed ($\kappa > 0$), strain-stiffened states

4) nonlinear shear modulus $dG/d\gamma$

$$\frac{dG}{d\gamma} \simeq \frac{1}{V}\sum_{\ell mn} \frac{(\boldsymbol{\psi}_\ell \cdot \boldsymbol{F}_\gamma)(\boldsymbol{\psi}_m \cdot \boldsymbol{F}_\gamma)(\boldsymbol{\psi}_n \cdot \boldsymbol{F}_\gamma)\,(\boldsymbol{\mathcal{U}}''' \therefore \boldsymbol{\psi}_\ell \boldsymbol{\psi}_m \boldsymbol{\psi}_n)}{\omega_\ell^2 \omega_m^2 \omega_n^2} + \mathcal{O}(\boldsymbol{\mathcal{H}}^{-2})$$



for soft modes ψ: $(i)\ \boldsymbol{\psi} \cdot \boldsymbol{F}_\gamma \sim \omega_{\rm soft} \sim \kappa^{1/3}$

$(ii)\ \boldsymbol{\mathcal{U}}''' \therefore \boldsymbol{\psi}\boldsymbol{\psi}\boldsymbol{\psi} \sim u_\star \sim \kappa^{1/3}$

$$\Rightarrow \frac{dG}{d\gamma} \sim \frac{\kappa^{4/3}}{\kappa^2} \sim \kappa^{-2/3}$$

properties of perturbed ($\kappa > 0$), strain-stiffened states

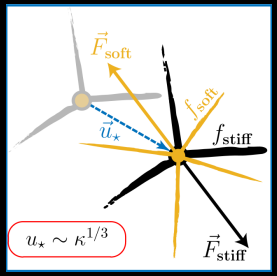
4) nonlinear shear modulus $dG/d\gamma$

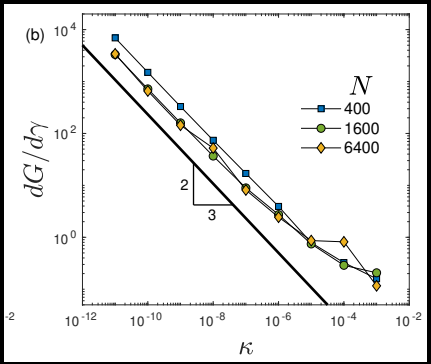
$$\frac{dG}{d\gamma} \simeq \frac{1}{V}\sum_{\ell mn} \frac{(\boldsymbol{\psi}_\ell \cdot \boldsymbol{F}_\gamma)(\boldsymbol{\psi}_m \cdot \boldsymbol{F}_\gamma)(\boldsymbol{\psi}_n \cdot \boldsymbol{F}_\gamma)\,(\boldsymbol{\mathcal{U}}''' \vdots \boldsymbol{\psi}_\ell \boldsymbol{\psi}_m \boldsymbol{\psi}_n)}{\omega_\ell^2 \omega_m^2 \omega_n^2} + \mathcal{O}(\boldsymbol{\mathcal{H}}^{-2})$$

for soft modes ψ: $(i)\ \boldsymbol{\psi} \cdot \boldsymbol{F}_\gamma \sim \omega_{\text{soft}} \sim \kappa^{1/3}$

$(ii)\ \boldsymbol{\mathcal{U}}''' \vdots \boldsymbol{\psi}\boldsymbol{\psi}\boldsymbol{\psi} \sim u_\star \sim \kappa^{1/3}$

$$\Rightarrow \frac{dG}{d\gamma} \sim \frac{\kappa^{4/3}}{\kappa^2} \sim \kappa^{-2/3}$$

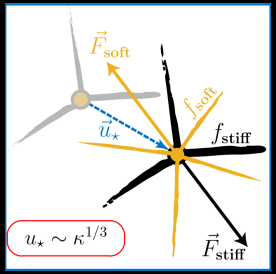




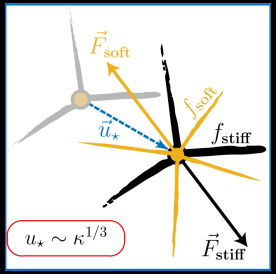
properties of perturbed ($\kappa > 0$), strain-stiffened states

5) nonaffine displacements $\vec{u}_{\text{na}}$

$$u_{\text{na}}^2 \simeq \sum_{\ell} \frac{(\boldsymbol{\psi}_\ell \cdot \boldsymbol{F}_\gamma)^2}{\omega_\ell^4}$$



properties of perturbed ($\kappa > 0$), strain-stiffened states

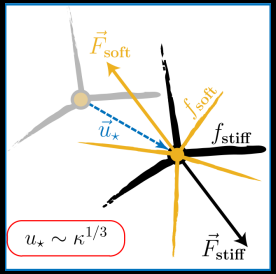


5) nonaffine displacements $\vec{u}_{\rm na}$

$$u_{\rm na}^2 \simeq \sum_\ell \frac{(\boldsymbol{\psi}_\ell \cdot \boldsymbol{F}_\gamma)^2}{\omega_\ell^4}$$

for soft modes ψ: $\boldsymbol{\psi} \cdot \boldsymbol{F}_\gamma \sim \omega_{\rm soft} \sim \kappa^{1/3}$

properties of perturbed ($\kappa > 0$), strain-stiffened states



5) nonaffine displacements $\vec{u}_{\rm na}$

$$u_{\rm na}^2 \simeq \sum_\ell \frac{(\boldsymbol{\psi}_\ell \cdot \boldsymbol{F}_\gamma)^2}{\omega_\ell^4}$$

for soft modes ψ: $\boldsymbol{\psi} \cdot \boldsymbol{F}_\gamma \sim \omega_{\rm soft} \sim \kappa^{1/3}$

$$\Rightarrow u_{\rm na}^2 \sim \frac{\kappa^{2/3}}{\kappa^{4/3}} \sim \kappa^{-2/3}$$

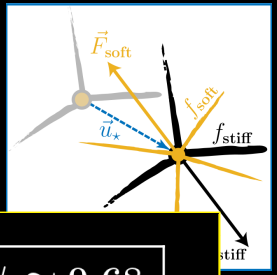
properties of perturbed ($\kappa > 0$), strain-stiffened states

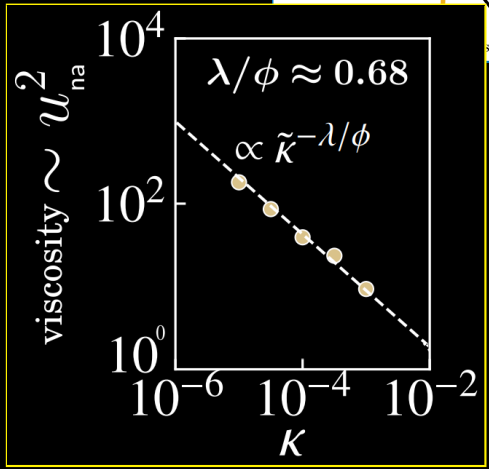
5) nonaffine displacements $\vec{u}_{\text{na}}$

$$u_{\text{na}}^2 \simeq \sum_\ell \frac{(\boldsymbol{\psi}_\ell \cdot \boldsymbol{F}_\gamma)^2}{\omega_\ell^4}$$

for soft modes ψ: $\boldsymbol{\psi} \cdot \boldsymbol{F}_\gamma \sim \omega_{\text{soft}} \sim \kappa^{1/3}$

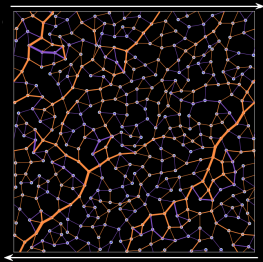
$$\Rightarrow u_{\text{na}}^2 \sim \frac{\kappa^{2/3}}{\kappa^{4/3}} \sim \kappa^{-2/3}$$





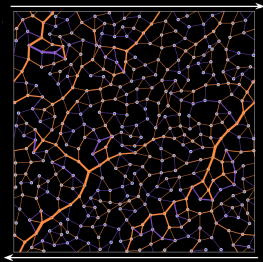
Shivers, Sharma, MacKintosh, PRL **131**, 178201 (2023)

properties of perturbed ($\kappa > 0$), strain-stiffened states – summary

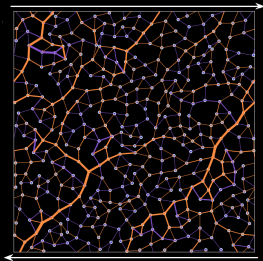


properties of perturbed ($\kappa > 0$), strain-stiffened states – summary

(i) state-of-self-stress destroyed by $\sqrt{\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle}} \sim \kappa^{1/3}$



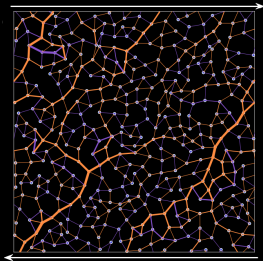
properties of perturbed ($\kappa > 0$), strain-stiffened states – summary



(i) state-of-self-stress destroyed by $\sqrt{\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle}} \sim \kappa^{1/3}$

(ii) floppy modes acquire finite frequency $\omega_{\rm soft} \sim \kappa^{1/3}$

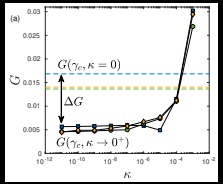
properties of perturbed ($\kappa > 0$), strain-stiffened states – summary



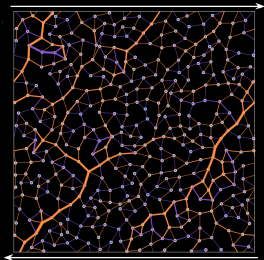
(i) state-of-self-stress destroyed by $\sqrt{\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle}} \sim \kappa^{1/3}$

(ii) floppy modes acquire finite frequency $\omega_{\text{soft}} \sim \kappa^{1/3}$

(iii) shear modulus $G \sim \kappa^0$



properties of perturbed ($\kappa > 0$), strain-stiffened states – summary

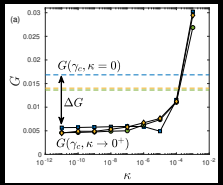


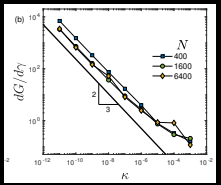
(i) state-of-self-stress destroyed by $\sqrt{\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle}} \sim \kappa^{1/3}$

(ii) floppy modes acquire finite frequency $\omega_{\text{soft}} \sim \kappa^{1/3}$

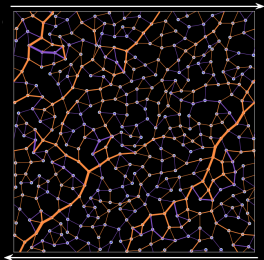
(iii) shear modulus $G \sim \kappa^0$

(iv) nonlinear modulus $\frac{dG}{d\gamma} \sim \kappa^{-2/3}$





properties of perturbed ($\kappa > 0$), strain-stiffened states – summary



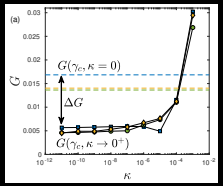
(i) state-of-self-stress destroyed by $\sqrt{\frac{\langle f|\mathcal{S}\mathcal{S}^T|f\rangle}{\langle f|f\rangle}} \sim \kappa^{1/3}$

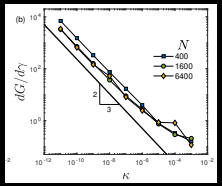
(ii) floppy modes acquire finite frequency $\omega_{\text{soft}} \sim \kappa^{1/3}$

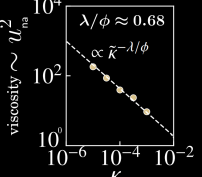
(iii) shear modulus $G \sim \kappa^0$

(iv) nonlinear modulus $\frac{dG}{d\gamma} \sim \kappa^{-2/3}$

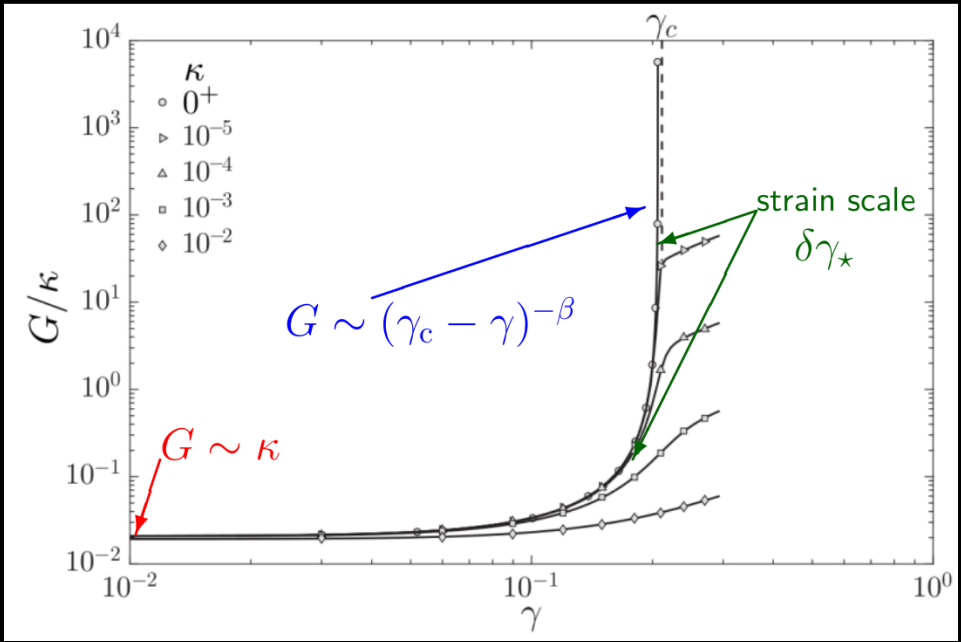
(v) nonaffine displacements $u_{\text{n.a.}} \sim\sim \kappa^{-2/3}$







recap:


γ_c
κ
0^+
10^{-5}
10^{-4}
10^{-3}
10^{-2}
strain scale
$\delta\gamma_\star$
$G \sim (\gamma_c - \gamma)^{-\beta}$
$G \sim \kappa$
G/κ
γ
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-2}
10^{-1}
10^0

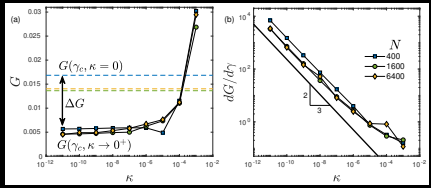
scaling theory for the shear modulus G:

scaling theory for the shear modulus G:

(i) we start with the ansatz: $G(\gamma, \kappa) \sim \mathcal{F}\left(\dfrac{\gamma_c - \gamma}{\delta\gamma_\star(\kappa)}\right)$

scaling theory for the shear modulus G:

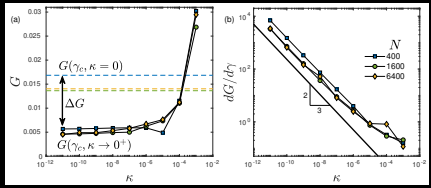
(i) we start with the ansatz: $G(\gamma,\kappa) \sim \mathcal{F}\left(\dfrac{\gamma_c-\gamma}{\delta\gamma_\star(\kappa)}\right)$



(ii) since $\dfrac{dG}{d\gamma} \sim \dfrac{1}{\kappa^{2/3}}$ and $\dfrac{dG}{d\gamma} = \dfrac{d\mathcal{F}/dx}{\delta\gamma_\star(\kappa)}$ then the **strain scale** $\delta\gamma_\star \sim \kappa^{2/3}$ (& $d\mathcal{F}/dx$ is finite)

scaling theory for the shear modulus G:

(i) we start with the ansatz: $G(\gamma,\kappa) \sim \mathcal{F}\left(\dfrac{\gamma_c-\gamma}{\delta\gamma_\star(\kappa)}\right)$

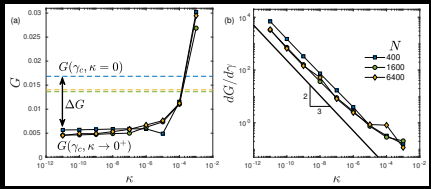


(ii) since $\dfrac{dG}{d\gamma} \sim \dfrac{1}{\kappa^{2/3}}$ and $\dfrac{dG}{d\gamma} = \dfrac{d\mathcal{F}/dx}{\delta\gamma_\star(\kappa)}$ then the **strain scale** $\delta\gamma_\star \sim \kappa^{2/3}$ (& $d\mathcal{F}/dx$ is finite)

(iii) since $G(\gamma_c) \sim \kappa^0$ is finite, $\mathcal{F}(0) = G(\gamma_c)$

scaling theory for the shear modulus G:

(i) we start with the ansatz: $G(\gamma,\kappa) \sim \mathcal{F}\left(\dfrac{\gamma_c-\gamma}{\delta\gamma_\star(\kappa)}\right)$



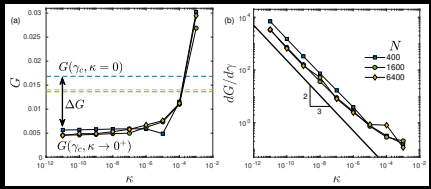
(ii) since $\dfrac{dG}{d\gamma} \sim \dfrac{1}{\kappa^{2/3}}$ and $\dfrac{dG}{d\gamma} = \dfrac{d\mathcal{F}/dx}{\delta\gamma_\star(\kappa)}$ then the **strain scale** $\delta\gamma_\star \sim \kappa^{2/3}$ (& $d\mathcal{F}/dx$ is finite)

(iii) since $G(\gamma_c) \sim \kappa^0$ is finite, $\mathcal{F}(0) = G(\gamma_c)$

(iv) since $\left.\dfrac{d\mathcal{F}}{dx}\right|_{x=0}$ is finite, then $G(\gamma_c) - G(\gamma) \sim \dfrac{\gamma_c-\gamma}{\kappa^{2/3}}$ for $\gamma_c-\gamma \lesssim \kappa^{2/3}$

scaling theory for the shear modulus G:

(i) we start with the ansatz: $G(\gamma,\kappa) \sim \mathcal{F}\left(\dfrac{\gamma_c-\gamma}{\delta\gamma_\star(\kappa)}\right)$



(ii) since $\dfrac{dG}{d\gamma} \sim \dfrac{1}{\kappa^{2/3}}$ and $\dfrac{dG}{d\gamma} = \dfrac{d\mathcal{F}/dx}{\delta\gamma_\star(\kappa)}$ then the **strain scale** $\delta\gamma_\star \sim \kappa^{2/3}$ (& $d\mathcal{F}/dx$ is finite)

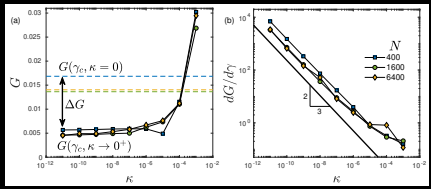
(iii) since $G(\gamma_c) \sim \kappa^0$ is finite, $\mathcal{F}(0) = G(\gamma_c)$

(iv) since $\left.\dfrac{d\mathcal{F}}{dx}\right|_{x=0}$ is finite, then $G(\gamma_c) - G(\gamma) \sim \dfrac{\gamma_c-\gamma}{\kappa^{2/3}}$ for $\gamma_c-\gamma \lesssim \kappa^{2/3}$

(v) since $G \sim \kappa$ for $\gamma \ll \gamma_c$ then $\mathcal{F}(x) \sim x^{-3/2}$, or $G \sim \dfrac{\kappa}{(\gamma_c-\gamma)^{3/2}}$

scaling theory for the shear modulus G:

(i) we start with the ansatz: $G(\gamma,\kappa) \sim \mathcal{F}\left(\dfrac{\gamma_c-\gamma}{\delta\gamma_\star(\kappa)}\right)$



(ii) since $\dfrac{dG}{d\gamma} \sim \dfrac{1}{\kappa^{2/3}}$ and $\dfrac{dG}{d\gamma} = \dfrac{d\mathcal{F}/dx}{\delta\gamma_\star(\kappa)}$ then the **strain scale** $\delta\gamma_\star \sim \kappa^{2/3}$ (& $d\mathcal{F}/dx$ is finite)

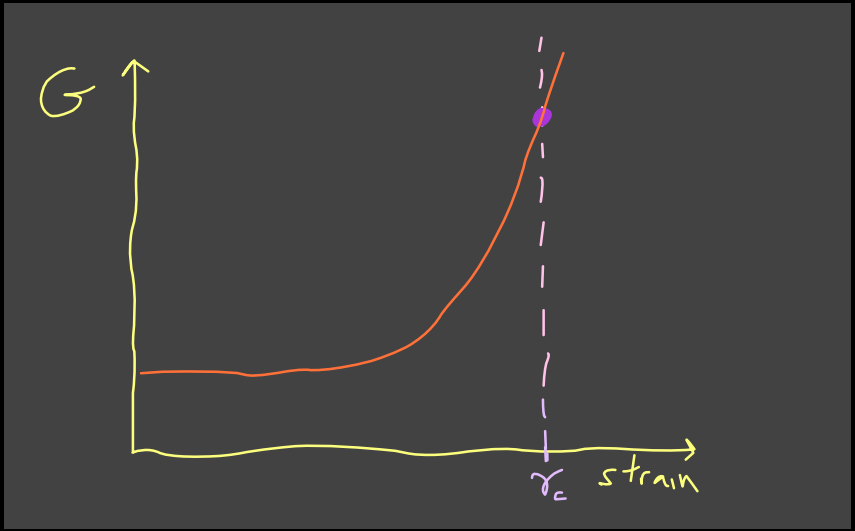
(iii) since $G(\gamma_c) \sim \kappa^0$ is finite, $\mathcal{F}(0) = G(\gamma_c)$

(iv) since $\left.\dfrac{d\mathcal{F}}{dx}\right|_{x=0}$ is finite, then $G(\gamma_c) - G(\gamma) \sim \dfrac{\gamma_c-\gamma}{\kappa^{2/3}}$ for $\gamma_c - \gamma \lesssim \kappa^{2/3}$

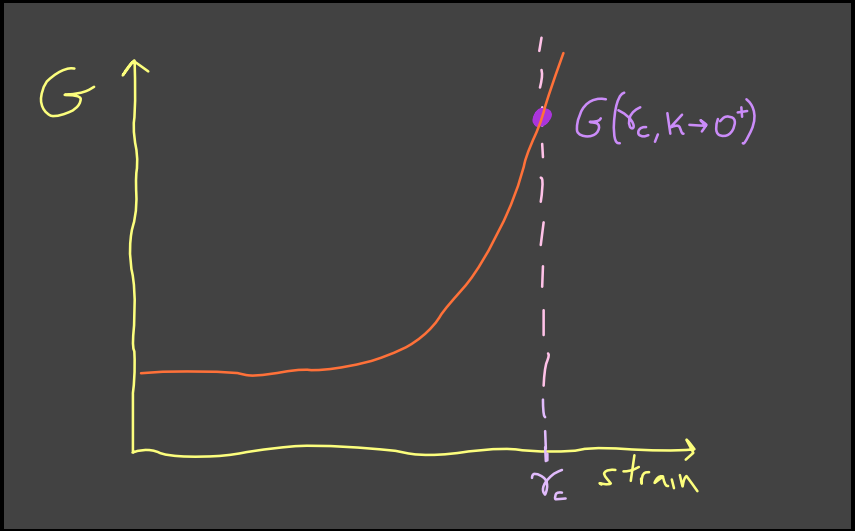
(v) since $G \sim \kappa$ for $\gamma \ll \gamma_c$ then $\mathcal{F}(x) \sim x^{-3/2}$, or $G \sim \dfrac{\kappa}{(\gamma_c-\gamma)^{3/2}}$

predictions from scaling theory away from γ_{c}

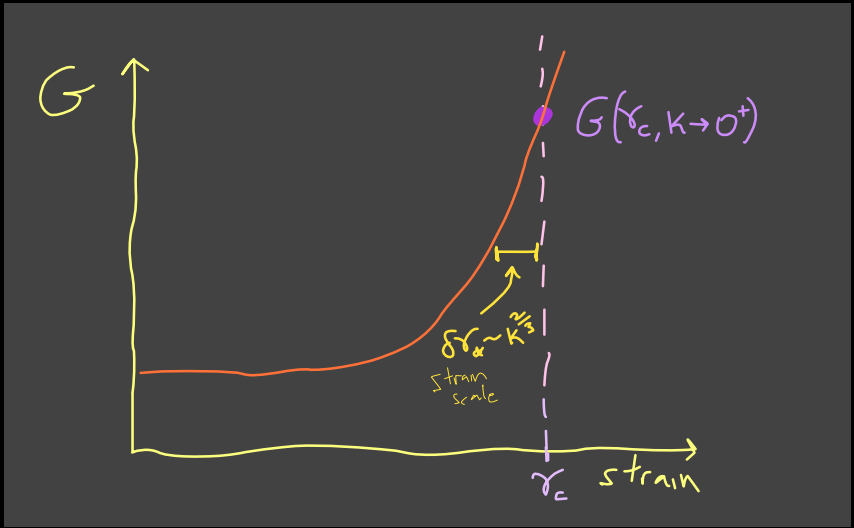
predictions from scaling theory away from γ_c



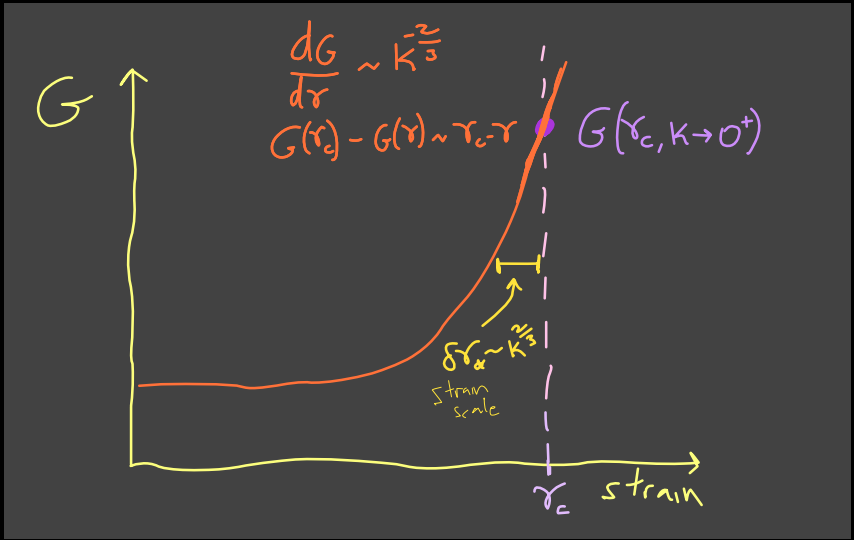
predictions from scaling theory away from γ_c



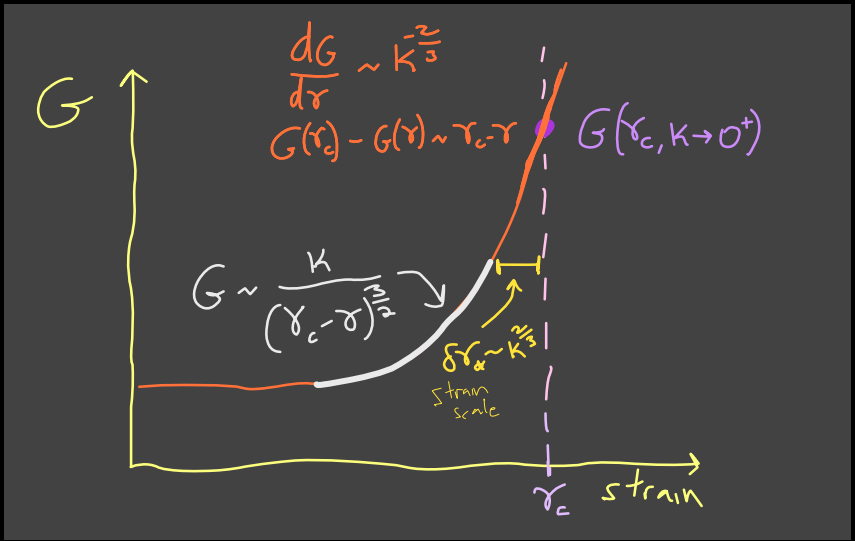
predictions from scaling theory away from γ_c



predictions from scaling theory away from γ_c

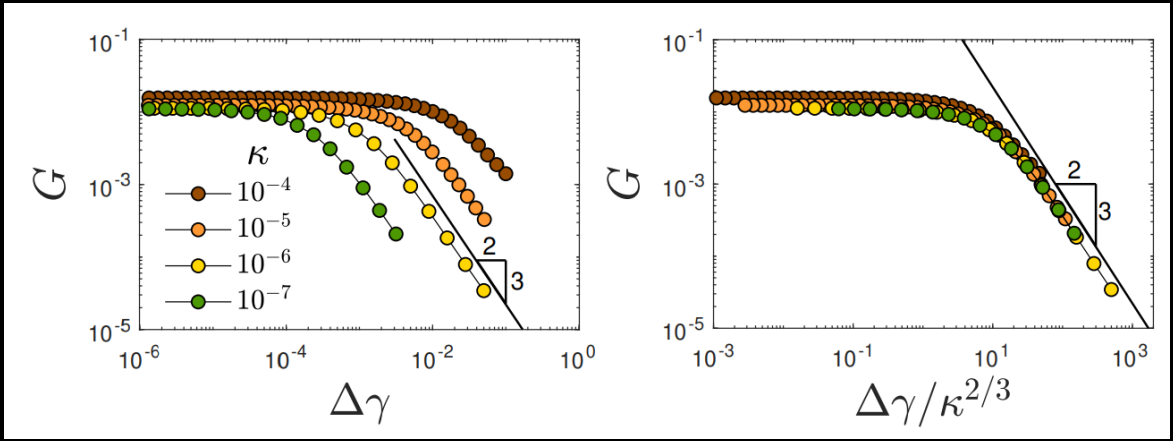


predictions from scaling theory away from γ_c



numerical validation of predictions away from γ_c

numerical validation of predictions away from γ_c



Zhang, Bouchbinder, and EL, J. Chem. Phys. **163**, 111102 (2025)

scaling theory of strain stiffening – summary

scaling theory of strain stiffening – summary

$\rightarrow$ **2-step procedure:** strain with $\kappa = 0$, then turn on $\kappa > 0$

scaling theory of strain stiffening – summary

→ **2-step procedure:** strain with $\kappa = 0$, then turn on $\kappa > 0$

→ we argue and validate that $G \sim \kappa^0$ and $dG/d\gamma \sim \kappa^{-2/3}$

scaling theory of strain stiffening – summary

→ **2-step procedure:** strain with $\kappa = 0$, then turn on $\kappa > 0$

→ we argue and validate that $G \sim \kappa^0$ and $dG/d\gamma \sim \kappa^{-2/3}$

→ simplest scaling ansatz $G(\gamma, \kappa) \sim \mathcal{F}\left(\dfrac{\gamma_c - \gamma}{\delta\gamma_\star(\kappa)}\right)$

scaling theory of strain stiffening – summary

→ **2-step procedure:** strain with $\kappa = 0$, then turn on $\kappa > 0$

→ we argue and validate that $G \sim \kappa^0$ and $dG/d\gamma \sim \kappa^{-2/3}$

→ simplest scaling ansatz $G(\gamma, \kappa) \sim \mathcal{F}\left(\dfrac{\gamma_c - \gamma}{\delta\gamma_\star(\kappa)}\right)$

→ predictions: (i) **linear** variation of $G \sim \gamma_c - \gamma$ with strain below the critical strain γ_c

(ii) strain scale $\delta\gamma_\star \sim \kappa^{2/3}$

(iii) scaling away from the critical strain $G \sim \dfrac{\kappa}{(\gamma_c - \gamma)^{3/2}}$

acknowledgments

Matthieu Wyart

Gustavo Düring

Eran Bouchbinder

JOHNS HOPKINS
KRIEGER SCHOOL
of ARTS & SCIENCES

PONTIFICIA
UNIVERSIDAD
CATÓLICA
DE CHILE

WEIZMANN
INSTITUTE
OF SCIENCE

references

→ packings physics:

Stefan Kooij and **EL**, Phys. Rev. E **100**, 042609 (2019)

→ suspension viscosity:

EL, Gustavo Düring, and Matthieu Wyart, PNAS **109**, 4798-4803 (2012)

Gustavo Düring, **EL**, and Matthieu Wyart, Phys. Rev. E **89**, 022305 (2014)

Gustavo Düring, **EL**, and Matthieu Wyart, Phys. Rev. E **94**, 022601 (2016)

→ strain stiffening

EL and Eran Bouchbinder, Extreme Mech. Lett. **65**, 102104 (2023)

Zibin Zhang, Eran Bouchbinder, and **EL**, J. Chem. Phys. **163**, 111102 (2025)